MTH2008: Real Analysis

Summary of the module

By taking this course at the University of Exeter, you will develop a wide range of mathematical readiness. This course will give you a deeper understanding of convergence, continuity and differentiability, together with powerful extensions of these that play a major role in mathematics. Many areas in applied methods were built using fundamental results of Real Analysis. You will be introduced to the pioneering work of Cauchy, Riemann and many other notable mathematicians.

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1 The Real Numbers

IN THIS CHAPTER we begin the study of the real number system.
SECTION 1.1 deals with the axioms that define the real numbers, definitions based on them, and some basic properties that follow from them.
SECTION 1.2 introduces basic ideas of set theory in the context of sets of real numbers. In this section we prove two fundamental theorems: the Heine–Borel and Bolzano–Weierstrass theorems.

1.1 The Real Number System

Having taken calculus, you know a lot about the real number system; however, you probably do not know that all its properties follow from a few basic ones. Although we will not carry out the development of the real number system from these basic properties, it is useful to state them as a starting point for the study of real analysis and also to focus on one property, completeness, that is probably new to you.

Field Properties

The real number system (which we will often call simply the *reals*) is first of all a set $\{a, b, c, \dots\}$ on which the operations of addition and multiplication are defined so that every pair of real numbers has a unique sum and product, both real numbers, with the following properties.

(A) $a + b = b + a$ and $ab = ba$ (commutative laws).

(B) $(a + b) + c = a + (b + c)$ and $(ab)c = a(bc)$ (associative laws).

(C) $a(b + c) = ab + ac$ (distributive law).

(D) There are distinct real numbers 0 (additive identity) and 1 (multiplicative identity) such that $a + 0 = a$ and $a1 = a$ for all a.

(E) For each a there is a real number $-a$ such that $a + (-a) = 0$, and if $a \neq 0$, there is a real number $1/a$ such that $a(1/a) = 1$.

The manipulative properties of the real numbers, such as the relations

$$(a + b)^2 = a^2 + 2ab + b^2,$$
$$(3a + 2b)(4c + 2d) = 12ac + 6ad + 8bc + 4bd,$$
$$(-a) = (-1)a, \quad a(-b) = (-a)b = -ab,$$

and

$$\frac{a}{b} + \frac{c}{d} = \frac{ad + bc}{bd} \quad (b, d \neq 0),$$

all follow from **(A)–(E)**. We assume that you are familiar with these properties.

A set on which two operations are defined so as to have properties **(A)–(E)** is called a *field*. The real number system is by no means the only field. The *rational numbers* (which are the real numbers that can be written as $r = p/q$, where p and q are integers and $q \neq 0$) also form a field under addition and multiplication. The simplest possible field consists of two elements, which we denote by 0 and 1, with addition defined by

$$0 + 0 = 1 + 1 = 0, \quad 1 + 0 = 0 + 1 = 1, \tag{1.1}$$

and multiplication defined by

$$0 \cdot 0 = 0 \cdot 1 = 1 \cdot 0 = 0, \quad 1 \cdot 1 = 1 \tag{1.2}$$

The Order Relation

The real number system is ordered by the relation $<$, which has the following properties.

(F) For each pair of real numbers a and b, exactly one of the following is true:

$$a = b, \quad a < b, \quad \text{or} \quad b < a.$$

(G) If $a < b$ and $b < c$, then $a < c$. (The relation $<$ is *transitive*.)

(H) If $a < b$, then $a + c < b + c$ for any c, and if $0 < c$, then $ac < bc$.

A field with an order relation satisfying **(F)–(H)** is an *ordered* field. Thus, the real numbers form an ordered field. The rational numbers also form an ordered field, but it is impossible to define an order on the field with two elements defined by (1.1) and (1.2) so as to make it into an ordered field.

We assume that you are familiar with other standard notation connected with the order relation: thus, $a > b$ means that $b < a$; $a \geq b$ means that either $a = b$ or $a > b$; $a \leq b$ means that either $a = b$ or $a < b$; the *absolute value of* a, denoted by $|a|$, equals a if $a \geq 0$ or $-a$ if $a \leq 0$. (Sometimes we call $|a|$ the *magnitude* of a.)

You probably know the following theorem from calculus, but we include the proof for your convenience.

Theorem 1.1 — The Triangle Inequality. If a and b are any two real numbers, then

$$|a + b| \leq |a| + |b|. \tag{1.3}$$

Proof. There are four possibilities:

(a) If $a \geq 0$ and $b \geq 0$, then $a + b \geq 0$, so $|a + b| = a + b = |a| + |b|$.

(b) If $a \leq 0$ and $b \leq 0$, then $a + b \leq 0$, so $|a + b| = -a + (-b) = |a| + |b|$.

(c) If $a \geq 0$ and $b \leq 0$, then $a + b = |a| - |b|$.

(d) If $a \leq 0$ and $b \geq 0$, then $a + b = -|a| + |b|$.

Eq. 1.3 holds in cases **(c)** and **(d)**, since

$$|a + b| = \begin{cases} |a| - |b| & \text{if } |a| \geq |b|, \\ |b| - |a| & \text{if } |b| \geq |a|. \end{cases}$$

■

The triangle inequality appears in various forms in many contexts. It is the most important inequality in mathematics. We will use it often.

Corollary 1.2 If a and b are any two real numbers, then

$$|a - b| \geq \big||a| - |b|\big| \tag{1.4}$$

and

$$|a+b| \geq \big||a|-|b|\big|. \tag{1.5}$$

Proof. Replacing a by $a-b$ in (1.3) yields $|a| \leq |a-b|+|b|$, and so

$$|a-b| \geq |a|-|b|. \tag{1.6}$$

Interchanging a and b here yields $|b-a| \geq |b|-|a|$, which is equivalent to

$$|a-b| \geq |b|-|a|, \tag{1.7}$$

since $|b-a| = |a-b|$. Since

$$\big||a|-|b|\big| = \begin{cases} |a|-|b| & \text{if} \quad |a| > |b|, \\ |b|-|a| & \text{if} \quad |b| > |a|, \end{cases}$$

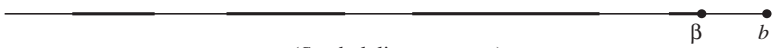
(1.6) and (1.7) imply (1.4). Replacing b by $-b$ in (1.4) yields (1.5), since $|-b| = |b|$. ■

Supremum of a Set

A set S of real numbers is *bounded above* if there is a real number b such that $x \leq b$ whenever $x \in S$. In this case, b is an *upper bound* of S. If b is an upper bound of S, then so is any larger number, because of property **(G)**. If β is an upper bound of S, but no number less than β is, then β is a *supremum* of S, and we write

$$\beta = \sup S.$$

With the real numbers associated in the usual way with the points on a line, these definitions can be interpreted geometrically as follows: b is an upper bound of S if no point of S is to the right of b; $\beta = \sup S$ if no point of S is to the right of β, but there is at least one point of S to the right of any number less than β (Figure 1.1).



(S = dark line segments)

Figure 1.1

■ **Example 1.3** If S is the set of negative numbers, then any nonnegative number is an upper bound of S, and $\sup S = 0$. If S_1 is the set of negative integers, then any number a such that $a \geq -1$ is an upper bound of S_1, and $\sup S_1 = -1$. ■

This example shows that a supremum of a set may or may not be in the set, since S_1 contains its supremum, but S does not.

A *nonempty* set is a set that has at least one member. The *empty set*, denoted by $\emptyset$, is the set that has no members. Although it may seem foolish to speak of such a set, we will see that it is a useful idea.

The Completeness Axiom

It is one thing to define an object and another to show that there really is an object that satisfies the definition. (For example, does it make sense to define the smallest positive real number?) This observation is particularly

appropriate in connection with the definition of the supremum of a set. For example, the empty set is bounded above by every real number, so it has no supremum. (Think about this.) More importantly, we will see in Example 1.8 that properties **(A)**–**(H)** do not guarantee that every nonempty set that is bounded above has a supremum. Since this property is indispensable to the rigorous development of calculus, we take it as an axiom for the real numbers.

(I) If a nonempty set of real numbers is bounded above, then it has a supremum.

Property **(I)** is called *completeness*, and we say that the real number system is a *complete ordered field*. It can be shown that the real number system is essentially the only complete ordered field; that is, if an alien from another planet were to construct a mathematical system with properties **(A)**–**(I)**, the alien's system would differ from the real number system only in that the alien might use different symbols for the real numbers and $+$, $\cdot$, and $<$.

Theorem 1.4 If a nonempty set S of real numbers is bounded above, then $\sup S$ is the unique real number β such that

(a) $x \le \beta$ for all x in S;

(b) if $\epsilon > 0$ (no matter how small), there is an x_0 in S with $x_0 > \beta - \epsilon$.

Proof. We first show that $\beta = \sup S$ has properties **(a)** and **(b)**. Since β is an upper bound of S, it must satisfy **(a)**. Since any real number a less than β can be written as $\beta - \epsilon$ with $\epsilon = \beta - a > 0$, **(b)** is just another way of saying that no number less than β is an upper bound of S. Hence, $\beta = \sup S$ satisfies **(a)** and **(b)**.

Now we show that there cannot be more than one real number with properties **(a)** and **(b)**. Suppose that $\beta_1 < \beta_2$ and β_2 has property **(b)**; thus, if $\epsilon > 0$, there is an x_0 in S such that $x_0 > \beta_2 - \epsilon$. Then, by taking $\epsilon = \beta_2 - \beta_1$, we see that there is an x_0 in S such that

$$x_0 > \beta_2 - (\beta_2 - \beta_1) = \beta_1,$$

so β_1 cannot have property **(a)**. Therefore, there cannot be more than one real number that satisfies both **(a)** and **(b)**. ■

Some Notation

We will often define a set S by writing $S = \{x \mid \cdots\}$, which means that S consists of all x that satisfy the conditions to the right of the vertical bar; thus, in Example 1.3,

$$S = \{x \mid x < 0\} \quad \text{and} \quad S_1 = \{x \mid x \text{ is a negative integer}\}. \tag{1.8}$$

We will sometimes abbreviate "x is a member of S" by $x \in S$, and "x is not a member of S" by $x \notin S$. For example, if S is defined by (1.8), then $-1 \in S$ but $0 \notin S$.

The Archimedean Property

The property of the real numbers described in the next theorem is called the *Archimedean property*. Intuitively, it states that it is possible to exceed any

positive number, no matter how large, by adding an arbitrary positive number, no matter how small, to itself sufficiently many times.

Theorem 1.5 — Archimedean Property. If ρ and ϵ are positive, then $n\epsilon > \rho$ for some integer n.

Proof. The proof is by contradiction. If the statement is false, ρ is an upper bound of the set $S = \{x \mid x = n\epsilon, n \text{ is an integer}\}$. Therefore, S has a supremum β, by property **(I)**. Therefore,

$$n\epsilon \leq \beta \quad \text{for all integers } n. \tag{1.9}$$

Since $n+1$ is an integer whenever n is, (1.9) implies that $(n+1)\epsilon \leq \beta$ and therefore $n\epsilon \leq \beta - \epsilon$ for all integers n. Hence, $\beta - \epsilon$ is an upper bound of S. Since $\beta - \epsilon < \beta$, this contradicts the definition of β. ■

Density of the Rationals and Irrationals

Definition 1.6 A set D is *dense in the reals* if every open interval (a,b) contains a member of D.

Theorem 1.7 The rational numbers are dense in the reals; that is, if a and b are real numbers with $a < b$, there is a rational number p/q such that $a < p/q < b$.

Proof. From Theorem 1.5, with $\rho = 1$ and $\epsilon = b - a$, there is a positive integer q such that $q(b-a) > 1$. There is also an integer j with $j > qa$. This is obvious if $a \leq 0$, and it follows from Theorem 1.5 with $\epsilon = 1$ and $\rho = qa$ if $a > 0$. Let p be the smallest integer such that $p > qa$. Then $p - 1 \leq qa$, so $qa < p \leq qa + 1$. Since $1 < q(b-a)$, this implies that $qa < p < qa + q(b-a) = qb$, and hence $qa < p < qb$. Therefore, $a < p/q < b$. ■

■ **Example 1.8** The rational number system is not complete; that is, a set of rational numbers may be bounded above (by rationals), but not have a rational upper bound less than any other rational upper bound. To see this, let

$$S = \left\{ r \mid r \text{ is rational and } r^2 < 2 \right\}.$$

If $r \in S$, then $r < \sqrt{2}$. Theorem 1.7 implies that if $\epsilon > 0$ there is a rational number r_0 such that $\sqrt{2} - \epsilon < r_0 < \sqrt{2}$, so Theorem 1.4 implies that $\sqrt{2} = \sup S$. However, $\sqrt{2}$ is *irrational*; that is, it cannot be written as the ratio of integers – we leave it as an exercise to the reader to show that $\sqrt{2}$ is irrational. Therefore, if r_1 is any rational upper bound of S, then $\sqrt{2} < r_1$. By Theorem 1.7, there is a rational number r_2 such that $\sqrt{2} < r_2 < r_1$. Since r_2 is also a rational upper bound of S, this shows that S has no rational supremum. ■

Since the rational numbers have properties **(A)**–**(H)**, but not **(I)**, this example shows that **(I)** does not follow from **(A)**–**(H)**.

Theorem 1.9 The set of irrational numbers is dense in the reals ; that is, if a and b are real numbers with $a < b$, there is an irrational number t such that $a < t < b$.

Proof. From Theorem 1.7, there are rational numbers r_1 and r_2 such that

$$a < r_1 < r_2 < b. \tag{1.10}$$

Let

$$t = r_1 + \frac{1}{\sqrt{2}}(r_2 - r_1).$$

Then t is irrational (why?) and $r_1 < t < r_2$, so $a < t < b$, from (1.10). ■

Infimum of a Set

A set S of real numbers is *bounded below* if there is a real number a with $x \geq a$ whenever $x \in S$. In this case, a is a *lower bound* of S. If a is a lower bound of S, so is any smaller number, because of property **(G)**. If α is a lower bound of S, but no number greater than α is, then α is an *infimum* of S, and we write

$$\alpha = \inf S.$$

Geometrically, this means that there are no points of S to the left of α, but there is at least one point of S to the left of any number greater than α.

Theorem 1.10 If a nonempty set S of real numbers is bounded below, then $\inf S$ is the unique real number α such that

(a) $x \geq \alpha$ for all x in S;

(b) if $\epsilon > 0$ (no matter how small), there is an x_0 in S such that $x_0 < \alpha + \epsilon$.

Proof. Exercise. ■

A set S is *bounded* if there are numbers a and b such that $a \leq x \leq b$ for all x in S. A bounded nonempty set has a unique supremum and a unique infimum, and

$$\inf S \leq \sup S \tag{1.11}$$

The Extended Real Number System

A nonempty set S of real numbers is *unbounded above* if it has no upper bound, or *unbounded below* if it has no lower bound. It is convenient to adjoin to the real number system two fictitious points, $+\infty$ (which we usually write more simply as ∞) and $-\infty$, and to define the order relationships between them and any real number x by

$$-\infty < x < \infty. \tag{1.12}$$

We call ∞ and $-\infty$ *points at infinity*. If S is a nonempty set of reals, we write

$$\sup S = \infty \tag{1.13}$$

to indicate that S is unbounded above, and

$$\inf S = -\infty \tag{1.14}$$

to indicate that S is unbounded below.

■ **Example 1.11** If $S = \{x \mid x < 2\}$, then $\sup S = 2$ and $\inf S = -\infty$. If $S = \{x \mid x \geq -2\}$, then $\sup S = \infty$ and $\inf S = -2$. If S is the set of all integers, then $\sup S = \infty$ and $\inf S = -\infty$. ■

The real number system with ∞ and $-\infty$ adjoined is called the *extended real number system*, or simply the *extended reals*. A member of the extended reals differing from $-\infty$ and ∞ is *finite*; that is, an ordinary real number is finite. However, the word "finite" in "finite real number" is redundant and used only for emphasis, since we would never refer to ∞ or $-\infty$ as real numbers.

The arithmetic relationships among ∞, $-\infty$, and the real numbers are defined as follows.

(a) If a is any real number, then

$$\begin{aligned} a + \infty &= \quad \infty + a = \quad \infty, \\ a - \infty &= -\infty + a = -\infty, \\ \frac{a}{\infty} &= \frac{a}{-\infty} = 0. \end{aligned}$$

(b) If $a > 0$, then

$$a\,\infty = \infty\, a = \infty, \quad \text{and} \quad a\,(-\infty) = (-\infty)\, a = -\infty.$$

(c) If $a < 0$, then

$$a\,\infty = \infty\, a = -\infty \quad \text{and} \quad a\,(-\infty) = (-\infty)\, a = \infty.$$

We also define

$$\infty + \infty = \infty\infty = (-\infty)(-\infty) = \infty$$

and

$$-\infty - \infty = \infty(-\infty) = (-\infty)\infty = -\infty.$$

Finally, we define

$$|\infty| = |-\infty| = \infty.$$

The introduction of ∞ and $-\infty$, along with the arithmetic and order relationships defined above, leads to simplifications in the statements of theorems. For example, the inequality (1.11), first stated only for bounded sets, holds for any nonempty set S if it is interpreted properly in accordance with (1.12) and the definitions of (1.13) and (1.14).

It is not useful to define $\infty - \infty$, $0 \cdot \infty$, ∞/∞, and $0/0$. They are called *indeterminate forms*, and left undefined. You probably studied indeterminate forms in calculus; we will look at them more carefully in Section 2.4.

1.2 The Real Line

One of our objectives is to develop rigorously the concepts of limit, continuity, differentiability, and integrability, which you have seen in calculus. To do

this requires a better understanding of the real numbers than is provided in calculus. The purpose of this section is to develop this understanding. Since the utility of the concepts introduced here will not become apparent until we are well into the study of limits and continuity, you should reserve judgment on their value until they are applied. As this occurs, you should reread the applicable parts of this section. This applies especially to the concept of an open covering and to the Heine–Borel and Bolzano–Weierstrass theorems, which will seem mysterious at first.

We assume that you are familiar with the geometric interpretation of the real numbers as points on a line. We will not prove that this interpretation is legitimate, for two reasons: (1) the proof requires an excursion into the foundations of Euclidean geometry, which is not the purpose of this book; (2) although we will use geometric terminology and intuition in discussing the reals, we will base all proofs on properties **(A)**–**(I)** (Section 1.1) and their consequences, not on geometric arguments.

Henceforth, we will use the terms *real number system* and *real line* synonymously and denote both by the symbol $\mathbb{R}$; also, we will often refer to a real number as a *point* (on the real line).

Some Set Theory

In this section we are interested in sets of points on the real line; however, we will consider other kinds of sets in subsequent sections. The following definition applies to arbitrary sets, with the understanding that the members of all sets under consideration in any given context come from a specific collection of elements, called the *universal set*. In this section the universal set is the real numbers.

Definition 1.12 Let S and T be sets.

(a) S *contains* T, and we write $S \supset T$ or $T \subset S$, if every member of T is also in S. In this case, T is a *subset* of S.

(b) $S - T$ is the set of elements that are in S but not in T.

(c) S *equals* T, and we write $S = T$, if S contains T and T contains S; thus, $S = T$ if and only if S and T have the same members.

(d) S *strictly contains* T if S contains T but T does not contain S; that is, if every member of T is also in S, but at least one member of S is not in T (Figure 1.2).

(e) The *complement* of S, denoted by S^c, is the set of elements in the universal set that are not in S.

(f) The *union* of S and T, denoted by $S \cup T$, is the set of elements in at least one of S and T (Figure 1.2 **(b)**).

(g) The *intersection* of S and T, denoted by $S \cap T$, is the set of elements in both S and T (Figure 1.2 **(c)**). If $S \cap T = \emptyset$ (the empty set), then S and T are *disjoint sets* (Figure 1.2 **(d)**).

(h) A set with only one member x_0 is a *singleton set*, denoted by $\{x_0\}$.

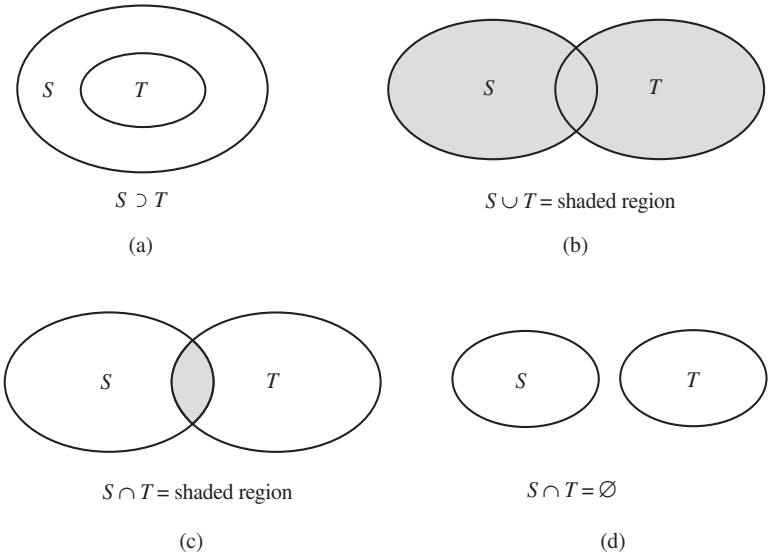


Figure 1.2

■ **Example 1.13** Let

$$S = \{x \mid 0 < x < 1\}, \quad T = \{x \mid 0 < x < 1 \text{ and } x \text{ is rational}\},$$

and

$$U = \{x \mid 0 < x < 1 \text{ and } x \text{ is irrational}\}.$$

Then $S \supset T$ and $S \supset U$, and the inclusion is strict in both cases. The unions of pairs of these sets are

$$S \cup T = S, \quad S \cup U = S, \quad \text{and} \quad T \cup U = S,$$

and their intersections are

$$S \cap T = T, \quad S \cap U = U, \quad \text{and} \quad T \cap U = \emptyset.$$

Also,

$$S - U = T \quad \text{and} \quad S - T = U.$$

Every set S contains the empty set $\emptyset$, for to say that $\emptyset$ is not contained in S is to say that some member of $\emptyset$ is not in S, which is absurd since $\emptyset$ has no members. If S is any set, then

$$(S^c)^c = S \quad \text{and} \quad S \cap S^c = \emptyset.$$

If S is a set of real numbers, then $S \cup S^c = \mathbb{R}$.

The definitions of union and intersection have generalizations: If $\mathcal{F}$ is an arbitrary collection of sets, then $\cup \{S \mid S \in \mathcal{F}\}$ is the set of all elements that are members of at least one of the sets in $\mathcal{F}$, and $\cap \{S \mid S \in \mathcal{F}\}$ is the set of all elements that are members of every set in $\mathcal{F}$. The union and intersection of finitely many sets $S_1, \ldots, S_n$ are also written as $\bigcup_{k=1}^{n} S_k$ and $\bigcap_{k=1}^{n} S_k$. The union and intersection of an infinite sequence $\{S_k\}_{k=1}^{\infty}$ of sets are written as $\bigcup_{k=1}^{\infty} S_k$ and $\bigcap_{k=1}^{\infty} S_k$.

■ **Example 1.14** If $\mathcal{F}$ is the collection of sets

$$S_\rho = \{x \mid \rho < x \leq 1+\rho\}, \quad 0 < \rho \leq 1/2,$$

then

$$\bigcup \{S_\rho \mid S_\rho \in \mathcal{F}\} = \{x \mid 0 < x \leq 3/2\}$$

and

$$\bigcap \{S_\rho \mid S_\rho \in \mathcal{F}\} = \{x \mid 1/2 < x \leq 1\}.$$

■ **Example 1.15** If, for each positive integer k, the set S_k is the set of real numbers that can be written as $x = m/k$ for some integer m, then $\bigcup_{k=1}^{\infty} S_k$ is the set of rational numbers and $\bigcap_{k=1}^{\infty} S_k$ is the set of integers.

Open and Closed Sets

If a and b are in the extended reals and $a < b$, then the *open interval* (a, b) is defined by

$$(a, b) = \{x \mid a < x < b\}.$$

The open intervals (a, ∞) and $(-\infty, b)$ are *semi-infinite* if a and b are finite, and $(-\infty, \infty)$ is the entire real line.

Definition 1.16 (a) If x_0 is a real number and $\epsilon > 0$, then the open interval $(x_0 - \epsilon, x_0 + \epsilon)$ is an *ϵ-neighborhood* of x_0.

(b) If a set S contains an ϵ-neighborhood of x_0, then S is a *neighborhood* of x_0, and x_0 is an *interior point* of S (Figure 1.3).

(c) The set of interior points of S is the *interior* of S, denoted by S^0.

(d) If every point of S is an interior point (that is, $S^0 = S$), then S is *open*.

(e) A set S is *closed* if S^c is open.

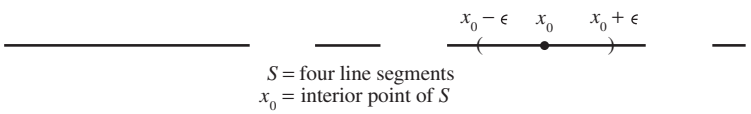


Figure 1.3

The idea of neighborhood is fundamental and occurs in many other contexts, some of which we will see later in this book. Whatever the context, the idea is the same: some definition of "closeness" is given (for example, two real numbers are "close" if their difference is "small"), and a neighborhood of a point x_0 is a set that contains all points sufficiently close to x_0.

■ **Example 1.17** An open interval (a, b) is an open set, because if $x_0 \in (a, b)$ and $\epsilon \leq \min\{x_0 - a, b - x_0\}$, then

$$(x_0 - \epsilon, x_0 + \epsilon) \subset (a, b).$$

The entire line $\mathbb{R} = (-\infty, \infty)$ is open, and therefore $\emptyset\,(= \mathbb{R}^c)$ is closed. However, $\emptyset$ is also open, for to deny this is to say that $\emptyset$ contains a point that is not an interior point, which is absurd because $\emptyset$ contains no points. Since $\emptyset$ is open, $\mathbb{R}\,(= \emptyset^c)$ is closed. Thus, $\mathbb{R}$ and $\emptyset$ are both open and closed. We leave it as an exercise to the reader to show that these are the only subsets of $\mathbb{R}$ with this property. ■

A *deleted neighborhood* of a point x_0 is a set that contains every point of some neighborhood of x_0 except for x_0 itself. For example,

$$S = \{x \mid 0 < |x - x_0| < \epsilon\}$$

is a deleted neighborhood of x_0. We also say that it is a *deleted ϵ-neighborhood* of x_0.

Theorem 1.18

(a) The union of open sets is open.

(b) The intersection of closed sets is closed.

These statements apply to arbitrary collections, finite or infinite, of open and closed sets.

Proof. **(a)** Let $\mathcal{G}$ be a collection of open sets and

$$S = \cup\{G \mid G \in \mathcal{G}\}.$$

If $x_0 \in S$, then $x_0 \in G_0$ for some G_0 in $\mathcal{G}$, and since G_0 is open, it contains some ϵ-neighborhood of x_0. Since $G_0 \subset S$, this ϵ-neighborhood is in S, which is consequently a neighborhood of x_0. Thus, S is a neighborhood of each of its points, and therefore open, by definition.

(b) Let $\mathcal{F}$ be a collection of closed sets and $T = \cap\{F \mid F \in \mathcal{F}\}$. Then by de Morgan's formula $T^c = \cup\{F^c \mid F \in \mathcal{F}\}$ and, since each F^c is open, T^c is open, from **(a)**. Therefore, T is closed, by definition. ■

■ **Example 1.19** If $-\infty < a < b < \infty$, the set

$$[a, b] = \{x \mid a \le x \le b\}$$

is closed, since its complement is the union of the open sets $(-\infty, a)$ and (b, ∞). We say that $[a, b]$ is a *closed interval*. The set

$$[a, b) = \{x \mid a \le x < b\}$$

is a *half-closed* or *half-open interval* if $-\infty < a < b < \infty$, as is

$$(a, b] = \{x \mid a < x \le b\};$$

however, neither of these sets is open or closed. (Why not?) *Semi-infinite closed intervals* are sets of the form

$$[a, \infty) = \{x \mid a \le x\} \quad \text{and} \quad (-\infty, a] = \{x \mid x \le a\},$$

where a is finite. They are closed sets, since their complements are the open intervals $(-\infty, a)$ and (a, ∞), respectively. ■

Example 1.17 shows that a set may be both open and closed, and Example 1.19 shows that a set may be neither. Thus, open and closed are not opposites in this context, as they are in everyday speech.

■ **Example 1.20** From Theorem 1.18 and Example 1.17, the union of any collection of open intervals is an open set. (In fact, it can be shown that every nonempty open subset of $\mathbb{R}$ is the union of open intervals.) From Theorem 1.18 and Example 1.19, the intersection of any collection of closed intervals is closed. ■

It can be shown that the intersection of finitely many open sets is open, and that the union of finitely many closed sets is closed. However, the intersection of infinitely many open sets need not be open, and the union of infinitely many closed sets need not be closed – it is left as an exercise to the reader to show these latter two statements.

Definition 1.21 Let S be a subset of $\mathcal{R}$. Then

(a) x_0 is a *limit point* of S if every deleted neighborhood of x_0 contains a point of S.

(b) x_0 is a *boundary point* of S if every neighborhood of x_0 contains at least one point in S and one not in S. The set of boundary points of S is the *boundary* of S, denoted by ∂S. The *closure* of S, denoted by $\overline{S}$, is $\overline{S} = S \cup \partial S$.

(c) x_0 is an *isolated point* of S if $x_0 \in S$ and there is a neighborhood of x_0 that contains no other point of S.

(d) x_0 is *exterior* to S if x_0 is in the interior of S^c. The collection of such points is the *exterior* of S.

■ **Example 1.22** Let $S = (-\infty, -1] \cup (1,2) \cup \{3\}$. Then

(a) The set of limit points of S is $(-\infty, -1] \cup [1,2]$.

(b) $\partial S = \{-1, 1, 2, 3\}$ and $\overline{S} = (-\infty, -1] \cup [1,2] \cup \{3\}$.

(c) 3 is the only isolated point of S.

(d) The exterior of S is $(-1,1) \cup (2,3) \cup (3,\infty)$.

■ **Example 1.23** For $n \geq 1$, let

$$I_n = \left[\frac{1}{2n+1}, \frac{1}{2n}\right] \quad \text{and} \quad S = \bigcup_{n=1}^{\infty} I_n.$$

Then

(a) The set of limit points of S is $S \cup \{0\}$.

(b) $\partial S = \left\{x \,\middle|\, x = 0 \text{ or } x = 1/n \ (n \geq 2)\right\}$ and $\overline{S} = S \cup \{0\}$.

(c) S has no isolated points.

(d) The exterior of S is

$$(-\infty,0)\cup\left[\bigcup_{n=1}^{\infty}\left(\frac{1}{2n+2},\frac{1}{2n+1}\right)\right]\cup\left(\frac{1}{2},\infty\right).$$

■ **Example 1.24** Let S be the set of rational numbers. Since every interval contains a rational number (Theorem 1.7), every real number is a limit point of S; thus, $\overline{S}=\mathbb{R}$. Since every interval also contains an irrational number (Theorem 1.9), every real number is a boundary point of S; thus $\partial S=\mathbb{R}$. The interior and exterior of S are both empty, and S has no isolated points. S is neither open nor closed. ■

The next theorem says that S is closed if and only if $S=\overline{S}$.

Theorem 1.25 A set S is closed if and only if no point of S^c is a limit point of S.

Proof. Suppose that S is closed and $x_0\in S^c$. Since S^c is open, there is a neighborhood of x_0 that is contained in S^c and therefore contains no points of S. Hence, x_0 cannot be a limit point of S. For the converse, if no point of S^c is a limit point of S then every point in S^c must have a neighborhood contained in S^c. Therefore, S^c is open and S is closed. ■

Theorem 1.25 is usually stated as follows.

Corollary 1.26 A set is closed if and only if it contains all its limit points.

Theorem 1.25 and Corollary 1.26 are equivalent. However, we stated the theorem as we did because students sometimes incorrectly conclude from the corollary that a closed set must have limit points. The corollary does not say this. If S has no limit points, then the set of limit points is empty and therefore contained in S. Hence, a set with no limit points is closed according to the corollary, in agreement with Theorem 1.25. For example, any finite set is closed. More generally, S is closed if there is a $\delta>0$ such $|x-y|\geq\delta$ for every pair $\{x,y\}$ of distinct points in S.

Open Coverings

A collection $\mathcal{H}$ of open sets is an *open covering* of a set S if every point in S is contained in a set H belonging to $\mathcal{H}$; that is, if $S\subset\cup\{H\,|\,H\in\mathcal{H}\}$.

■ **Example 1.27** The sets

$$S_1=[0,1],\ S_2=\{1,2,\ldots,n,\ldots\},$$
$$S_3=\left\{1,\frac{1}{2},\ldots,\frac{1}{n},\ldots\right\},\quad\text{and}\quad S_4=(0,1)$$

are covered by the families of open intervals

$$\mathcal{H}_1=\left\{\left(x-\frac{1}{N},x+\frac{1}{N}\right)\,\middle|\,0<x<1\right\},\quad (N=\text{positive integer}),$$
$$\mathcal{H}_2=\left\{\left(n-\frac{1}{4},n+\frac{1}{4}\right)\,\middle|\,n=1,2,\ldots\right\},$$
$$\mathcal{H}_3=\left\{\left(\frac{1}{n+\frac{1}{2}},\frac{1}{n-\frac{1}{2}}\right)\,\middle|\,n=1,2,\ldots\right\},$$
$$\mathcal{H}_4=\{(0,\rho)\,|\,0<\rho<1\},$$

respectively.

Theorem 1.28 — Heine–Borel Theorem. If $\mathcal{H}$ is an open covering of a closed and bounded subset S of the real line, then S has an open covering $\widetilde{\mathcal{H}}$ consisting of finitely many open sets belonging to $\mathcal{H}$.

Proof. Since S is bounded, it has an infimum α and a supremum β, and, since S is closed, α and β belong to S. Define

$$S_t = S \cap [\alpha, t] \quad \text{for } t \geq \alpha,$$

and let

$$F = \left\{ t \,\middle|\, \alpha \leq t \leq \beta \text{ and finitely many sets from } \mathcal{H} \text{ cover } S_t \right\}.$$

Since $S_\beta = S$, the theorem will be proved if we can show that $\beta \in F$. To do this, we use the completeness of the reals.

Since $\alpha \in S$, S_α is the singleton set $\{\alpha\}$, which is contained in some open set H_α from $\mathcal{H}$ because $\mathcal{H}$ covers S; therefore, $\alpha \in F$. Since F is nonempty and bounded above by β, it has a supremum γ. First, we wish to show that $\gamma = \beta$. Since $\gamma \leq \beta$ by definition of F, it suffices to rule out the possibility that $\gamma < \beta$. We consider two cases.

CASE 1. Suppose that $\gamma < \beta$ and $\gamma \notin S$. Then, since S is closed, γ is not a limit point of S (Theorem 1.25). Consequently, there is an $\epsilon > 0$ such that

$$[\gamma - \epsilon, \gamma + \epsilon] \cap S = \emptyset,$$

so $S_{\gamma-\epsilon} = S_{\gamma+\epsilon}$. However, the definition of γ implies that $S_{\gamma-\epsilon}$ has a finite subcovering from $\mathcal{H}$, while $S_{\gamma+\epsilon}$ does not. This is a contradiction.

CASE 2. Suppose that $\gamma < \beta$ and $\gamma \in S$. Then there is an open set H_γ in $\mathcal{H}$ that contains γ and, along with γ, an interval $[\gamma - \epsilon, \gamma + \epsilon]$ for some positive ϵ. Since $S_{\gamma-\epsilon}$ has a finite covering $\{H_1, \ldots, H_n\}$ of sets from $\mathcal{H}$, it follows that $S_{\gamma+\epsilon}$ has the finite covering $\{H_1, \ldots, H_n, H_\gamma\}$. This contradicts the definition of γ.

Now we know that $\gamma = \beta$, which is in S. Therefore, there is an open set H_β in $\mathcal{H}$ that contains β and along with β, an interval of the form $[\beta - \epsilon, \beta + \epsilon]$, for some positive ϵ. Since $S_{\beta-\epsilon}$ is covered by a finite collection of sets $\{H_1, \ldots, H_k\}$, S_β is covered by the finite collection $\{H_1, \ldots, H_k, H_\beta\}$. Since $S_\beta = S$, we are finished. ∎

Henceforth, we will say that a closed and bounded set is *compact*. The Heine–Borel theorem says that any open covering of a compact set S contains a finite collection that also covers S. This theorem and its converse (left as an exercise to the reader to prove) show that we could just as well define a set S of reals to be compact if it has the Heine–Borel property; that is, if every open covering of S contains a finite subcovering. The same is true of $\mathbb{R}^n$, which we study in Section 5.1. This definition generalizes to more abstract spaces (called *topological spaces*) for which the concept of boundedness need not be defined.

▪ **Example 1.29** Since S_1 in Example 1.27 is compact, the Heine–Borel theorem implies that S_1 can be covered by a finite number of intervals from $\mathcal{H}_1$. This is easily verified, since, for example, the $2N$ intervals from $\mathcal{H}_1$ centered at the points $x_k = k/2N$ $(0 \leq k \leq 2N - 1)$ cover S_1.

The Heine–Borel theorem does not apply to the other sets in Example 1.27 since they are not compact: S_2 is unbounded and S_3 and S_4 are not closed, since they do not contain all their limit points (Corollary 1.26). The conclusion of the Heine–Borel theorem does not hold for these sets and the open coverings that we have given for them. Each point in S_2 is contained in exactly one set from $\mathcal{H}_2$, so removing even one of these sets leaves a point of S_2 uncovered. If $\widetilde{\mathcal{H}}_3$ is any finite collection of sets from $\mathcal{H}_3$, then

$$\frac{1}{n} \notin \cup \left\{ H \,\middle|\, H \in \widetilde{\mathcal{H}}_3 \right\}$$

for n sufficiently large. Any finite collection $\{(0,\rho_1),\ldots,(0,\rho_n)\}$ from $\mathcal{H}_4$ covers only the interval $(0,\rho_{\max})$, where

$$\rho_{\max} = \max\{\rho_1,\ldots,\rho_n\} < 1.$$

The Bolzano–Weierstrass Theorem

As an application of the Heine–Borel theorem, we prove the following theorem of Bolzano and Weierstrass.

Theorem 1.30 — Bolzano–Weierstrass Theorem. Every bounded infinite set of real numbers has at least one limit point.

Proof. We will show that a bounded nonempty set without a limit point can contain only a finite number of points. If S has no limit points, then S is closed (Theorem 1.25) and every point x of S has an open neighborhood N_x that contains no point of S other than x. The collection

$$\mathcal{H} = \{ N_x \,|\, x \in S \}$$

is an open covering for S. Since S is also bounded, Theorem 1.28 implies that S can be covered by a finite collection of sets from $\mathcal{H}$, say $N_{x_1},\ldots,N_{x_n}$. Since these sets contain only $x_1,\ldots,x_n$ from S, it follows that $S = \{x_1,\ldots,x_n\}$. ∎

2 Differential Calculus of Functions of One Variable

IN THIS CHAPTER we study the differential calculus of functions of one variable.

SECTION 2.1 introduces the concept of function and discusses arithmetic operations on functions, limits, one-sided limits, limits at $\pm\infty$, and monotonic functions.

SECTION 2.2 defines continuity and discusses removable discontinuities, composite functions, bounded functions, the intermediate value theorem, uniform continuity, and additional properties of monotonic functions.

SECTION 2.3 introduces the derivative and its geometric interpretation. Topics covered include the interchange of differentiation and arithmetic operations, the chain rule, one-sided derivatives, extreme values of a differentiable function, Rolle's theorem, the intermediate value theorem for derivatives, and the mean value theorem and its consequences.

2.1 Functions and Limits

In this section we study limits of real-valued functions of a real variable. You studied limits in calculus. However, we will look more carefully at the definition of limit and prove theorems usually not proved in calculus.

A rule f that assigns to each member of a nonempty set D a unique member of a set Y is a *function from D to Y*. We write the relationship between a member x of D and the member y of Y that f assigns to x as

$$y = f(x).$$

The set D is the *domain* of f, denoted by D_f. The members of Y are the possible *values* of f. If $y_0 \in Y$ and there is an x_0 in D such that $f(x_0) = y_0$ then we say that f *attains*, or *assumes*, the value y_0. The set of values attained by f is the *range* of f. A *real-valued function of a real variable* is a function whose domain and range are both subsets of the reals. Although we are concerned only with real-valued functions of a real variable in this section, our definitions are not restricted to this situation. In later sections we will consider situations where the range or domain, or both, are subsets of vector spaces.

■ **Example 2.1** The functions f, g, and h defined on $(-\infty,\infty)$ by

$$f(x) = x^2, \quad g(x) = \sin x, \quad \text{and} \quad h(x) = e^x$$

have ranges $[0,\infty)$, $[-1,1]$, and $(0,\infty)$, respectively.

■ **Example 2.2** The equation

$$[f(x)]^2 = x \tag{2.1}$$

does not define a function except on the singleton set $\{0\}$. If $x < 0$, no real number satisfies (2.1), while if $x > 0$, two real numbers satisfy (2.1). However, the conditions

$$[f(x)]^2 = x \quad \text{and} \quad f(x) \geq 0$$

define a function f on $D_f = [0,\infty)$ with values $f(x) = \sqrt{x}$. Similarly, the conditions

$$[g(x)]^2 = x \quad \text{and} \quad g(x) \leq 0$$

define a function g on $D_g = [0, \infty)$ with values $g(x) = -\sqrt{x}$. The ranges of f and g are $[0, \infty)$ and $(-\infty, 0]$, respectively. ■

It is important to understand that the definition of a function includes the specification of its domain and that there is a difference between f, the *name* of the function, and $f(x)$, the *value* of f at x. However, strict observance of these points leads to annoying verbosity, such as "the function f with domain $(-\infty, \infty)$ and values $f(x) = x$." We will avoid this in two ways: (1) by agreeing that if a function f is introduced without explicitly defining D_f, then D_f will be understood to consist of all points x for which the rule defining $f(x)$ makes sense, and (2) by bearing in mind the distinction between f and $f(x)$, but not emphasizing it when it would be a nuisance to do so. For example, we will write "consider the function $f(x) = \sqrt{1 - x^2}$," rather than "consider the function f defined on $[-1, 1]$ by $f(x) = \sqrt{1 - x^2}$," or "consider the function $g(x) = 1/\sin x$," rather than "consider the function g defined for $x \neq k\pi$ (k = integer) by $g(x) = 1/\sin x$." We will also write $f = c$ (constant) to denote the function f defined by $f(x) = c$ for all x.

Our definition of function is somewhat intuitive, but adequate for our purposes. Moreover, it is the working form of the definition, even if the idea is introduced more rigorously to begin with. For a more precise definition, we first define the Cartesian *product* $X \times Y$ of two nonempty sets X and Y to be the set of all ordered pairs (x, y) such that $x \in X$ and $y \in Y$; thus,

$$X \times Y = \{(x, y) \mid x \in X, y \in Y\}.$$

A nonempty subset f of $X \times Y$ is a *function* if no x in X occurs more than once as a first member among the elements of f. Put another way, if (x, y) and (x, y_1) are in f, then $y = y_1$. The set of x's that occur as first members of f is the of f. If x is in the domain of f, then the unique y in Y such that $(x, y) \in f$ is the *value of f at x*, and we write $y = f(x)$. The set of all such values, a subset of Y, is the *range of f*.

Definition 2.3 If $D_f \cap D_g \neq \emptyset$, then $f + g$, $f - g$, and fg are defined on $D_f \cap D_g$ by

$$(f + g)(x) = f(x) + g(x),$$
$$(f - g)(x) = f(x) - g(x),$$

and

$$(fg)(x) = f(x)g(x).$$

The quotient f/g is defined by

$$\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}$$

for x in $D_f \cap D_g$ such that $g(x) \neq 0$.

■ **Example 2.4** If $f(x) = \sqrt{4 - x^2}$ and $g(x) = \sqrt{x - 1}$, then $D_f = [-2, 2]$ and $D_g = [1, \infty)$, so $f + g$, $f - g$, and fg are defined on $D_f \cap D_g = [1, 2]$

by

$$(f+g)(x) = \sqrt{4-x^2} + \sqrt{x-1},$$
$$(f-g)(x) = \sqrt{4-x^2} - \sqrt{x-1},$$

and

$$(fg)(x) = (\sqrt{4-x^2})(\sqrt{x-1}) = \sqrt{(4-x^2)(x-1)}. \tag{2.2}$$

The quotient f/g is defined on $(1,2]$ by

$$\left(\frac{f}{g}\right)(x) = \sqrt{\frac{4-x^2}{x-1}}.$$

Although the last expression in (2.2) is also defined for $-\infty < x < -2$, it does not represent fg for such x, since f and g are not defined on $(-\infty, -2]$.

■ **Example 2.5** If c is a real number, the function cf defined by $(cf)(x) = cf(x)$ can be regarded as the product of f and a constant function. Its domain is D_f. The sum and product of n (≥ 2) functions $f_1, \ldots, f_n$ are defined by

$$(f_1 + f_2 + \cdots + f_n)(x) = f_1(x) + f_2(x) + \cdots + f_n(x)$$

and

$$(f_1 f_2 \cdots f_n)(x) = f_1(x) f_2(x) \cdots f_n(x) \tag{2.3}$$

on $D = \bigcap_{i=1}^{n} D_{f_i}$, provided that D is nonempty. If $f_1 = f_2 = \cdots = f_n$, then (2.3) defines the nth *power of* f:

$$(f^n)(x) = (f(x))^n.$$

From these definitions, we can build the set of all *polynomials*

$$p(x) = a_0 + a_1 x + \cdots + a_n x^n,$$

starting from the constant functions and $f(x) = x$. The quotient of two polynomials is a *rational function*

$$r(x) = \frac{a_0 + a_1 x + \cdots + a_n x^n}{b_0 + b_1 x + \cdots + b_m x^m} \quad (b_m \neq 0).$$

The domain of r is the set of points where the denominator is nonzero.

Limits

The essence of the concept of limit for real-valued functions of a real variable is this: If L is a real number, then $\lim_{x\to x_0} f(x) = L$ means that the value $f(x)$ can be made as close to L as we wish by taking x sufficiently close to x_0. This is made precise in the following definition.

Definition 2.6 We say that $f(x)$ *approaches the limit* L *as* x *approaches* x_0, and write

$$\lim_{x\to x_0} f(x) = L,$$

if f is defined on some deleted neighborhood of x_0 and, for every $\epsilon > 0$, there is a $\delta > 0$ such that

$$|f(x) - L| < \epsilon \tag{2.4}$$

if

$$0 < |x - x_0| < \delta. \tag{2.5}$$

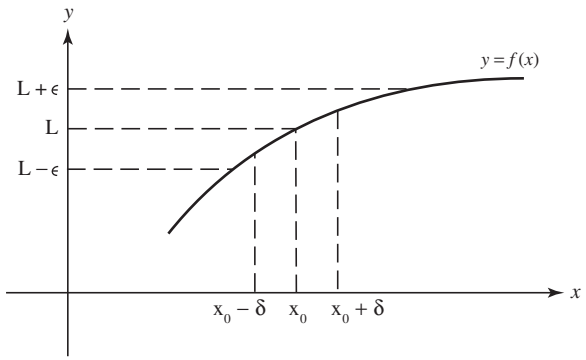


Figure 2.1

Figure 2.1 depicts the graph of a function for which $\lim_{x\to x_0} f(x)$ exists.

■ **Example 2.7** If c and x are arbitrary real numbers and $f(x) = cx$, then

$$\lim_{x\to x_0} f(x) = cx_0.$$

To prove this, we write

$$|f(x) - cx_0| = |cx - cx_0| = |c||x - x_0|.$$

If $c \neq 0$, this yields

$$|f(x) - cx_0| < \epsilon \tag{2.6}$$

if

$$|x - x_0| < \delta,$$

where δ is any number such that $0 < \delta \leq \epsilon/|c|$. If $c = 0$, then $f(x) - cx_0 = 0$ for all x, so (2.6) holds for all x. ■

We emphasize that Definition 2.6 does not involve $f(x_0)$, or even require that it be defined, since (2.5) excludes the case where $x = x_0$.

■ **Example 2.8** If

$$f(x) = x \sin \frac{1}{x}, \quad x \neq 0,$$

then

$$\lim_{x\to 0} f(x) = 0$$

even though f is not defined at $x_0 = 0$, because if

$$0 < |x| < \delta = \epsilon,$$

then

$$|f(x) - 0| = \left| x \sin \frac{1}{x} \right| \leq |x| < \epsilon.$$

On the other hand, the function

$$g(x) = \sin \frac{1}{x}, \quad x \neq 0,$$

has no limit as x approaches 0, since it assumes all values between -1 and 1 in every neighborhood of the origin. ■

The next theorem says that a function cannot have more than one limit at a point.

Theorem 2.9 If $\lim_{x\to x_0} f(x)$ exists, then it is unique ; that is, if

$$\lim_{x\to x_0} f(x) = L_1 \quad \text{and} \quad \lim_{x\to x_0} f(x) = L_2, \tag{2.7}$$

then $L_1 = L_2$.

Proof. Suppose that (2.7) holds and let $\epsilon > 0$. From Definition 2.6, there are positive numbers δ_1 and δ_2 such that

$$|f(x) - L_i| < \epsilon \quad \text{if} \quad 0 < |x - x_0| < \delta_i, \quad i = 1,2.$$

If $\delta = \min(\delta_1, \delta_2)$, then

$$\begin{aligned} |L_1 - L_2| &= |L_1 - f(x) + f(x) - L_2| \\ &\le |L_1 - f(x)| + |f(x) - L_2| < 2\epsilon \quad \text{if} \quad 0 < |x - x_0| < \delta. \end{aligned}$$

We have now established an inequality that does not depend on x; that is, $|L_1 - L_2| < 2\epsilon$. Since this holds for any positive ϵ, $L_1 = L_2$. ■

Definition 2.6 is not changed by replacing (2.4) with

$$|f(x) - L| < K\epsilon, \tag{2.8}$$

where K is a positive constant, because if either of (2.4) or (2.8) can be made to hold for any $\epsilon > 0$ by making $|x - x_0|$ sufficiently small and positive, then so can the other. This may seem to be a minor point, but it is often convenient to work with (2.8) rather than (2.4), as we will see in the proof of the following theorem.

Theorem 2.10 If

$$\lim_{x\to x_0} f(x) = L_1 \quad \text{and} \quad \lim_{x\to x_0} g(x) = L_2, \tag{2.9}$$

then

$$\lim_{x\to x_0} (f+g)(x) = L_1 + L_2, \tag{2.10}$$

$$\lim_{x\to x_0} (f-g)(x) = L_1 - L_2, \tag{2.11}$$

$$\lim_{x\to x_0} (fg)(x) = L_1 L_2, \tag{2.12}$$

and, if $L_2 \neq 0$,

$$\lim_{x\to x_0} \left(\frac{f}{g}\right)(x) = \frac{L_1}{L_2}. \tag{2.13}$$

Proof. From (2.9) and Definition 2.6, if $\epsilon > 0$, there is a $\delta_1 > 0$ such that

$$|f(x) - L_1| < \epsilon \tag{2.14}$$

if $0 < |x - x_0| < \delta_1$, and a $\delta_2 > 0$ such that

$$|g(x) - L_2| < \epsilon \tag{2.15}$$

if $0 < |x - x_0| < \delta_2$. Suppose that

$$0 < |x - x_0| < \delta = \min(\delta_1, \delta_2), \tag{2.16}$$

so that (2.14) and (2.15) both hold. Then

$$\begin{aligned}|(f \pm g)(x) - (L_1 \pm L_2)| &= |(f(x) - L_1) \pm (g(x) - L_2)| \\ &\leq |f(x) - L_1| + |g(x) - L_2| < 2\epsilon,\end{aligned}$$

which proves (2.10) and (2.11).

To prove (2.12), we assume (2.16) and write

$$\begin{aligned}|(fg)(x) - L_1L_2| &= |f(x)g(x) - L_1L_2| \\ &= |f(x)(g(x) - L_2) + L_2(f(x) - L_1)| \\ &\leq |f(x)||g(x) - L_2| + |L_2||f(x) - L_1| \\ &\leq (|f(x)| + |L_2|)\epsilon \quad \text{(from (2.14) and (2.15))} \\ &\leq (|f(x) - L_1| + |L_1| + |L_2|)\epsilon \\ &\leq (\epsilon + |L_1| + |L_2|)\epsilon \quad \text{from (2.14)} \\ &\leq (1 + |L_1| + |L_2|)\epsilon\end{aligned}$$

if $\epsilon < 1$ and x satisfies (2.16). This proves (2.12).

To prove (2.13), we first observe that if $L_2 \neq 0$, there is a $\delta_3 > 0$ such that

$$|g(x) - L_2| < \frac{|L_2|}{2},$$

so

$$|g(x)| > \frac{|L_2|}{2} \tag{2.17}$$

if $0 < |x - x_0| < \delta_3$. To see this, let $L = L_2$ and $\epsilon = |L_2|/2$ in (2.4). Now suppose that

$$0 < |x - x_0| < \min(\delta_1, \delta_2, \delta_3),$$

so that (2.14), (2.15), and (2.17) all hold. Then

$$\begin{aligned}\left|\left(\frac{f}{g}\right)(x) - \frac{L_1}{L_2}\right| &= \left|\frac{f(x)}{g(x)} - \frac{L_1}{L_2}\right| \\ &= \frac{|L_2f(x) - L_1g(x)|}{|g(x)L_2|} \\ &\leq \frac{2}{|L_2|^2}|L_2f(x) - L_1g(x)| \\ &= \frac{2}{|L_2|^2}|L_2[f(x) - L_1] + L_1[L_2 - g(x)]| \quad \text{(from (2.17))} \\ &\leq \frac{2}{|L_2|^2}[|L_2||f(x) - L_1| + |L_1||L_2 - g(x)|] \\ &\leq \frac{2}{|L_2|^2}(|L_2| + |L_1|)\epsilon \quad \text{(from (2.14) and (2.15)).}\end{aligned}$$

This proves (2.13). ■

Successive applications of the various parts of Theorem 2.10 permit us to find limits without the ϵ–δ arguments required by Definition 2.6.

■ **Example 2.11** Use Theorem 2.10 to find

$$\lim_{x\to 2}\frac{9 - x^2}{x + 1} \quad \text{and} \quad \lim_{x\to 2}(9 - x^2)(x + 1).$$

Solution If c is a constant, then $\lim_{x\to x_0} c = c$, and, from Example 2.7, $\lim_{x\to x_0} x = x_0$. Therefore, from Theorem 2.10,

$$\lim_{x\to 2}(9-x^2) = \lim_{x\to 2} 9 - \lim_{x\to 2} x^2 = \lim_{x\to 2} 9 - (\lim_{x\to 2} x)^2 = 9 - 2^2 = 5,$$

and

$$\lim_{x\to 2}(x+1) = \lim_{x\to 2} x + \lim_{x\to 2} 1 = 2 + 1 = 3.$$

Therefore,

$$\lim_{x\to 2} \frac{9-x^2}{x+1} = \frac{\lim_{x\to 2}(9-x^2)}{\lim_{x\to 2}(x+1)} = \frac{5}{3}$$

and

$$\lim_{x\to 2}(9-x^2)(x+1) = \lim_{x\to 2}(9-x^2)\lim_{x\to 2}(x+1) = 5\cdot 3 = 15. \quad \blacksquare$$

One-Sided Limits

The function

$$f(x) = 2x \sin\sqrt{x}$$

satisfies the inequality $|f(x)| < \epsilon$ if $0 < x < \delta = \epsilon/2$. However, this does not mean that $\lim_{x\to 0} f(x) = 0$, since f is not defined for negative x, as it must be to satisfy the conditions of Definition 2.6 with $x_0 = 0$ and $L = 0$. The function

$$g(x) = x + \frac{|x|}{x}, \quad x \neq 0,$$

can be rewritten as

$$g(x) = \begin{cases} x+1, & x > 0, \\ x-1, & x < 0; \end{cases}$$

hence, every open interval containing $x_0 = 0$ also contains points x_1 and x_2 such that $|g(x_1) - g(x_2)|$ is as close to 2 as we please. Therefore, $\lim_{x\to x_0} g(x)$ does not exist.

Although $f(x)$ and $g(x)$ do not approach limits as x approaches zero, they each exhibit a definite sort of limiting behavior for small positive values of x, as does $g(x)$ for small negative values of x. The kind of behavior we have in mind is defined precisely as follows.

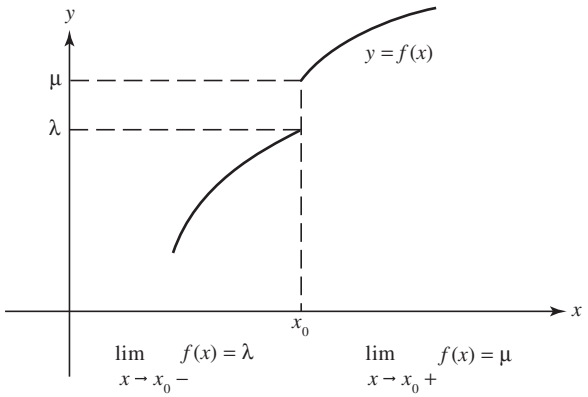


Figure 2.2

Definition 2.12

(a) We say that $f(x)$ *approaches the left-hand limit* L *as* x *approaches* x_0 *from the left*, and write

$$\lim_{x\to x_0-} f(x) = L,$$

if f is defined on some open interval (a, x_0) and, for each $\epsilon > 0$, there is a $\delta > 0$ such that

$$|f(x) - L| < \epsilon \quad \text{if} \quad x_0 - \delta < x < x_0.$$

(b) We say that $f(x)$ *approaches the right-hand limit* L *as* x *approaches* x_0 *from the right*, and write

$$\lim_{x\to x_0+} f(x) = L,$$

if f is defined on some open interval (x_0, b) and, for each $\epsilon > 0$, there is a $\delta > 0$ such that

$$|f(x) - L| < \epsilon \quad \text{if} \quad x_0 < x < x_0 + \delta.$$

Figure 2.2 shows the graph of a function that has distinct left- and right-hand limits at a point x_0.

■ **Example 2.13** Let

$$f(x) = \frac{x}{|x|}, \quad x \neq 0.$$

If $x < 0$, then $f(x) = -x/x = -1$, so

$$\lim_{x\to 0-} f(x) = -1.$$

If $x > 0$, then $f(x) = x/x = 1$, so

$$\lim_{x\to 0+} f(x) = 1.$$

■ **Example 2.14** Let

$$g(x) = \frac{x + |x|(1+x)}{x} \sin\frac{1}{x}, \quad x \neq 0.$$

If $x < 0$, then

$$g(x) = -x \sin\frac{1}{x},$$

so

$$\lim_{x\to 0-} g(x) = 0,$$

since

$$|g(x) - 0| = \left| x \sin\frac{1}{x} \right| \leq |x| < \epsilon$$

if $-\epsilon < x < 0$; that is, Definition 2.12 **(a)** is satisfied with $\delta = \epsilon$. If $x > 0$, then

$$g(x) = (2 + x) \sin\frac{1}{x},$$

which takes on every value between -2 and 2 in every interval $(0, \delta)$. Hence, $g(x)$ does not approach a right-hand limit at x approaches 0 from the right.

This shows that a function may have a limit from one side at a point but fail to have a limit from the other side.

▪ **Example 2.15** We leave it to you to verify that

$$\begin{aligned}\lim_{x\to 0+}\left(\frac{|x|}{x}+x\right) &= 1,\\ \lim_{x\to 0-}\left(\frac{|x|}{x}+x\right) &= -1,\\ \lim_{x\to 0+} x\sin\sqrt{x} &= 0,\end{aligned}$$

and $\lim_{x\to 0-}\sin\sqrt{x}$ does not exist. ■

Left- and right-hand limits are also called *one-sided limits*. We will often simplify the notation by writing

$$\lim_{x\to x_0-} f(x) = f(x_0-) \quad \text{and} \quad \lim_{x\to x_0+} f(x) = f(x_0+).$$

The following theorem states the connection between limits and one-sided limits. We leave the proof as an exercise to the reader.

Theorem 2.16 A function f has a limit at x_0 if and only if it has left- and right-hand limits at x_0, and they are equal. More specifically,

$$\lim_{x\to x_0} f(x) = L$$

if and only if

$$f(x_0+) = f(x_0-) = L.$$

With only minor modifications of their proofs (replacing the inequality $0 < |x - x_0| < \delta$ by $x_0 - \delta < x < x_0$ or $x_0 < x < x_0 + \delta$), it can be shown that the assertions of Theorems 2.9 and 2.10 remain valid if "$\lim_{x\to x_0}$" is replaced by "$\lim_{x\to x_0-}$" or "$\lim_{x\to x_0+}$" throughout – try proving this yourself.

Limits at $\pm\infty$

Limits and one-sided limits have to do with the behavior of a function f near a limit point of D_f. It is equally reasonable to study f for large positive values of x if D_f is unbounded above or for large negative values of x if D_f is unbounded below.

Definition 2.17 We say that $f(x)$ *approaches the limit* L *as* x *approaches* ∞, and write

$$\lim_{x\to\infty} f(x) = L,$$

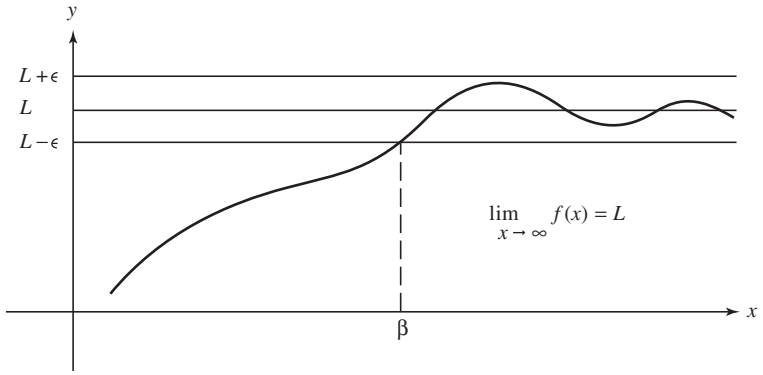
if f is defined on an interval (a,∞) and, for each $\epsilon > 0$, there is a number β such that

$$|f(x) - L| < \epsilon \qquad \text{if} \quad x > \beta.$$ ■

Figure 2.3 provides an illustration of the situation described in the above definition, namely Definition 2.17.

We leave it to you to define the statement "$\lim_{x\to-\infty} f(x) = L$" and to show that Theorems 2.9 and 2.10 remain valid if x_0 is replaced throughout by ∞ or $-\infty$.

Figure 2.3



■ **Example 2.18** Let

$$f(x) = 1 - \frac{1}{x^2}, \quad g(x) = \frac{2|x|}{1+x}, \quad \text{and} \quad h(x) = \sin x.$$

Then

$$\lim_{x\to\infty} f(x) = 1,$$

since

$$|f(x) - 1| = \frac{1}{x^2} < \epsilon \quad \text{if} \quad x > \frac{1}{\sqrt{\epsilon}},$$

and

$$\lim_{x\to\infty} g(x) = 2,$$

since

$$|g(x) - 2| = \left|\frac{2x}{1+x} - 2\right| = \frac{2}{1+x} < \frac{2}{x} < \epsilon \quad \text{if} \quad x > \frac{2}{\epsilon}.$$

However, $\lim_{x\to\infty} h(x)$ does not exist, since h assumes all values between -1 and 1 in any semi-infinite interval (τ, ∞).

We leave it to you to show that $\lim_{x\to-\infty} f(x) = 1$, $\lim_{x\to-\infty} g(x) = -2$, and $\lim_{x\to-\infty} h(x)$ does not exist. ■

We will sometimes denote $\lim_{x\to\infty} f(x)$ and $\lim_{x\to-\infty} f(x)$ by $f(\infty)$ and $f(-\infty)$, respectively.

Infinite Limits

The functions

$$f(x) = \frac{1}{x}, \quad g(x) = \frac{1}{x^2}, \quad p(x) = \sin\frac{1}{x},$$

and

$$q(x) = \frac{1}{x^2}\sin\frac{1}{x}$$

do not have limits, or even one-sided limits, at $x_0 = 0$. They fail to have limits in different ways:

- $f(x)$ increases beyond bound as x approaches 0 from the right and decreases beyond bound as x approaches 0 from the left;
- $g(x)$ increases beyond bound as x approaches zero;
- $p(x)$ oscillates with ever-increasing frequency as x approaches zero;

- $q(x)$ oscillates with ever-increasing amplitude and frequency as x approaches 0.

The kind of behavior exhibited by f and g near $x_0 = 0$ is sufficiently common and simple to lead us to define *infinite limits*.

Definition 2.19 We say that $f(x)$ *approaches* ∞ *as* x *approaches* x_0 *from the left*, and write

$$\lim_{x\to x_0-} f(x) = \infty \quad \text{or} \quad f(x_0-) = \infty,$$

if f is defined on an interval (a, x_0) and, for each real number M, there is a $\delta > 0$ such that

$$f(x) > M \quad \text{if} \quad x_0 - \delta < x < x_0.$$

■ **Example 2.20** We leave it to you to define the other kinds of infinite limits and show that

$$\lim_{x\to 0-} \frac{1}{x} = -\infty, \quad \lim_{x\to 0+} \frac{1}{x} = \infty;$$
$$\lim_{x\to 0-} \frac{1}{x^2} = \lim_{x\to 0+} \frac{1}{x^2} = \lim_{x\to 0} \frac{1}{x^2} = \infty;$$
$$\lim_{x\to\infty} x^2 = \lim_{x\to-\infty} x^2 = \infty;$$

and

$$\lim_{x\to\infty} x^3 = \infty, \quad \lim_{x\to-\infty} x^3 = -\infty.$$ ■

Throughout this book, "$\lim_{x\to x_0} f(x)$ exists" will mean that

$$\lim_{x\to x_0} f(x) = L, \quad \text{where } L \text{ is } \textit{finite}.$$

To leave open the possibility that $L = \pm\infty$, we will say that $\lim_{x\to x_0} f(x)$ *exists in the extended reals*. This convention also applies to one-sided limits and limits as x tends to $\pm\infty$.

We mentioned earlier that Theorems 2.9 and 2.10 remain valid if "$\lim_{x\to x_0}$" is replaced by "$\lim_{x\to x_0-}$" or "$\lim_{x\to x_0+}$." They are also valid with x_0 replaced by $\pm\infty$. Moreover, the counterparts of (2.10), (2.11), and (2.12) in all these versions of Theorem 2.10 remain valid if either or both of L_1 and L_2 are infinite, provided that their right sides are not indeterminate. Equation (2.13) and its counterparts remain valid if L_1/L_2 is not indeterminate and $L_2 \neq 0$.

■ **Example 2.21** Results like Theorem 2.10 yield

$$\begin{aligned}\lim_{x\to\infty}\sinh x &= \lim_{x\to\infty}\frac{e^x-e^{-x}}{2} = \frac{1}{2}\left(\lim_{x\to\infty}e^x - \lim_{x\to\infty}e^{-x}\right)\\ &= \frac{1}{2}(\infty - 0) = \infty,\\ \lim_{x\to-\infty}\sinh x &= \lim_{x\to-\infty}\frac{e^x-e^{-x}}{2} = \frac{1}{2}\left(\lim_{x\to-\infty}e^x - \lim_{x\to-\infty}e^{-x}\right)\\ &= \frac{1}{2}(0 - \infty) = -\infty,\end{aligned}$$

and

$$\lim_{x\to\infty}\frac{e^{-x}}{x} = \frac{\lim\limits_{x\to\infty}e^{-x}}{\lim\limits_{x\to\infty}x} = \frac{0}{\infty} = 0.$$

■ **Example 2.22** If

$$f(x) = e^{2x} - e^x,$$

we cannot obtain $\lim_{x\to\infty} f(x)$ by writing

$$\lim_{x\to\infty} f(x) = \lim_{x\to\infty} e^{2x} - \lim_{x\to\infty} e^x,$$

because this produces the indeterminate form $\infty - \infty$. However, by writing

$$f(x) = e^{2x}(1 - e^{-x}),$$

we find that

$$\lim_{x\to\infty} f(x) = \left(\lim_{x\to\infty} e^{2x}\right)\left(\lim_{x\to\infty} 1 - \lim_{x\to\infty} e^{-x}\right) = \infty(1-0) = \infty.$$

■ **Example 2.23** Let

$$g(x) = \frac{2x^2 - x + 1}{3x^2 + 2x - 1}.$$

Trying to find $\lim_{x\to\infty} g(x)$ by applying a version of Theorem 2.10 to this fraction as it is written leads to an indeterminate form (try it!). However, by rewriting it as

$$g(x) = \frac{2 - 1/x + 1/x^2}{3 + 2/x - 1/x^2}, \quad x \neq 0,$$

we find that

$$\lim_{x\to\infty} g(x) = \frac{\lim\limits_{x\to\infty} 2 - \lim\limits_{x\to\infty} 1/x + \lim\limits_{x\to\infty} 1/x^2}{\lim\limits_{x\to\infty} 3 + \lim\limits_{x\to\infty} 2/x - \lim\limits_{x\to\infty} 1/x^2} = \frac{2 - 0 + 0}{3 + 0 - 0} = \frac{2}{3}.$$

Monotonic Function

A function f is *nondecreasing* on an interval I if

$$f(x_1) \leq f(x_2) \quad \text{whenever } x_1 \text{ and } x_2 \text{ are in } I \text{ and } x_1 < x_2, \tag{2.18}$$

or *nonincreasing* on I if

$$f(x_1) \geq f(x_2) \quad \text{whenever } x_1 \text{ and } x_2 \text{ are in } I \text{ and } x_1 < x_2. \tag{2.19}$$

In either case, if $\leq$ can be replaced by $<$ in (2.18), we say f is *increasing* on I. If $\geq$ can be replaced by $>$ in (2.19), f is *decreasing* on I. In either of these two cases, f is *strictly monotonic* on I.

■ **Example 2.24** The function

$$f(x) = \begin{cases} x, & 0 \le x < 1, \\ 2, & 1 \le x \le 2, \end{cases}$$

is nondecreasing on $I = [0,2]$ (Figure 2.4), and $-f$ is nonincreasing on $I = [0,2]$.

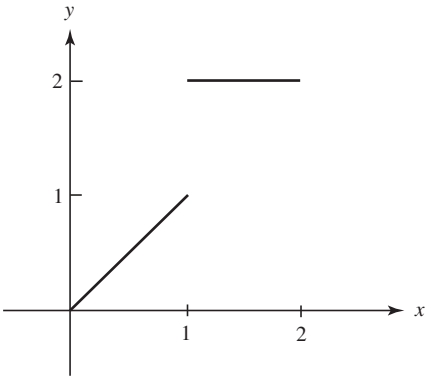


Figure 2.4

The function $g(x) = x^2$ is increasing on $[0,\infty)$ (Figure 2.5),

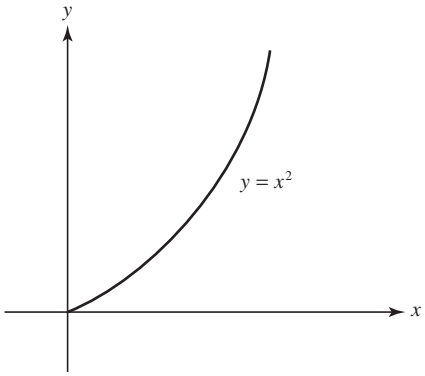


Figure 2.5

and $h(x) = -x^3$ is decreasing on $(-\infty,\infty)$ (Figure 2.6). ■

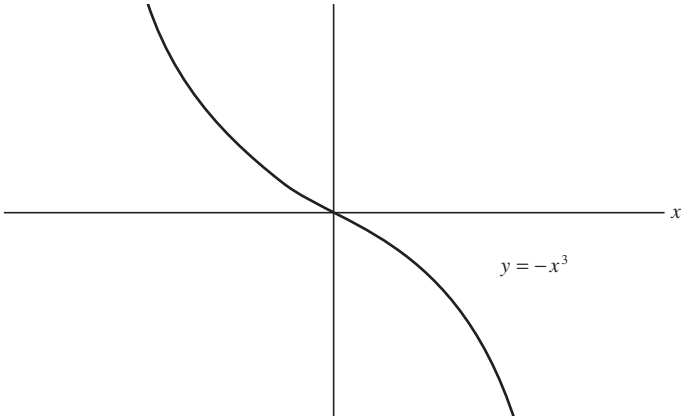


Figure 2.6

Theorem 2.25 Suppose that f is monotonic on (a,b) and define

$$\alpha = \inf_{a<x<b} f(x) \quad \text{and} \quad \beta = \sup_{a<x<b} f(x).$$

(a) If f is nondecreasing, then $f(a+) = \alpha$ and $f(b-) = \beta$.

(b) If f is nonincreasing, then $f(a+) = \beta$ and $f(b-) = \alpha$
(Here $a+ = -\infty$ if $a = -\infty$ and $b- = \infty$ if $b = \infty$.)

(c) If $a < x_0 < b$, then $f(x_0+)$ and $f(x_0-)$ exist and are finite. Moreover,

$$f(x_0-) \le f(x_0) \le f(x_0+)$$

if f is nondecreasing, and

$$f(x_0-) \ge f(x_0) \ge f(x_0+)$$

if f is nonincreasing.

Proof. **(a)** We first show that $f(a+) = \alpha$. If $M > \alpha$, there is an x_0 in (a,b) such that $f(x_0) < M$. Since f is nondecreasing, $f(x) < M$ if $a < x < x_0$. Therefore, if $\alpha = -\infty$, then $f(a+) = -\infty$. If $\alpha > -\infty$, let $M = \alpha + \epsilon$, where $\epsilon > 0$. Then $\alpha \le f(x) < \alpha + \epsilon$, so

$$|f(x) - \alpha| < \epsilon \quad \text{if} \quad a < x < x_0. \tag{2.20}$$

If $a = -\infty$, this implies that $f(-\infty) = \alpha$. If $a > -\infty$, let $\delta = x_0 - a$. Then (2.20) is equivalent to

$$|f(x) - \alpha| < \epsilon \quad \text{if} \quad a < x < a + \delta,$$

which implies that $f(a+) = \alpha$.

We now show that $f(b-) = \beta$. If $M < \beta$, there is an x_0 in (a,b) such that $f(x_0) > M$. Since f is nondecreasing, $f(x) > M$ if $x_0 < x < b$. Therefore, if $\beta = \infty$, then $f(b-) = \infty$. If $\beta < \infty$, let $M = \beta - \epsilon$, where $\epsilon > 0$. Then $\beta - \epsilon < f(x) \le \beta$, so

$$|f(x) - \beta| < \epsilon \quad \text{if} \quad x_0 < x < b. \tag{2.21}$$

If $b = \infty$, this implies that $f(\infty) = \beta$. If $b < \infty$, let $\delta = b - x_0$. Then (2.21) is equivalent to

$$|f(x) - \beta| < \epsilon \quad \text{if} \quad b - \delta < x < b,$$

which implies that $f(b-) = \beta$.

(b) Left as an exercise to the reader.

(c) Suppose that f is nondecreasing. Applying **(a)** to f on (a, x_0) and (x_0, b) separately shows that

$$f(x_0-) = \sup_{a<x<x_0} f(x) \quad \text{and} \quad f(x_0+) = \inf_{x_0<x<b} f(x).$$

However, if $x_1 < x_0 < x_2$, then

$$f(x_1) \le f(x_0) \le f(x_2);$$

hence,

$$f(x_0-) \le f(x_0) \le f(x_0+).$$

We leave the case where f is nonincreasing as an exercise for the reader. ■

2.2 Continuity

In this section we study continuous functions of a real variable. We will prove some important theorems about continuous functions that, although intuitively plausible, are beyond the scope of the elementary calculus course. They are accessible now because of our better understanding of the real number system, especially of those properties that stem from the completeness axiom.

The definitions of

$$f(x_0-) = \lim_{x\to x_0-} f(x), \quad f(x_0+) = \lim_{x\to x_0+} f(x), \quad \text{and} \quad \lim_{x\to x_0} f(x)$$

do not involve $f(x_0)$ or even require that it be defined. However, the case where $f(x_0)$ is defined and equal to one or more of these quantities is important.

Definition 2.26

(a) We say that f is *continuous at* x_0 if f is defined on an open interval (a,b) containing x_0 and $\lim_{x\to x_0} f(x) = f(x_0)$.

(b) We say that f is *continuous from the left at* x_0 if f is defined on an open interval (a,x_0) and $f(x_0-) = f(x_0)$.

(c) We say that f is *continuous from the right at* x_0 if f is defined on an open interval (x_0,b) and $f(x_0+) = f(x_0)$. ■

The following theorem provides a method for determining whether these definitions are satisfied. The proof, which we leave to the reader, rests on Definitions 2.6, 2.12, and 2.26.

Theorem 2.27

(a) A function f is continuous at x_0 if and only if f is defined on an open interval (a,b) containing x_0 and for each $\epsilon > 0$ there is a $\delta > 0$ such that

$$|f(x) - f(x_0)| < \epsilon \tag{2.22}$$

whenever $|x - x_0| < \delta$.

(b) A function f is continuous from the right at x_0 if and only if f is defined on an interval $[x_0,b)$ and for each $\epsilon > 0$ there is a $\delta > 0$ such that (2.22) holds whenever $x_0 \le x < x_0 + \delta$.

(c) A function f is continuous from the left at x_0 if and only if f is defined on an interval $(a,x_0]$ and for each $\epsilon > 0$ there is a $\delta > 0$ such that (2.22) holds whenever $x_0 - \delta < x \le x_0$.

From Definition 2.26 and Theorem 2.27, f is continuous at x_0 if and only if

$$f(x_0-) = f(x_0+) = f(x_0)$$

or, equivalently, if and only if it is continuous from the right and left at x_0.

■ **Example 2.28** Let f be defined on $[0,2]$ by

$$f(x) = \begin{cases} x^2, & 0 \le x < 1, \\ x+1, & 1 \le x \le 2 \end{cases}$$

see Figure 2.7, then

$$\begin{aligned} f(0+) &= 0 = f(0), \\ f(1-) &= 1 \neq f(1) = 2, \\ f(1+) &= 2 = f(1), \\ f(2-) &= 3 = f(2). \end{aligned}$$

Therefore, f is continuous from the right at 0 and 1 and continuous from the left at 2, but not at 1. If $0 < x$ and $x_0 < 1$, then

$$|f(x) - f(x_0)| = |x^2 - x_0^2| = |x - x_0|\,|x + x_0| \le 2|x - x_0| < \epsilon,$$

for all x satisfying $|x - x_0| < \epsilon/2$. Hence, f is continuous at each x_0 in $(0,1)$. If $1 < x$ and $x_0 < 2$, then

$$|f(x) - f(x_0)| = |(x+1) - (x_0+1) = |x - x_0| < \epsilon,$$

for all x satisfying $|x - x_0| < \epsilon$. Hence, f is continous at each x_0 in $(1,2)$.

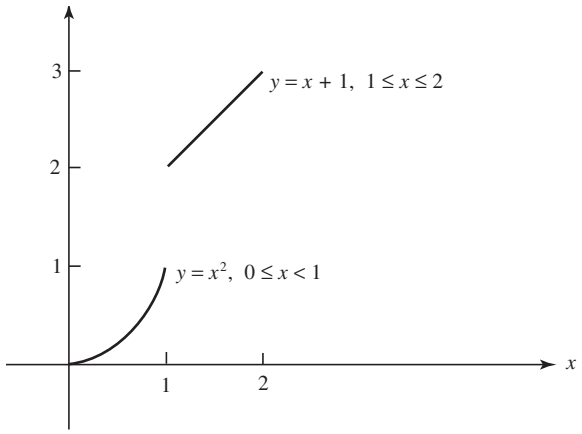


Figure 2.7

Definition 2.29 A function f is *continuous on an open interval* (a,b) if it is continuous at every point in (a,b). If, in addition,

$$f(b-) = f(b), \text{ or} \tag{2.23}$$
$$f(a+) = f(a) \tag{2.24}$$

then f is *continuous on* $(a,b]$ *or* $[a,b)$, respectively. If f is continuous on (a,b) and (2.23) and (2.24) both hold, then f *is continuous on* $[a,b]$. More generally, if S is a subset of D_f consisting of finitely or infinitely many disjoint intervals, then f is *continuous on* S if f is continuous on every interval in S. (Henceforth, in connection with functions of one variable, whenever we say "f is continuous on S" we mean that S is a set of this kind.)

■ **Example 2.30** Let $f(x) = \sqrt{x}$, $0 \le x < \infty$. Then

$$|f(x) - f(0)| = \sqrt{x} < \epsilon \quad \text{if} \quad 0 \le x < \epsilon^2,$$

so $f(0+) = f(0)$. If $x_0 > 0$ and $x \ge 0$, then

$$|f(x) - f(x_0)| = |\sqrt{x} - \sqrt{x_0}| = \frac{|x - x_0|}{\sqrt{x} + \sqrt{x_0}}$$
$$\le \frac{|x - x_0|}{\sqrt{x_0}} < \epsilon \quad \text{if} \quad |x - x_0| < \epsilon\sqrt{x_0},$$

so $\lim_{x \to x_0} f(x) = f(x_0)$. Hence, f is continuous on $[0, \infty)$.

■ **Example 2.31** The function

$$g(x) = \frac{1}{\sin \pi x}$$

is continuous on $S = \bigcup_{n=-\infty}^{\infty} (n, n+1)$. However, g is not continuous at any $x_0 = n$ (integer), since it is not defined at such points. ■

The function f defined in Example 2.28 (see also Figure 2.7) is continuous on $[0,1)$ and $[1,2]$, but not on any open interval containing 1. The discontinuity of f there is of the simplest kind, described in the following definition.

Definition 2.32 A function f is *piecewise continuous* on $[a,b]$ if

(a) $f(x_0+)$ exists for all x_0 in $[a,b)$;

(b) $f(x_0-)$ exists for all x_0 in $(a,b]$;

(c) $f(x_0+) = f(x_0-) = f(x_0)$ for all but finitely many points x_0 in (a,b).

If **(c)** fails to hold at some x_0 in (a,b), f has a *jump discontinuity at* x_0. Also, f has a *jump discontinuity at* a if $f(a+) \ne f(a)$ or *at* b if $f(b-) \ne f(b)$.

■ **Example 2.33** The function

$$f(x) = \begin{cases} 1, & x = 0, \\ x, & 0 < x < 1, \\ 2, & x = 1, \\ x, & 1 < x \le 2, \\ -1, & 2 < x < 3, \\ 0, & x = 3, \end{cases}$$

(Figure 2.8) is the graph of a piecewise continuous function on $[0,3]$, with jump discontinuities at $x_0 = 0$, 1, 2, and 3.

The reason for the adjective "jump" can be seen in Figures 2.7 and 2.8, where the graphs exhibit a definite jump at each point of discontinuity. The next example shows that not all discontinuities are of this kind.

■ **Example 2.34** The function

$$f(x) = \begin{cases} \sin \dfrac{1}{x}, & x \ne 0, \\ \\ 0, & x = 0, \end{cases}$$

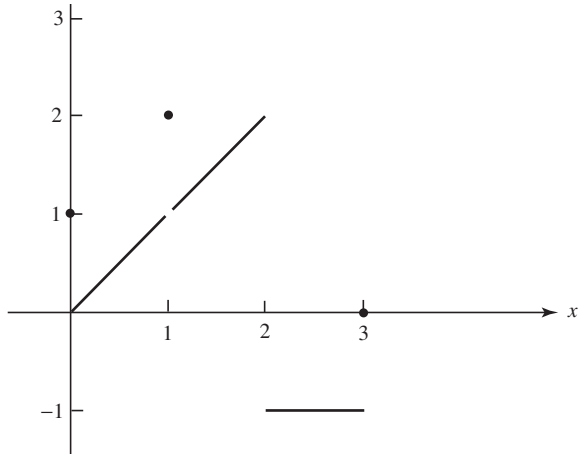


Figure 2.8

is continuous at all x_0 except $x_0 = 0$. As x approaches 0 from either side, $f(x)$ oscillates between -1 and 1 with ever-increasing frequency, so neither $f(0+)$ nor $f(0-)$ exists. Therefore, the discontinuity of f at 0 is not a jump discontinuity, and if $\rho > 0$, then f is not piecewise continuous on any interval of the form $[-\rho, 0]$, $[-\rho, \rho]$, or $[0, \rho]$. ■

Theorems 2.10 and 2.27 imply our next result.

Theorem 2.35 If f and g are continuous on a set S, then so are $f + g$, $f - g$, and fg. In addition, f/g is continuous at each x_0 in S such that $g(x_0) \neq 0$.

■ **Example 2.36** Since the constant functions and the function $f(x) = x$ are continuous for all x, successive applications of the various parts of Theorem 2.35 imply that the function

$$r(x) = \frac{9 - x^2}{x + 1}$$

is continuous for all x except $x = -1$ (see Example 2.11). More generally, by starting from Theorem 2.35 and using

induction, it can be shown that if $f_1, f_2, \ldots, f_n$ are continuous on a set S, then so are $f_1 + f_2 + \cdots + f_n$ and $f_1 f_2 \cdots f_n$. Therefore, any rational function

$$r(x) = \frac{a_0 + a_1 x + \cdots + a_n x^n}{b_0 + b_1 x + \cdots + b_m x^m} \quad (b_m \neq 0)$$

is continuous for all values of x except those for which its denominator vanishes.

Removable Discontinuities

Let f be defined on a deleted neighborhood of x_0 and discontinuous (perhaps even undefined) at x_0. We say that f has a *removable discontinuity* at x_0 if $\lim_{x \to x_0} f(x)$ exists. In this case, the function

$$g(x) = \begin{cases} f(x) & \text{if } x \in D_f \text{ and } x \neq x_0, \\ \lim\limits_{x \to x_0} f(x) & \text{if } x = x_0, \end{cases}$$

is continuous at x_0.

■ **Example 2.37** The function

$$f(x) = x \sin \frac{1}{x}$$

is not defined at $x_0 = 0$, and therefore certainly not continuous there, but $\lim_{x\to 0} f(x) = 0$ (Example 2.8). Therefore, f has a removable discontinuity at 0.

The function

$$f_1(x) = \sin \frac{1}{x}$$

is undefined at 0 and its discontinuity there is not removable, since $\lim_{x\to 0} f_1(x)$ does not exist (Example 2.34).

Composite Functions

We have seen that the investigation of limits and continuity can be simplified by regarding a given function as the result of addition, subtraction, multiplication, and division of simpler functions. Another operation useful in this connection is *composition* of functions; that is, substitution of one function into another.

Definition 2.38 Suppose that f and g are functions with domains D_f and D_g. If D_g has a nonempty subset T such that $g(x) \in D_f$ whenever $x \in T$, then the *composite function* $f \circ g$ is defined on T by

$$(f \circ g)(x) = f(g(x)).$$

■ **Example 2.39** If

$$f(x) = \log x \quad \text{and} \quad g(x) = \frac{1}{1-x^2},$$

then

$$D_f = (0, \infty) \quad \text{and} \quad D_g = \left\{ x \,\middle|\, x \neq \pm 1 \right\}.$$

Since $g(x) > 0$ if $x \in T = (-1, 1)$, the composite function $f \circ g$ is defined on $(-1, 1)$ by

$$(f \circ g)(x) = \log \frac{1}{1-x^2}.$$

We leave it to you to verify that $g \circ f$ is defined on $(0, 1/e) \cup (1/e, e) \cup (e, \infty)$ by

$$(g \circ f)(x) = \frac{1}{1 - (\log x)^2}.$$ ■

The next theorem says that the composition of continuous functions is continuous.

Theorem 2.40 Suppose that g is continuous at x_0, $g(x_0)$ is an interior point of D_f, and f is continuous at $g(x_0)$. Then $f \circ g$ is continuous at x_0.

Proof. Suppose that $\epsilon > 0$. Since $g(x_0)$ is an interior point of D_f and f is continuous at $g(x_0)$, there is a $\delta_1 > 0$ such that $f(t)$ is defined and

$$|f(t) - f(g(x_0))| < \epsilon \quad \text{if} \quad |t - g(x_0)| < \delta_1. \tag{2.25}$$

Since g is continuous at x_0, there is a $\delta > 0$ such that $g(x)$ is defined and

$$|g(x) - g(x_0)| < \delta_1 \quad \text{if} \quad |x - x_0| < \delta. \tag{2.26}$$

Now (2.25) and (2.26) imply that

$$|f(g(x)) - f(g(x_0))| < \epsilon \quad \text{if} \quad |x - x_0| < \delta.$$

Therefore, $f \circ g$ is continuous at x_0. ■

■ **Example 2.41** In Examples 2.30 and 2.36 we saw that the function

$$f(x) = \sqrt{x}$$

is continuous for $x > 0$, and the function

$$g(x) = \frac{9 - x^2}{x + 1}$$

is continuous for $x \neq -1$. Since $g(x) > 0$ if $x < -3$ or $-1 < x < 3$, Theorem 2.40 implies that the function

$$(f \circ g)(x) = \sqrt{\frac{9 - x^2}{x + 1}}$$

is continuous on $(-\infty, -3) \cup (-1, 3)$. It is also continuous from the left at -3 and 3.

Bounded Functions

A function f is *bounded below* on a set S if there is a real number m such that

$$f(x) \geq m \text{ for all } x \in S.$$

In this case, the set

$$V = \{f(x) \mid x \in S\}$$

has an infimum α, and we write

$$\alpha = \inf_{x \in S} f(x).$$

If there is a point x_1 in S such that $f(x_1) = \alpha$, we say that α is the *minimum of f on S*, and write

$$\alpha = \min_{x \in S} f(x).$$

Similarly, f is *bounded above on S* if there is a real number M such that $f(x) \leq M$ for all x in S. In this case, V has a supremum β, and we write

$$\beta = \sup_{x \in S} f(x).$$

If there is a point x_2 in S such that $f(x_2) = \beta$, we say that β is the *maximum of f on S*, and write

$$\beta = \max_{x \in S} f(x).$$

If f is bounded above and below on a set S, we say that f is *bounded* on S.

Figure 2.9 illustrates the geometric meaning of these definitions for a function f bounded on an interval $S = [a, b]$. The graph of f lies in the strip bounded by the lines $y = M$ and $y = m$, where M is any upper bound and m is any lower bound for f on $[a, b]$. The narrowest strip containing the graph is the one bounded above by $y = \beta = \sup_{a \leq x \leq b} f(x)$ and below by $y = \alpha = \inf_{a \leq x \leq b} f(x)$.

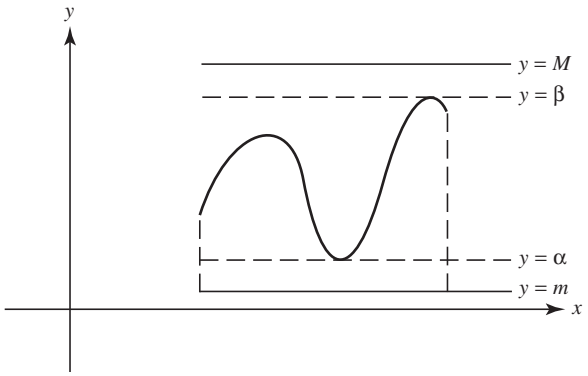


Figure 2.9

■ **Example 2.42** The function

$$g(x) = \begin{cases} \frac{1}{2}, & x = 0 \quad \text{or} \quad x = 1, \\ 1 - x, & 0 < x < 1, \end{cases} +$$

(Figure 2.10 **(a)**) is bounded on $[0, 1]$, and

$$\sup_{0 \le x \le 1} g(x) = 1, \quad \inf_{0 \le x \le 1} g(x) = 0.$$

Therefore, g has no maximum or minimum on $[0, 1]$, since it does not assume either of the values 0 and 1.

The function

$$h(x) = 1 - x, \quad 0 \le x \le 1,$$

which differs from g only at 0 and 1 (Figure 2.10 **(b)**), has the same supremum and infimum as g, but it attains these values at $x = 0$ and $x = 1$, respectively; therefore,

$$\max_{0 \le x \le 1} h(x) = 1 \quad \text{and} \quad \min_{0 \le x \le 1} h(x) = 0.$$

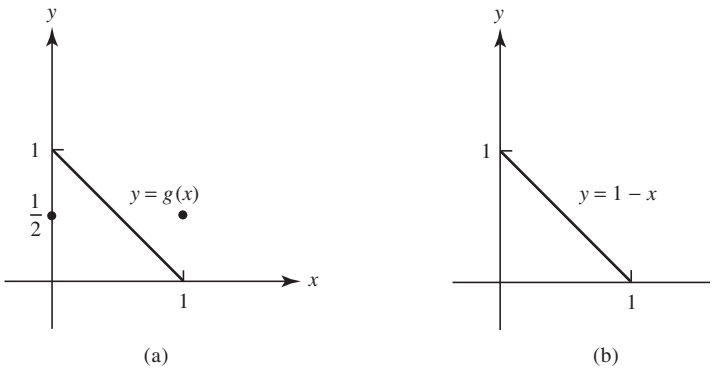


Figure 2.10

■ **Example 2.43** The function

$$f(x) = e^{x(x-1)} \sin \frac{1}{x(x-1)}, \quad 0 < x < 1,$$

oscillates between $\pm e^{x(x-1)}$ infinitely often in every interval of the form $(0, \rho)$ or $(1 - \rho, 1)$, where $0 < \rho < 1$, and

$$\sup_{0 < x < 1} f(x) = 1, \quad \inf_{0 < x < 1} f(x) = -1.$$

However, f does not assume these values, so f has no maximum or minimum on $(0,1)$.

Theorem 2.44 If f is continuous on a finite closed interval $[a,b]$, then f is bounded on $[a,b]$.

Proof. Suppose that $t \in [a,b]$. Since f is continuous at t, there is an open interval I_t containing t such that

$$|f(x) - f(t)| < 1 \quad \text{if} \quad x \in I_t \cap [a,b]. \tag{2.27}$$

(To see this, set $\epsilon = 1$ in (2.22), Theorem 2.27.) The collection $\mathcal{H} = \{I_t \mid a \le t \le b\}$ is an open covering of $[a,b]$. Since $[a,b]$ is compact, the Heine–Borel theorem implies that there are finitely many points $t_1, t_2, \ldots, t_n$ such that the intervals $I_{t_1}, I_{t_2}, \ldots, I_{t_n}$ cover $[a,b]$. According to (2.27) with $t = t_i$,

$$|f(x) - f(t_i)| < 1 \quad \text{if} \quad x \in I_{t_i} \cap [a,b].$$

Therefore,

$$\begin{aligned} |f(x)| &= |(f(x) - f(t_i)) + f(t_i)| \le |f(x) - f(t_i)| + |f(t_i)| \\ &\le 1 + |f(t_i)| \quad \text{if} \quad x \in I_{t_i} \cap [a,b]. \end{aligned} \tag{2.28}$$

Let

$$M = 1 + \max_{1 \le i \le n} |f(t_i)|.$$

Since $[a,b] \subset \bigcup_{i=1}^{n} (I_{t_i} \cap [a,b])$, (2.28) implies that $|f(x)| \le M$ if $x \in [a,b]$. ∎

This proof illustrates the utility of the Heine–Borel theorem, which allows us to choose M as the largest of a *finite* set of numbers.

Theorem 2.44 and the completeness of the reals imply that

if f is continuous on a finite closed interval $[a,b]$, then f has an infimum and a supremum on $[a,b]$. The next theorem shows that f actually assumes these values at some points in $[a,b]$.

Theorem 2.45 — Extreme Value Theorem. Suppose that f is continuous on a finite closed interval $[a,b]$. Let

$$\alpha = \inf_{a \le x \le b} f(x) \quad \text{and} \quad \beta = \sup_{a \le x \le b} f(x).$$

Then α and β are respectively the minimum and maximum of f on $[a,b]$; that is, there are points x_1 and x_2 in $[a,b]$ such that

$$f(x_1) = \alpha \quad \text{and} \quad f(x_2) = \beta.$$

Proof. We show that x_1 exists and leave it as an exercise to the reader to show that x_2 exists.

Suppose that there is no x_1 in $[a,b]$ such that $f(x_1) = \alpha$. Then $f(x) > \alpha$ for all $x \in [a,b]$. We will show that this leads to a contradiction.

Suppose that $t \in [a,b]$. Then $f(t) > \alpha$, so

$$f(t) > \frac{f(t) + \alpha}{2} > \alpha.$$

Since f is continuous at t, there is an open interval I_t about t such that

$$f(x) > \frac{f(t)+\alpha}{2} \quad \text{if} \quad x \in I_t \cap [a,b]. \tag{2.29}$$

The collection $\mathcal{H} = \{I_t \mid a \le t \le b\}$ is an open covering of $[a,b]$. Since $[a,b]$ is compact, the Heine–Borel theorem implies that there are finitely many points $t_1, t_2, \ldots, t_n$ such that the intervals $I_{t_1}, I_{t_2}, \ldots, I_{t_n}$ cover $[a,b]$. Define

$$\alpha_1 = \min_{1\le i\le n} \frac{f(t_i)+\alpha}{2}.$$

Then, since $[a,b] \subset \bigcup_{i=1}^{n}(I_{t_i} \cap [a,b])$, (2.29) implies that

$$f(t) > \alpha_1, \quad a \le t \le b.$$

But $\alpha_1 > \alpha$, so this contradicts the definition of α. Therefore, $f(x_1) = \alpha$ for some x_1 in $[a,b]$. ■

▪ **Example 2.46** We used the compactness of $[a,b]$ in the proof of Theorem 2.45 when we invoked the Heine–Borel theorem. To see that compactness is essential to the proof, consider the function

$$g(x) = 1 - (1-x)\sin\frac{1}{x},$$

which is continuous and has supremum 2 on the noncompact interval $(0,1]$, but does not assume its supremum on $(0,1]$, since

$$\begin{aligned} g(x) &\le 1 + (1-x)\left|\sin\frac{1}{x}\right| \\ &\le 1 + (1-x) < 2 \quad \text{if} \quad 0 < x \le 1. \end{aligned}$$

As another example, consider the function

$$f(x) = e^{-x},$$

which is continuous and has infimum 0, which it does not attain, on the noncompact interval $(0,\infty)$. ■

The next theorem shows that if f is continuous on a finite closed interval $[a,b]$, then f assumes every value between $f(a)$ and $f(b)$ as x varies from a to b (Figure 2.11, page 41).

Theorem 2.47 — Intermediate Value Theorem. Suppose that f is continuous on $[a,b]$, $f(a) \ne f(b)$, and μ is between $f(a)$ and $f(b)$. Then $f(c) = \mu$ for some c in (a,b).

Proof. Suppose that $f(a) < \mu < f(b)$. The set

$$S = \{x \mid a \le x \le b \quad \text{and} \quad f(x) \le \mu\}$$

is bounded and nonempty. Let $c = \sup S$. We will show that $f(c) = \mu$. If $f(c) > \mu$, then $c > a$ and, since f is continuous at c, there is an $\epsilon > 0$ such that $f(x) > \mu$ if $c - \epsilon < x \le c$. Therefore, $c - \epsilon$ is an upper bound for S, which contradicts the definition of c as the supremum of S. If $f(c) < \mu$, then $c < b$ and there is an $\epsilon > 0$ such that $f(x) < \mu$ for $c \le x < c + \epsilon$, so c is not an upper bound for S. This is also a contradiction. Therefore, $f(c) = \mu$.

The proof for the case where $f(b) < \mu < f(a)$ can be obtained by applying this result to $-f$.

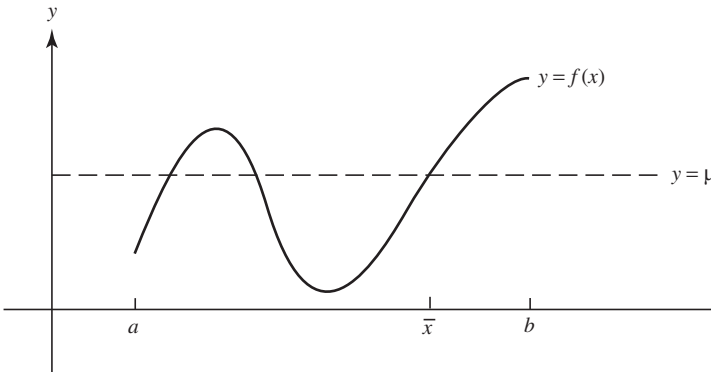


Figure 2.11

Uniform Continuity

Theorem 2.27 and Definition 2.29 imply that a function f is continuous on a subset S of its domain if for each $\epsilon > 0$ and each x_0 in S, there is a $\delta > 0$, *which may depend upon* x_0 *as well as* ϵ, such that

$$|f(x) - f(x_0)| < \epsilon \quad \text{if} \quad |x - x_0| < \delta \quad \text{and} \quad x \in D_f.$$

The next definition introduces another kind of continuity on a set S.

Definition 2.48 A function f is *uniformly continuous* on a subset S of its domain if, for every $\epsilon > 0$, there is a $\delta > 0$ such that

$$|f(x) - f(x')| < \epsilon \text{ whenever } |x - x'| < \delta \text{ and } x, x' \in S.$$

We emphasize that in this definition δ depends only on ϵ and S and not on the particular choice of x and x', provided that they are both in S.

■ **Example 2.49** The function

$$f(x) = 2x$$

is uniformly continuous on $(-\infty, \infty)$, since

$$|f(x) - f(x')| = 2|x - x'| < \epsilon \quad \text{if} \quad |x - x'| < \epsilon/2.$$

■ **Example 2.50** If $0 < r < \infty$, then the function

$$g(x) = x^2$$

is uniformly continuous on $[-r, r]$. To see this, note that

$$|g(x) - g(x') = |x^2 - (x')^2| = |x - x'|\,|x + x'| \le 2r|x - x'|,$$

so

$$|g(x) - g(x')| < \epsilon \quad \text{if} \quad |x - x'| < \delta = \frac{\epsilon}{2r} \quad \text{and} \quad -r \le x, x' \le r. \quad ■$$

Often a concept is clarified by considering its negation: a function f is *not* uniformly continuous on S if there is an $\epsilon_0 > 0$ such that if δ is any positive number, there are points x and x' in S such that

$$|x - x'| < \delta \quad \text{but} \quad |f(x) - f(x')| \ge \epsilon_0.$$

■ **Example 2.51** The function $g(x) = x^2$ is uniformly continuous on $[-r, r]$ for any finite r (Example 2.50), but not on $(-\infty, \infty)$. To see this, we will show that if $\delta > 0$ there are real numbers x and x' such that

$$|x - x'| = \delta/2 \quad \text{and} \quad |g(x) - g(x')| \geq 1.$$

To this end, we write

$$|g(x) - g(x')| = |x^2 - (x')^2| = |x - x'|\,|x + x'|.$$

If $|x - x'| = \delta/2$ and $x, x' > 1/\delta$, then

$$|x - x'|\,|x + x'| > \frac{\delta}{2}\left(\frac{1}{\delta} + \frac{1}{\delta}\right) = 1.$$

■ **Example 2.52** We have seen that the function

$$f(x) = \cos\frac{1}{x}$$

is continuous on $(0, 1]$. However, f is not uniformly continuous on $(0, 1]$, since

$$\left| f\left(\frac{1}{n\pi}\right) - f\left(\frac{1}{(n+1)\pi}\right)\right| = 2, \quad n = 1, 2, \ldots. \qquad \blacksquare$$

Examples 2.51 and 2.52 show that a function may be continuous but not uniformly continuous on an interval. The next theorem shows that this cannot happen if the interval is closed and bounded, and therefore compact.

Theorem 2.53 If f is continuous on a closed and bounded interval $[a, b]$, then f is uniformly continuous on $[a, b]$.

Proof. Suppose that $\epsilon > 0$. Since f is continuous on $[a, b]$, for each t in $[a, b]$ there is a positive number δ_t such that

$$|f(x) - f(t)| < \frac{\epsilon}{2} \quad \text{if} \quad |x - t| < 2\delta_t \quad \text{and} \quad x \in [a, b]. \tag{2.30}$$

If $I_t = (t - \delta_t, t + \delta_t)$, the collection

$$\mathcal{H} = \{I_t \mid t \in [a, b]\}$$

is an open covering of $[a, b]$. Since $[a, b]$ is compact, the Heine–Borel theorem implies that there are finitely many points $t_1, t_2, \ldots, t_n$ in $[a, b]$ such that I_{t_1}, $I_{t_2}, \ldots, I_{t_n}$ cover $[a, b]$. Now define

$$\delta = \min\{\delta_{t_1}, \delta_{t_2}, \ldots, \delta_{t_n}\}. \tag{2.31}$$

We will show that if

$$|x - x'| < \delta \quad \text{and} \quad x, x' \in [a, b], \tag{2.32}$$

then $|f(x) - f(x')| < \epsilon$.

From the triangle inequality,

$$\begin{aligned} |f(x) - f(x')| &= |\,(f(x) - f(t_r)) + (f(t_r) - f(x'))\,| \\ &\leq |f(x) - f(t_r)| + |f(t_r) - f(x')|. \end{aligned} \tag{2.33}$$

Since $I_{t_1}, I_{t_2}, \ldots, I_{t_n}$ cover $[a,b]$, x must be in one of these intervals. Suppose that $x \in I_{t_r}$; that is,

$$|x - t_r| < \delta_{t_r}. \tag{2.34}$$

From (2.30) with $t = t_r$,

$$|f(x) - f(t_r)| < \frac{\epsilon}{2}. \tag{2.35}$$

From (2.32), (2.34), and the triangle inquality,

$$|x' - t_r| = |(x' - x) + (x - t_r)| \le |x' - x| + |x - t_r| < \delta + \delta_{t_r} \le 2\delta_{t_r}.$$

Therefore, (2.30) with $t = t_r$ and x replaced by x' implies that

$$|f(x') - f(t_r)| < \frac{\epsilon}{2}.$$

This, (2.33), and (2.35) imply that $|f(x) - f(x')| < \epsilon$. ■

This proof again shows the utility of the Heine–Borel theorem, which allowed us to define δ in (2.31) as the smallest of a *finite* set of positive numbers, so that δ is sure to be positive. (An infinite set of positive numbers may fail to have a smallest positive member; for example, consider the open interval $(0,1)$.)

Corollary 2.54 If f is continuous on a set T, then f is uniformly continuous on any finite closed interval contained in T.

Applied to Example 2.52, Corollary 2.54 implies that the function $g(x) = \cos 1/x$ is uniformly continuous on $[\rho, 1]$ if $0 < \rho < 1$.

More About Monotonic Functions

Theorem 2.25 implies that if f is monotonic on an interval I, then f is either continuous or has a jump discontinuity at each x_0 in I. This and Theorem 2.47 provide the key to the proof of the following theorem.

Theorem 2.55 If f is monotonic and nonconstant on $[a,b]$, then f is continuous on $[a,b]$ if and only if its range $R_f = \{f(x) \mid x \in [a,b]\}$ is the closed interval with endpoints $f(a)$ and $f(b)$.

Proof. We assume that f is nondecreasing, and leave the case where f is nonincreasing as an exercise for the reader. Theorem 2.25 **(a)** implies that the set $\widetilde{R}_f = \{f(x) \mid x \in (a,b)\}$ is a subset of the open interval $(f(a+), f(b-))$. Therefore,

$$R_f = \{f(a)\} \cup \widetilde{R}_f \cup \{f(b)\} \subset \{f(a)\} \cup (f(a+), f(b-)) \cup \{f(b)\}. \tag{2.36}$$

Now suppose that f is continuous on $[a,b]$. Then $f(a) = f(a+)$, $f(b-) = f(b)$, so (2.36) implies that $R_f \subset [f(a), f(b)]$. If $f(a) < \mu < f(b)$, then Theorem 2.47 implies that $\mu = f(x)$ for some x in (a,b). Hence, $R_f = [f(a), f(b)]$.

For the converse, suppose that $R_f = [f(a), f(b)]$. Since $f(a) \le f(a+)$ and $f(b-) \le f(b)$, (2.36) implies that $f(a) = f(a+)$ and $f(b-) = f(b)$. We know from Theorem 2.25 **(c)** that if f is nondecreasing and $a < x_0 < b$, then

$$f(x_0-) \le f(x_0) \le f(x_0+).$$

If either of these inequalities is strict, R_f cannot be an interval. Since this contradicts our assumption, $f(x_0-) = f(x_0) = f(x_0+)$. Therefore, f is continuous at x_0. We can now conclude that f is continuous on $[a,b]$. ■

Theorem 2.55 implies the following result.

Theorem 2.56 Suppose that f is increasing and continuous on $[a,b]$, and let $f(a) = c$ and $f(b) = d$. Then there is a unique function g defined on $[c,d]$ such that

$$g(f(x)) = x, \quad a \le x \le b, \tag{2.37}$$

and

$$f(g(y)) = y, \quad c \le y \le d. \tag{2.38}$$

Moreover, g is continuous and increasing on $[c,d]$.

Proof. We first show that there is a function g satisfying (2.37) and (2.38). Since f is continuous, Theorem 2.55 implies that for each y_0 in $[c,d]$ there is an x_0 in $[a,b]$ such that

$$f(x_0) = y_0, \tag{2.39}$$

and, since f is increasing, there is only one such x_0. Define

$$g(y_0) = x_0. \tag{2.40}$$

The definition of x_0 is illustrated in Figure 2.12: with $[c,d]$ drawn on the y-axis, find the intersection of the line $y = y_0$ with the curve $y = f(x)$ and drop a vertical from the intersection to the x-axis to find x_0.

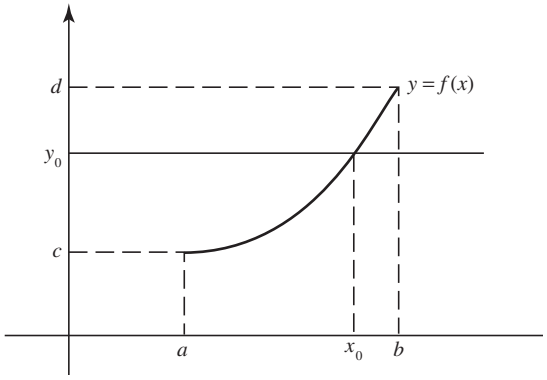


Figure 2.12

Substituting (2.40) into (2.39) yields

$$f(g(y_0)) = y_0,$$

and substituting (2.39) into (2.40) yields

$$g(f(x_0)) = x_0.$$

Dropping the subscripts in these two equations yields (2.37) and (2.38).

The uniqueness of g follows from our assumption that f is increasing, and therefore only one value of x_0 can satisfy (2.39) for each y_0.

To see that g is increasing, suppose that $y_1 < y_2$ and let x_1 and x_2 be the points in $[a,b]$ such that $f(x_1) = y_1$ and $f(x_2) = y_2$. Since f is increasing, $x_1 < x_2$. Therefore,

$$g(y_1) = x_1 < x_2 = g(y_2),$$

so g is increasing. Since $R_g = \{g(y) \mid y \in [c,d]\}$ is the interval $[g(c), g(d)] = [a,b]$, Theorem 2.55 with f and $[a,b]$ replaced by g and $[c,d]$ implies that g is continuous on $[c,d]$. ■

The function g of Theorem 2.56 is the *inverse* of f, denoted by f^{-1}. Since (2.37) and (2.38) are symmetric in f and g, we can also regard f as the inverse of g, and denote it by g^{-1}.

■ **Example 2.57** If

$$f(x) = x^2, \quad 0 \le x \le R,$$

then

$$f^{-1}(y) = g(y) = \sqrt{y}, \quad 0 \le y \le R^2.$$

■ **Example 2.58** If

$$f(x) = 2x + 4, \quad 0 \le x \le 2,$$

then

$$f^{-1}(y) = g(y) = \frac{y-4}{2}, \quad 4 \le y \le 8.$$

2.3 Differentiable Functions of One Variable

In calculus you studied differentiation, emphasizing rules for calculating derivatives. Here we consider the theoretical properties of differentiable functions. In doing this, we assume that you know how to differentiate elementary functions such as x^n, e^x, and $\sin x$, and we will use such functions in examples.

Definition 2.59 A function f is *differentiable* at an interior point x_0 of its domain if the difference quotient

$$\frac{f(x) - f(x_0)}{x - x_0}, \quad x \ne x_0,$$

approaches a limit as x approaches x_0, in which case the limit is called the *derivative of f at x_0*, and is denoted by $f'(x_0)$; thus,

$$f'(x_0) = \lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0}. \tag{2.41}$$

It is sometimes convenient to let $x = x_0 + h$ and write (2.41) as

$$f'(x_0) = \lim_{h \to 0} \frac{f(x_0 + h) - f(x_0)}{h}.$$

If f is defined on an open set S, we say that f is *differentiable on* S if f is differentiable at every point of S. If f is differentiable on S, then f' is a function on S. We say that f is *continuously differentiable* on S if f' is continuous on S. If f is differentiable on a neighborhood of x_0, it is reasonable to ask if f' is differentiable at x_0. If so, we denote the derivative of f' at x_0 by $f''(x_0)$. This is the *second derivative of f at x_0*, and it is also denoted by $f^{(2)}(x_0)$. Continuing inductively, if $f^{(n-1)}$ is defined on a neighborhood of x_0, then the nth *derivative of f at x_0*, denoted by $f^{(n)}(x_0)$, is the derivative of $f^{(n-1)}$ at x_0. For convenience we define the *zeroth derivative* of f to be f itself; thus

$$f^{(0)} = f.$$

We assume that you are familiar with the other standard notations for derivatives; for example,

$$f^{(2)} = f'', \quad f^{(3)} = f''',$$

and so on, and

$$\frac{d^n f}{dx^n} = f^{(n)}.$$

■ **Example 2.60** If n is a positive integer and

$$f(x) = x^n,$$

then

$$\frac{f(x) - f(x_0)}{x - x_0} = \frac{x^n - x_0^n}{x - x_0} = \frac{x - x_0}{x - x_0} \sum_{k=0}^{n-1} x^{n-k-1} x_0^k,$$

so

$$f'(x_0) = \lim_{x \to x_0} \sum_{k=0}^{n-1} x^{n-k-1} x_0^k = n x_0^{n-1}.$$

Since this holds for every x_0, we drop the subscript and write

$$f'(x) = n x^{n-1} \quad \text{or} \quad \frac{d}{dx}(x^n) = n x^{n-1}.$$ ■

To derive differentiation formulas for elementary functions such as $\sin x$, $\cos x$, and e^x directly from Definition 2.59 requires estimates based on the properties of these functions. Since this is done in calculus, we will not repeat it here.

Interpretations of the Derivative

If $f(x)$ is the position of a particle at time $x \neq x_0$, the difference quotient

$$\frac{f(x) - f(x_0)}{x - x_0}$$

is the average velocity of the particle between times x_0 and x. As x approaches x_0, the average applies to shorter and shorter intervals. Therefore, it makes sense to regard the limit (2.41), if it exists, as the particle's *instantaneous velocity at time* x_0. This interpretation may be useful even if x is not time, so we often regard $f'(x_0)$ as the *instantaneous rate of change of* $f(x)$ *at* x_0, regardless of the specific nature of the variable x. The derivative also has a geometric interpretation. The equation of the line through two points $(x_0, f(x_0))$ and $(x_1, f(x_1))$ on the curve $y = f(x)$ (Figure 2.13) is

$$y = f(x_0) + \frac{f(x_1) - f(x_0)}{x_1 - x_0}(x - x_0).$$

Varying x_1 generates lines through $(x_0, f(x_0))$ that rotate into the line

$$y = f(x_0) + f'(x_0)(x - x_0) \tag{2.42}$$

as x_1 approaches x_0. This is the *tangent* to the curve $y = f(x)$ at the point $(x_0, f(x_0))$. Figure 2.14 depicts the situation for various values of x_1.

Here is a less intuitive definition of the tangent line: If the function

$$T(x) = f(x_0) + m(x - x_0)$$

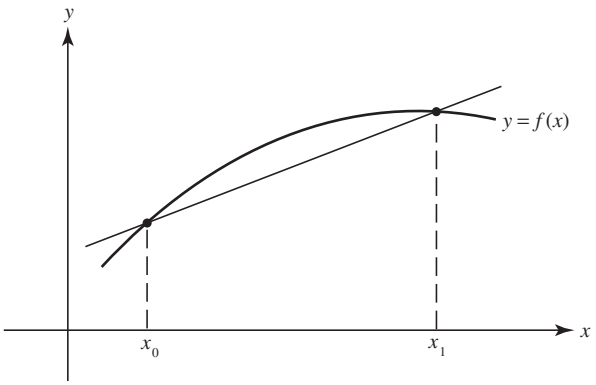


Figure 2.13

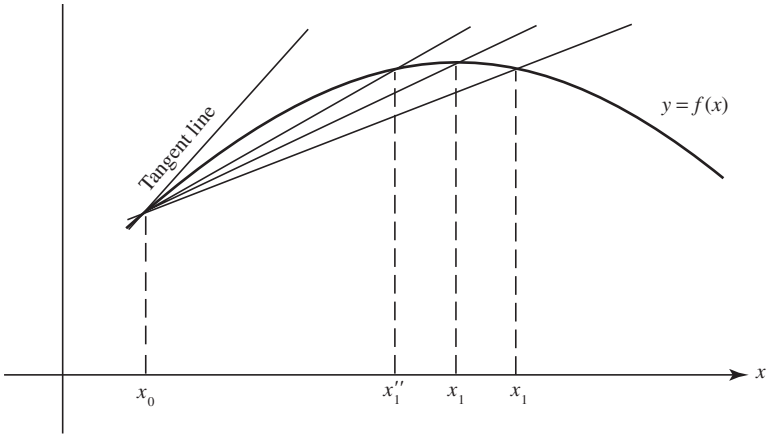


Figure 2.14

approximates f so well near x_0 that

$$\lim_{x\to x_0} \frac{f(x) - T(x)}{x - x_0} = 0,$$

we say that the line $y = T(x)$ is *tangent to the curve* $y = f(x)$ *at* $(x_0, f(x_0))$. This tangent line exists if and only if $f'(x_0)$ exists, in which case m is uniquely determined by $m = f'(x_0)$. Thus, (2.42) is the equation of the tangent line.

We will use the following lemma to study differentiable functions.

Lemma 2.61 If f is differentiable at x_0, then

$$f(x) = f(x_0) + [f'(x_0) + E(x)](x - x_0), \tag{2.43}$$

where E is defined on a neighborhood of x_0 and

$$\lim_{x\to x_0} E(x) = E(x_0) = 0.$$

Proof. Define

$$E(x) = \begin{cases} \dfrac{f(x) - f(x_0)}{x - x_0} - f'(x_0), & x \in D_f \text{ and } x \neq x_0, \\ 0, & x = x_0. \end{cases} \tag{2.44}$$

Solving (2.44) for $f(x)$ yields (2.43) if $x \neq x_0$, and (2.43) is obvious if $x = x_0$. Definition 2.59 implies that $\lim_{x\to x_0} E(x) = 0$. We defined $E(x_0) = 0$ to make E continuous at x_0. ∎

Since the right side of (2.43) is continuous at x_0, so is the left. This yields the following theorem.

Theorem 2.62 If f is differentiable at x_0, then f is continuous at x_0.

The converse of this theorem is false, since a function may be continuous at a point without being differentiable at the point.

■ **Example 2.63** The function

$$f(x) = |x|$$

can be written as

$$f(x) = \quad x, \quad x > 0, \tag{2.45}$$

or as

$$f(x) = -x, \quad x < 0. \tag{2.46}$$

From (2.45),

$$f'(x) = \quad 1, \quad x > 0,$$

and from (2.46),

$$f'(x) = -1, \quad x < 0.$$

Neither (2.45) nor (2.46) holds throughout any neighborhood of 0, so neither can be used alone to calculate $f'(0)$. In fact, since the one-sided limits

$$\lim_{x\to 0+} \frac{f(x)-f(0)}{x-0} = \lim_{x\to 0+} \frac{x}{x} \tag{2.47}$$

and

$$\lim_{x\to 0-} \frac{f(x)-f(0)}{x-0} = \lim_{x\to 0-} \frac{-x}{x} = -1 \tag{2.48}$$

are different,

$$\lim_{x\to 0} \frac{f(x)-f(0)}{x-0}$$

does not exist (Theorem 2.16); thus, f is not differentiable at 0, even though it is continuous at 0.

Interchanging Differentiation and Arithmetic Operations

The following theorem should be familiar from calculus.

Theorem 2.64 If f and g are differentiable at x_0, then so are $f+g$, $f-g$, and fg, with

(a) $(f+g)'(x_0) = f'(x_0) + g'(x_0)$;

(b) $(f-g)'(x_0) = f'(x_0) - g(x_0)$;

(c) $(fg)'(x_0) = f'(x_0)g(x_0) + f(x_0)g'(x_0)$.

The quotient f/g is differentiable at x_0 if $g(x_0) \neq 0$, with

(d) $$\left(\frac{f}{g}\right)'(x_0) = \frac{f'(x_0)g(x_0) - f(x_0)g'(x_0)}{[g(x_0)]^2}.$$

Proof. The proof is accomplished by forming the appropriate difference quotients and applying Definition 2.59 and Theorem 2.10. We will prove **(c)** and leave the rest as an exercise to the reader.

The trick is to add and subtract the right quantity in the numerator of the difference quotient for $(fg)'(x_0)$; thus,

$$\begin{aligned}\frac{f(x)g(x)-f(x_0)g(x_0)}{x-x_0} &= \frac{f(x)g(x)-f(x_0)g(x)+f(x_0)g(x)-f(x_0)g(x_0)}{x-x_0}\\ &= \frac{f(x)-f(x_0)}{x-x_0}g(x)+f(x_0)\frac{g(x)-g(x_0)}{x-x_0}.\end{aligned}$$

The difference quotients on the right approach $f'(x_0)$ and $g'(x_0)$ as x approaches x_0, and $\lim_{x\to x_0} g(x) = g(x_0)$ (Theorem 2.62). This proves **(c)**.
■

The Chain Rule

Here is the rule for differentiating a composite function.

Theorem 2.65 — The Chain Rule. Suppose that g is differentiable at x_0 and f is differentiable at $g(x_0)$. Then the composite function $h = f \circ g$, defined by

$$h(x) = f(g(x)),$$

is differentiable at x_0, with

$$h'(x_0) = f'(g(x_0))g'(x_0).$$

Proof. Since f is differentiable at $g(x_0)$, Lemma 2.61 implies that

$$f(t) - f(g(x_0)) = [f'(g(x_0)) + E(t)][t - g(x_0)],$$

where

$$\lim_{t\to g(x_0)} E(t) = E(g(x_0)) = 0. \tag{2.49}$$

Letting $t = g(x)$ yields

$$f(g(x)) - f(g(x_0)) = [f'(g(x_0)) + E(g(x))][g(x) - g(x_0)].$$

Since $h(x) = f(g(x))$, this implies that

$$\frac{h(x)-h(x_0)}{x-x_0} = [f'(g(x_0)) + E(g(x))]\frac{g(x)-g(x_0)}{x-x_0}. \tag{2.50}$$

Since g is continuous at x_0 (Theorem 2.62), (2.49) and Theorem 2.40 imply that

$$\lim_{x\to x_0} E(g(x)) = E(g(x_0)) = 0.$$

Therefore, (2.50) implies that

$$h'(x_0) = \lim_{x\to x_0}\frac{h(x)-h(x_0)}{x-x_0} = f'(g(x_0))g'(x_0),$$

as stated. ■

■ **Example 2.66** If

$$f(x) = \sin x \quad \text{and} \quad g(x) = \frac{1}{x}, \quad x \neq 0,$$

then

$$h(x) = f(g(x)) = \sin \frac{1}{x}, \quad x \neq 0,$$

and

$$h'(x) = f'(g(x))g(x) = \left(\cos \frac{1}{x}\right)\left(-\frac{1}{x^2}\right), \quad x \neq 0. \qquad \blacksquare$$

It may seem reasonable to justify the chain rule by writing

$$\begin{aligned} \frac{h(x) - h(x_0)}{x - x_0} &= \frac{f(g(x)) - f(g(x_0))}{x - x_0} \\ &= \frac{f(g(x)) - f(g(x_0))}{g(x) - g(x_0)} \, \frac{g(x) - g(x_0)}{x - x_0} \end{aligned}$$

and arguing that

$$\lim_{x \to x_0} \frac{f(g(x)) - f(g(x_0))}{g(x) - g(x_0)} = f'(g(x_0))$$

(because $\lim_{x \to x_0} g(x) = g(x_0)$) and

$$\lim_{x \to x_0} \frac{g(x) - g(x_0)}{x - x_0} = g'(x_0).$$

However, this is not a valid proof.

One-Sided Derivatives

One-sided limits of difference quotients such as (2.47) and (2.48) in Example 2.63 are called *one-sided* or *right- and left-hand derivatives*. That is, if f is defined on $[x_0, b)$, the *right-hand derivative of f at x_0* is defined to be

$$f'_+(x_0) = \lim_{x \to x_0+} \frac{f(x) - f(x_0)}{x - x_0}$$

if the limit exists, while if f is defined on $(a, x_0]$, the *left-hand derivative of f at x_0* is defined to be

$$f'_-(x_0) = \lim_{x \to x_0-} \frac{f(x) - f(x_0)}{x - x_0}$$

if the limit exists. Theorem 2.16 implies that f is differentiable at x_0 if and only if $f'_+(x_0)$ and $f'_-(x_0)$ exist and are equal, in which case

$$f'(x_0) = f'_+(x_0) = f'_-(x_0).$$

In Example 2.63, $f'_+(0) = 1$ and $f'_-(0) = -1$.

■ **Example 2.67** If

$$f(x) = \begin{cases} x^3, & x \leq 0, \\ x^2 \sin \dfrac{1}{x}, & x > 0, \end{cases} \tag{2.51}$$

then

$$f'(x) = \begin{cases} 3x^2, & x < 0, \\ 2x \sin \dfrac{1}{x} - \cos \dfrac{1}{x}, & x > 0. \end{cases} \tag{2.52}$$

Since neither formula in (2.51) holds for all x in any neighborhood of 0, we cannot simply differentiate either to obtain $f'(0)$; instead, we calculate

$$f'_+(0) = \lim_{x\to 0+} \frac{x^2 \sin\frac{1}{x} - 0}{x-0} = \lim_{x\to 0+} x \sin\frac{1}{x} = 0,$$

$$f'_-(0) = \lim_{x\to 0-} \frac{x^3 - 0}{x-0} = \lim_{x\to 0-} x^2 = 0;$$

hence, $f'(0) = f'_+(0) = f'_-(0) = 0$. ■

This example shows that there is a difference between a one-sided derivative and a one-sided limit of a derivative, since $f'_+(0) = 0$, but, from (2.52), $f'(0+) = \lim_{x\to 0+} f'(x)$ does not exist. It also shows that a derivative may exist in a neighborhood of a point x_0 ($= 0$ in this case), but be discontinuous at x_0.

Definition 2.68

(a) We say that f is *differentiable on the closed interval* $[a,b]$ if f is differentiable on the open interval (a,b) and $f'_+(a)$ and $f'_-(b)$ both exist.

(b) We say that f is *continuously differentiable on* $[a,b]$ if f is differentiable on $[a,b]$, f' is continuous on (a,b), $f'_+(a) = f'(a+)$, and $f'_-(b) = f'(b-)$.

Extreme Values

We say that $f(x_0)$ is a *local extreme value* of f if there is a $\delta > 0$ such that $f(x) - f(x_0)$ does not change sign on

$$(x_0 - \delta, x_0 + \delta) \cap D_f. \tag{2.53}$$

More specifically, $f(x_0)$ is a *local maximum value* of f if

$$f(x) \le f(x_0) \tag{2.54}$$

or a *local minimum value* of f if

$$f(x) \ge f(x_0) \tag{2.55}$$

for all x in the set (2.53). The point x_0 is called a *local extreme point* of f, or, more specifically, a *local maximum* or *local minimum point* of f.

■ **Example 2.69** If

$$f(x) = \begin{cases} 1, & -1 < x \le -\frac{1}{2} \\ |x|, & -\frac{1}{2} < x \le \frac{1}{2}, \\ \frac{1}{\sqrt{2}} \sin\frac{\pi x}{2}, & \frac{1}{2} < x \le 4 \end{cases}$$

(Figure 2.15), then 0, 3, and every x in $(-1, -\frac{1}{2})$ are local minimum points of f, while 1, 4, and every x in $(-1, -\frac{1}{2}]$ are local maximum points. ■

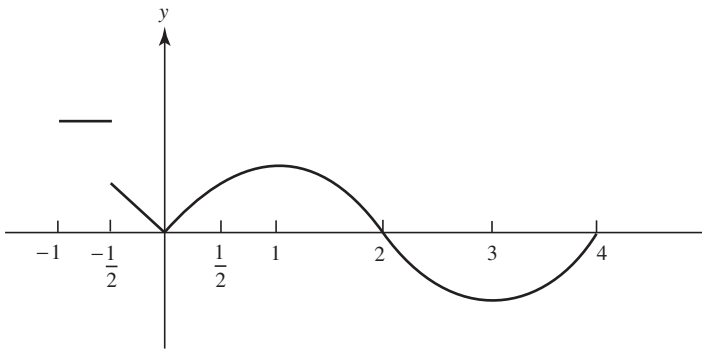


Figure 2.15

It is geometrically plausible that if the curve $y = f(x)$ has a tangent at a local extreme point of f, then the tangent must be horizontal; that is, have zero slope. (For example, in Figure 2.15, see $x = 1$, $x = 3$, and every x in $(-1, -1/2)$.) The following theorem shows that this must be so.

Theorem 2.70 If f is differentiable at a local extreme point $x_0 \in D_f^0$, then $f'(x_0) = 0$.

Proof. We will show that x_0 is not a local extreme point of f if $f'(x_0) \neq 0$. From Lemma 2.61,

$$\frac{f(x) - f(x_0)}{x - x_0} = f'(x_0) + E(x), \tag{2.56}$$

where $\lim_{x \to x_0} E(x) = 0$. Therefore, if $f'(x_0) \neq 0$, there is a $\delta > 0$ with

$$|E(x)| < |f'(x_0)| \quad \text{if} \quad |x - x_0| < \delta,$$

and the right side of (2.56) must have the same sign as $f'(x_0)$ for $|x - x_0| < \delta$. Since the same is true of the left side, $f(x) - f(x_0)$ must change sign in every neighborhood of x_0 (since $x - x_0$ does). Therefore, neither (2.54) nor (2.55) can hold for all x in any interval about x_0. ■

If $f'(x_0) = 0$, we say that x_0 is a *critical point* of f. Theorem 2.70 says that every local extreme point of f at which f is differentiable is a critical point of f. The converse is false. For example, 0 is a critical point of $f(x) = x^3$, but not a local extreme point.

Rolle's Theorem

The use of Theorem 2.70 for finding local extreme points is covered in calculus, so we will not pursue it here. However, we will use Theorem 2.70 to prove the following fundamental theorem, which says that if a curve $y = f(x)$ intersects a horizontal line at $x = a$ and $x = b$ and has a tangent at $(x, f(x))$ for every x in (a, b), then there is a point c in (a, b) such that the tangent to the curve at $(c, f(c))$ is horizontal (Figure 2.16).

Theorem 2.71 — Rolle's Theorem. Suppose that f is continuous on the closed interval $[a, b]$ and differentiable on the open interval (a, b), and $f(a) = f(b)$. Then $f'(c) = 0$ for some c in the open interval (a, b).

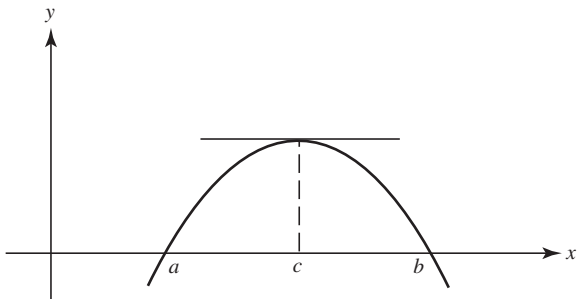


Figure 2.16

Proof. Since f is continuous on $[a,b]$, f attains a maximum and a minimum value on $[a,b]$ (Theorem 2.45). If these two extreme values are the same, then f is constant on (a,b), so $f'(x) = 0$ for all x in (a,b). If the extreme values differ, then at least one must be attained at some point c in the open interval (a,b), and $f'(c) = 0$, by Theorem 2.70. ■

Intermediate Values of Derivatives

A derivative may exist on an interval $[a,b]$ without being continuous on $[a,b]$. Nevertheless, an intermediate value theorem similar to Theorem 2.47 applies to derivatives.

Theorem 2.72 — Intermediate Value Theorem for Derivatives. Suppose that f is differentiable on $[a,b]$, $f'(a) \neq f'(b)$, and μ is between $f'(a)$ and $f'(b)$. Then $f'(c) = \mu$ for some c in (a,b).

Proof. Suppose first that

$$f'(a) < \mu < f'(b) \tag{2.57}$$

and define

$$g(x) = f(x) - \mu x.$$

Then

$$g'(x) = f'(x) - \mu, \quad a \leq x \leq b, \tag{2.58}$$

and (2.57) implies that

$$g'(a) < 0 \quad \text{and} \quad g'(b) > 0. \tag{2.59}$$

Since g is continuous on $[a,b]$, g attains a minimum at some point c in $[a,b]$. Lemma 2.61 and (2.59) imply that there is a $\delta > 0$ such that

$$g(x) < g(a), \quad a < x < a + \delta, \quad \text{and} \quad g(x) < g(b), \quad b - \delta < x < b$$

and therefore $c \neq a$ and $c \neq b$. Hence, $a < c < b$, and therefore $g'(c) = 0$, by Theorem 2.70. From (2.58), $f'(c) = \mu$.

The proof for the case where $f'(b) < \mu < f'(a)$ can be obtained by applying this result to $-f$. ■

Mean Value Theorems

Theorem 2.73 — Generalized Mean Value Theorem. If f and g are continuous on the closed interval $[a,b]$ and differentiable on the open interval (a,b), then

$$[g(b)-g(a)]f'(c)=[f(b)-f(a)]g'(c) \tag{2.60}$$

for some c in (a,b).

Proof. The function

$$h(x)=[g(b)-g(a)]f(x)-[f(b)-f(a)]g(x)$$

is continuous on $[a,b]$ and differentiable on (a,b), and

$$h(a)=h(b)=g(b)f(a)-f(b)g(a).$$

Therefore, Rolle's theorem implies that $h'(c)=0$ for some c in (a,b). Since

$$h'(c)=[g(b)-g(a)]f'(c)-[f(b)-f(a)]g'(c),$$

this implies (2.60). ■

The following special case of Theorem 2.73 is important enough to be stated separately.

Theorem 2.74 — Mean Value Theorem. If f is continuous on the closed interval $[a,b]$ and differentiable on the open interval (a,b), then

$$f'(c)=\frac{f(b)-f(a)}{b-a}$$

for some c in (a,b).

Proof. Apply Theorem 2.73 with $g(x)=x$.

■

Theorem 2.74 implies that the tangent to the curve $y=f(x)$ at $(c,f(c))$ is parallel to the line connecting the points $(a,f(a))$ and $(b,f(b))$ on the curve (Figure 2.17, page 55).

Consequences of the Mean Value Theorem

If f is differentiable on (a,b) and x_1, $x_2\in(a,b)$ then f is continuous on the closed interval with endpoints x_1 and x_2 and differentiable on its interior. Hence, the mean value theorem implies that

$$f(x_2)-f(x_1)=f'(c)(x_2-x_1)$$

for some c between x_1 and x_2. (This is true whether $x_1<x_2$ or $x_2<x_1$.) The next three theorems follow from this.

Theorem 2.75 If $f'(x)=0$ for all x in (a,b), then f is constant on (a,b).

Theorem 2.76 If f' exists and does not change sign on (a,b), then f is monotonic on (a,b) : increasing, nondecreasing, decreasing, or nonincreasing as

$$f'(x) > 0, \quad f'(x) \geq 0, \quad f'(x) < 0, \quad \text{or} \quad f'(x) \leq 0,$$

respectively, for all x in (a,b).

Theorem 2.77 If

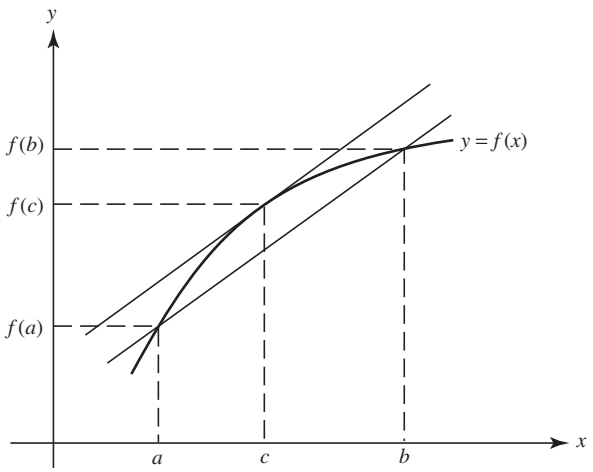
$$|f'(x)| \leq M, \quad a < x < b,$$

then

$$|f(x) - f(x')| \leq M|x - x'|, \quad x, x' \in (a,b). \tag{2.61}$$

A function that satisfies an inequality like (2.61) for all x and x' in an interval is said to satisfy a Lipschitz *condition* on the interval.

Figure 2.17



3 Infinite Sequences and Series

IN THIS CHAPTER we consider infinite sequences and series of constants and functions of a real variable.

SECTION 3.1 introduces infinite sequences of real numbers. The concept of a limit of a sequence is defined, as is the concept of divergence of a sequence to $\pm\infty$.

SECTION 3.2 defines a subsequence of an infinite sequence. We show that if a sequence converges to a limit or diverges to $\pm\infty$, then so do all subsequences of the sequence. Limit points and boundedness of a set of real numbers are discussed in terms of sequences of members of the set. Continuity and boundedness of a function are discussed in terms of the values of the function at sequences of points in its domain.

SECTION 3.3 introduces concepts of convergence and divergence to $\pm\infty$ for infinite series of constants.

SECTION 3.4 deals with pointwise and uniform convergence of sequences and series of functions. Cauchy's uniform convergence criteria for sequences and series are proved. We give sufficient conditions for the limit of a sequence of functions or the sum of an infinite series of functions to be continuous.

3.1 Sequences of Real Numbers

An *infinite sequence* (more briefly, a *sequence*) of real numbers is a real-valued function defined on a set of integers $\{n \mid n \geq k\}$. We call the values of the function the *terms* of the sequence. We denote a sequence by listing its terms in order; thus,

$$\{s_n\}_k^\infty = \{s_k, s_{k+1}, \dots\}. \tag{3.1}$$

For example,

$$\left\{\frac{1}{n^2+1}\right\}_0^\infty = \left\{1, \frac{1}{2}, \frac{1}{5}, \dots, \frac{1}{n^2+1}, \dots\right\},$$
$$\{(-1)^n\}_0^\infty = \{1, -1, 1, \dots, (-1)^n, \dots\},$$

and

$$\left\{\frac{1}{n-2}\right\}_3^\infty = \left\{1, \frac{1}{2}, \frac{1}{3}, \dots, \frac{1}{n-2}, \dots\right\}.$$

The real number s_n is the nth *term* of the sequence. Usually we are interested only in the terms of a sequence and the order in which they appear, but not in the particular value of k in (3.1). Therefore, we regard the sequences

$$\left\{\frac{1}{n-2}\right\}_3^\infty \quad \text{and} \quad \left\{\frac{1}{n}\right\}_1^\infty$$

as identical.

We will usually write $\{s_n\}$ rather than $\{s_n\}_k^\infty$. In the absence of any indication to the contrary, we take $k = 0$ unless s_n is given by a rule that is invalid for some nonnegative integer, in which case k is understood to be the smallest positive integer such that s_n is defined for all $n \geq k$. For example, if

$$s_n = \frac{1}{(n-1)(n-5)},$$

then $k = 6$.

The interesting questions about a sequence $\{s_n\}$ concern the behavior of s_n for large n.

Definition 3.1 A sequence $\{s_n\}$ *converges to a limit* s if for every $\epsilon > 0$ there is an integer N such that

$$|s_n - s| < \epsilon \quad \text{if} \quad n \geq N. \tag{3.2}$$

In this case we say that $\{s_n\}$ is *convergent* and write

$$\lim_{n\to\infty} s_n = s.$$

A sequence that does not converge *diverges*, or is *divergent* ■

As we saw in Section 2.1 when discussing limits of functions, Definition 3.1 is not changed by replacing (3.2) with

$$|s_n - s| < K\epsilon \quad \text{if} \quad n \geq N,$$

where K is a positive constant.

■ **Example 3.2** If $s_n = c$ for $n \geq k$, then $|s_n - c| = 0$ for $n \geq k$, and $\lim_{n\to\infty} s_n = c$.

■ **Example 3.3** If

$$s_n = \frac{2n+1}{n+1},$$

then $\lim_{n\to\infty} s_n = 2$, since

$$|s_n - 2| = \left|\frac{2n+1}{n+1} - \frac{2n+2}{n+1}\right| = \frac{1}{n+1};$$

hence, if $\epsilon > 0$, then (3.2) holds with $s = 2$ if $N \geq 1/\epsilon$. ■

Definition 3.1 does not require that there be an integer N such that (3.2) holds for all ϵ; rather, it requires that for each positive ϵ there be an integer N that satisfies (3.2) for that particular ϵ. Usually, N depends on ϵ and must be increased if ϵ is decreased. The constant sequences (Example 3.2) are essentially the only ones for which N does not depend on ϵ.

We say that the terms of a sequence $\{s_n\}_k^\infty$ satisfy a given condition *for all* n if s_n satisfies the condition for all $n \geq k$, or *for large* n if there is an integer $N > k$ such that s_n satisfies the condition whenever $n \geq N$. For example, the terms of $\{1/n\}_1^\infty$ are positive for all n, while those of $\{1 - 7/n\}_1^\infty$ are positive for large n (take $N = 8$).

Sequences Diverging to $\pm\infty$

We say that

$$\lim_{n\to\infty} s_n = \infty$$

if for any real number a, $s_n > a$ for large n. Similarly,

$$\lim_{n\to\infty} s_n = -\infty$$

if for any real number a, $s_n < a$ for large n. However, we do not regard $\{s_n\}$ as convergent unless $\lim_{n\to\infty} s_n$ is finite, as required by Definition 3.1. To emphasize this distinction, we say that $\{s_n\}$ *diverges to* ∞ $(-\infty)$ if $\lim_{n\to\infty} s_n = \infty$ $(-\infty)$.

■ **Example 3.4** The sequence $\{n/2 + 1/n\}$ diverges to ∞, since, if a is any real number, then

$$\frac{n}{2} + \frac{1}{n} > a \quad \text{if} \quad n \geq 2a.$$

The sequence $\{n - n^2\}$ diverges to $-\infty$, since, if a is any real number, then

$$-n^2 + n = -n(n-1) < a \quad \text{if} \quad n > 1 + \sqrt{|a|}.$$

Therefore, we write

$$\lim_{n\to\infty} \left(\frac{n}{2} + \frac{1}{n}\right) = \infty \quad \text{and} \quad \lim_{n\to\infty} (-n^2 + n) = -\infty.$$

The sequence $\{(-1)^n n^3\}$ diverges, but not to $-\infty$ or ∞.

Sequences of Functional Values

The next theorem enables us to apply the theory of limits developed in Section 2.1 to some sequences. We leave the proof as an exercise for the reader.

Theorem 3.5 Let $\lim_{x\to\infty} f(x) = L$, where L is in the extended reals, and suppose that $s_n = f(n)$ for large n. Then

$$\lim_{n\to\infty} s_n = L.$$

■ **Example 3.6** Let

$$s_n = \frac{\log n}{n} \quad \text{and} \quad f(x) = \frac{\log x}{x}.$$

By L'Hospital's rule,

$$\lim_{x\to\infty} \frac{\log x}{x} = \lim_{x\to\infty} \frac{1/x}{1} = 0.$$

Hence, $\lim_{n\to\infty} \log n/n = 0$.

■ **Example 3.7** Let $s_n = (1 + 1/n)^n$ and

$$f(x) = \left(1 + \frac{1}{x}\right)^x = e^{x\log(1+1/x)}.$$

By L'Hospital's rule,

$$\lim_{x\to\infty} x \log\left(1 + \frac{1}{x}\right) = \lim_{x\to\infty} \frac{\log(1+1/x)}{1/x} = \lim_{x\to\infty} \frac{-\frac{1}{x^2}\frac{1}{1+1/x}}{-1/x^2} = 1;$$

hence,

$$\lim_{x\to\infty} \left(1 + \frac{1}{x}\right)^x = e^1 = e \quad \text{and} \quad \lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^n = e.$$

The last equation is sometimes used to define e.

■ **Example 3.8** Suppose that $s_n = \rho^n$ with $\rho > 0$, and let $f(x) = \rho^x = e^{x\log\rho}$. Since

$$\lim_{x\to\infty} e^{x\log\rho} = \begin{cases} 0, & \text{if } \log\rho < 0 \quad (0 < \rho < 1), \\ 1, & \text{if } \log\rho = 0 \quad (\rho = 1), \\ \infty, & \text{if } \log\rho > 0 \quad (\rho > 1), \end{cases}$$

it follows that

$$\lim_{n\to\infty} \rho^n = \begin{cases} 0, & 0 < \rho < 1, \\ 1, & \rho = 1, \\ \infty, & \rho > 1. \end{cases}$$

Therefore,

$$\lim_{n\to\infty} r^n = \begin{cases} 0, & -1 < r < 1, \\ 1, & r = 1, \\ \infty, & r > 1, \end{cases}$$

a result that we will use often.

3.2 Earlier Topics Revisited with Sequences

In Section 2.3 we used ϵ–δ definitions and arguments to develop the theory of limits, continuity, and differentiability; for example, f is continuous at x_0 if for each $\epsilon > 0$ there is a $\delta > 0$ such that $|f(x) - f(x_0)| < \epsilon$ when $|x - x_0| < \delta$. The same theory can be developed by methods based on sequences. Although we will not carry this out in detail, we will develop it enough to give some examples. First, we need another definition about sequences.

Definition 3.9 A sequence $\{t_k\}$ is a *subsequence* of a sequence $\{s_n\}$ if

$$t_k = s_{n_k}, \quad k \geq 0,$$

where $\{n_k\}$ is an increasing infinite sequence of integers in the domain of $\{s_n\}$. We denote the subsequence $\{t_k\}$ by $\{s_{n_k}\}$. ■

Note that $\{s_n\}$ is a subsequence of itself, as can be seen by taking $n_k = k$. All other subsequences of $\{s_n\}$ are obtained by deleting terms from $\{s_n\}$ and leaving those remaining in their original relative order.

■ **Example 3.10** If

$$\{s_n\} = \left\{\frac{1}{n}\right\} = \left\{1, \frac{1}{2}, \frac{1}{3}, \ldots, \frac{1}{n}, \ldots\right\},$$

then letting $n_k = 2k$ yields the subsequence

$$\{s_{2k}\} = \left\{\frac{1}{2k}\right\} = \left\{\frac{1}{2}, \frac{1}{4}, \ldots, \frac{1}{2k}, \ldots\right\},$$

and letting $n_k = 2k + 1$ yields the subsequence

$$\{s_{2k+1}\} = \left\{\frac{1}{2k+1}\right\} = \left\{\frac{1}{3}, \ldots, \frac{1}{2k+1}, \ldots\right\}. \quad ■$$

Since a subsequence $\{s_{n_k}\}$ is again a sequence (with respect to k), we may ask whether $\{s_{n_k}\}$ converges.

■ **Example 3.11** The sequence $\{s_n\}$ defined by

$$s_n = (-1)^n \left(1 + \frac{1}{n}\right)$$

does not converge, but $\{s_n\}$ has subsequences that do. For example,

$$\{s_{2k}\} = \left\{1 + \frac{1}{2k}\right\} \quad \text{and} \quad \lim_{k\to\infty} s_{2k} = 1,$$

while

$$\{s_{2k+1}\} = \left\{-1 - \frac{1}{2k+1}\right\} \quad \text{and} \quad \lim_{k\to\infty} s_{2k+1} = -1.$$

It can be shown that a subsequence $\{s_{n_k}\}$ of $\{s_n\}$ converges to 1 if and only if n_k is even for k sufficiently large, or to -1 if and only if n_k is odd for k sufficiently large. Otherwise, $\{s_{n_k}\}$ diverges. ■

The sequence in this example has subsequences that converge to different limits. The next theorem shows that if a sequence converges to a finite limit or diverges to $\pm\infty$, then all its subsequences do also.

Theorem 3.12 If

$$\lim_{n\to\infty} s_n = s \quad (-\infty \le s \le \infty), \tag{3.3}$$

then

$$\lim_{k\to\infty} s_{n_k} = s \tag{3.4}$$

for every subsequence $\{s_{n_k}\}$ of $\{s_n\}$.

Proof. We consider the case where s is finite and leave the rest as an exercise to the reader. If (3.3) holds and $\epsilon > 0$, there is an integer N such that

$$|s_n - s| < \epsilon \quad \text{if} \quad n \ge N.$$

Since $\{n_k\}$ is an increasing sequence, there is an integer K such that $n_k \ge N$ if $k \ge K$. Therefore,

$$|s_{n_k} - L| < \epsilon \quad \text{if} \quad k \ge K,$$

which implies (3.4). ■

Limit Points in Terms of Sequences

In Section 1.2 we defined *limit point* in terms of neighborhoods: $\overline{x}$ is a limit point of a set S if every neighborhood of $\overline{x}$ contains points of S distinct from $\overline{x}$. The next theorem shows that an equivalent definition can be stated in terms of sequences.

Theorem 3.13 A point $\overline{x}$ is a limit point of a set S if and only if there is a sequence $\{x_n\}$ of points in S such that $x_n \ne \overline{x}$ for $n \ge 1$, and

$$\lim_{n\to\infty} x_n = \overline{x}.$$

Proof. For sufficiency, suppose that the stated condition holds. Then, for each $\epsilon > 0$, there is an integer N such that $0 < |x_n - x| < \epsilon$ if $n \ge N$. Therefore, every ϵ-neighborhood of $\overline{x}$ contains infinitely many points of S. This means that $\overline{x}$ is a limit point of S.

For necessity, let $\overline{x}$ be a limit point of S. Then, for every integer $n \ge 1$, the interval $(\overline{x} - 1/n, \overline{x} + 1/n)$ contains a point x_n $(\ne \overline{x})$ in S. Since $|x_m - \overline{x}| \le 1/n$ if $m \ge n$, $\lim_{n\to\infty} x_n = \overline{x}$. ■

We will use the next theorem to show that continuity can be defined in terms of sequences.

Theorem 3.14

(a) If $\{x_n\}$ is bounded, then $\{x_n\}$ has a convergent subsequence.

(b) If $\{x_n\}$ is unbounded above, then $\{x_n\}$ has a subsequence $\{x_{n_k}\}$ such that
$$\lim_{k\to\infty} x_{n_k} = \infty.$$

(c) If $\{x_n\}$ is unbounded below, then $\{x_n\}$ has a subsequence $\{x_{n_k}\}$ such that
$$\lim_{k\to\infty} x_{n_k} = -\infty.$$

Proof. We prove **(a)** and leave parts **(b)** and **(c)** as an exercise for the reader. to you. Let S be the set of distinct numbers that occur as terms of $\{x_n\}$. (For example, if $\{x_n\} = \{(-1)^n\}$, $S = \{1,-1\}$; if $\{x_n\} = \{1,\frac{1}{2},1,\frac{1}{3},\dots,1,1/n,\dots\}$, $S = \{1,\frac{1}{2},\dots,1/n,\dots\}$.) If S contains only finitely many points, then some $\overline{x}$ in S occurs infinitely often in $\{x_n\}$; that is, $\{x_n\}$ has a subsequence $\{x_{n_k}\}$ such that $x_{n_k} = \overline{x}$ for all k. Then $\lim_{k\to\infty} x_{n_k} = \overline{x}$, and we are finished in this case.

If S is infinite, then, since S is bounded (by assumption), the Bolzano–Weierstrass theorem (Theorem 1.30) implies that S has a limit point $\overline{x}$. From Theorem 3.13, there is a sequence of points $\{y_j\}$ in S, distinct from $\overline{x}$, such that
$$\lim_{j\to\infty} y_j = \overline{x}. \tag{3.5}$$
Although each y_j occurs as a term of $\{x_n\}$, $\{y_j\}$ is not necessarily a subsequence of $\{x_n\}$, because if we write $y_j = x_{n_j}$, there is no reason to expect that $\{n_j\}$ is an increasing sequence as required in Definition 3.9. However, it is always possible to pick a subsequence $\{n_{j_k}\}$ of $\{n_j\}$ that is increasing, and then the sequence $\{y_{j_k}\} = \{s_{n_{j_k}}\}$ is a subsequence of both $\{y_j\}$ and $\{x_n\}$. Because of (3.5) and Theorem 3.12 this subsequence converges to $\overline{x}$.

Cauchy's Convergence Criterion

To determine from Definition 3.1 whether a sequence has a limit, it is necessary to guess what the limit is. (This is particularly difficult if the sequence diverges!) So we introduce the notion of Cauchy sequences to study convergence of sequences without knowing their limit.

Definition 3.15 A sequence $\{s_n\}$ of real numbers is said to be a **Cauchy sequence** (or, more simply, is said to be Cauchy) if for any $\varepsilon > 0$, there exists a $N \in \mathbb{N}$ such that if $n \geq N$ and $m \geq N$ (briefly $n, m \geq N$) then $|s_n - s_m| < \varepsilon$.

■ **Example 3.16** The sequence $\{1/n\}$ is a Cauchy sequence. Indeed, fix $\varepsilon > 0$. For any positive N, if $n, m \geq N$ then
$$\left|\frac{1}{n} - \frac{1}{m}\right| \leq \frac{1}{n} + \frac{1}{m} \leq \frac{2}{N}.$$
Take $N = [2/\varepsilon] + 1$. Then if $n, m \geq N$, we have
$$|s_n - s_m| \leq \frac{2}{N} < \varepsilon,$$

as required.

Lemma 3.17 Let $\{s_n\}$ be a convergent sequence of real numbers. Then $\{s_n\}$ is Cauchy.

Proof. Suppose that $s_n \to s$ as $n \to \infty$. Let $\varepsilon > 0$. Then there exists N such that if $n \geq N$, then $|s_n - s| < \varepsilon/2$. Now suppose that $n, m \geq N$, then

$$\begin{aligned} |s_n - s_m| &= |(s_n - s) - (s_m - s)| \\ &\leq |s_n - s| + |s_m - s| \quad \text{by the triangle inequality} \\ &< \varepsilon/2 + \varepsilon/2 \\ &= \varepsilon, \end{aligned}$$

whence $\{s_n\}$ is Cauchy. ■

Theorem 3.18 [Cauchy convergence criterion] Let $\{s_n\}$ be a Cauchy sequence. Then $\{s_n\}$ is convergent.

Proof. Let $\{s_n\}$ be a Cauchy sequence. Then $\{s_n\}$ is bounded (an exercise for the reader). By the Theorem 3.14 a), there is some convergent subsequence $s_{n_k} \to s$ as $k \to \infty$, for some $s \in \mathbb{R}$. We claim that $s_k \to s$ as $k \to \infty$. Indeed, let $\varepsilon > 0$. Then there exists N_1 such that if $k \geq N_1$, then $|s_{n_k} - s| < \varepsilon/2$. But there exists also N_2 such that if $n, m \geq N_2$ then $|s_n - s_m| < \varepsilon/2$. If $k \geq \max(N_1, N_2)$ then $n_k \geq N_2$ and $n_k \geq N_1$ hence

$$\begin{aligned} |s_k - s| &= |(s_k - s_{n_k}) + (s_{n_k} - s)| \\ &\leq |s_k - s_{n_k}| + |s_{n_k} - s| < \varepsilon/2 + \varepsilon/2 = \varepsilon, \end{aligned}$$

as claimed. ■

Continuity in Terms of Sequences

We now show that continuity can be defined and studied in terms of sequences.

Theorem 3.19 Let f be defined on a closed interval $[a, b]$ containing $\overline{x}$. Then f is continuous at $\overline{x}$ (from the right if $\overline{x} = a$, from the left if $\overline{x} = b$) if and only if

$$\lim_{n\to\infty} f(x_n) = f(\overline{x}) \tag{3.6}$$

whenever $\{x_n\}$ is a sequence of points in $[a, b]$ such that

$$\lim_{n\to\infty} x_n = \overline{x}. \tag{3.7}$$

Proof. Assume that $a < \overline{x} < b$; only minor changes in the proof are needed if $\overline{x} = a$ or $\overline{x} = b$. First, suppose that f is continuous at $\overline{x}$ and $\{x_n\}$ is a sequence of points in $[a, b]$ satisfying (3.7). If $\epsilon > 0$, there is a $\delta > 0$ such that

$$|f(x) - f(\overline{x})| < \epsilon \quad \text{if} \quad |x - \overline{x}| < \delta. \tag{3.8}$$

From (3.7), there is an integer N such that $|x_n - \overline{x}| < \delta$ if $n \geq N$. This and (3.8) imply that $|f(x_n) - f(\overline{x})| < \epsilon$ if $n \geq N$. This implies (3.6), which shows that the stated condition is necessary.

For sufficiency, suppose that f is discontinuous at $\overline{x}$. Then there is an $\epsilon_0 > 0$ such that, for each positive integer n, there is a point x_n that satisfies the inequality

$$|x_n - \overline{x}| < \frac{1}{n}$$

while

$$|f(x_n) - f(\overline{x})| \geq \epsilon_0.$$

The sequence $\{x_n\}$ therefore satisfies (3.7), but not (3.6). Hence, the stated condition cannot hold if f is discontinuous at $\overline{x}$. This proves sufficiency. ■

Armed with the theorems we have proved so far in this section, we could develop the theory of continuous functions by means of definitions and proofs based on sequences and subsequences. We give one example, a new proof of Theorem 2.44, and leave others for exercises.

Theorem 3.20 If f is continuous on a closed interval $[a,b]$, then f is bounded on $[a,b]$.

Proof. The proof is by contradiction. If f is not bounded on $[a,b]$, there is for each positive integer n a point x_n in $[a,b]$ such that $|f(x_n)| > n$. This implies that

$$\lim_{n\to\infty} |f(x_n)| = \infty. \tag{3.9}$$

Since $\{x_n\}$ is bounded, $\{x_n\}$ has a convergent subsequence $\{x_{n_k}\}$ (Theorem 3.14 **(a)**). If

$$\overline{x} = \lim_{k\to\infty} x_{n_k},$$

then $\overline{x}$ is a limit point of $[a,b]$, so $\overline{x} \in [a,b]$. If f is continuous on $[a,b]$, then

$$\lim_{k\to\infty} f(x_{n_k}) = f(\overline{x})$$

by Theorem 3.19, so

$$\lim_{k\to\infty} |f(x_{n_k})| = |f(\overline{x})|,$$

which contradicts (3.9). Therefore, f cannot be both continuous and unbounded on $[a,b]$ ■

3.3 Infinite Series of Constants

The theory of sequences developed in the last two sections can be combined with the familiar notion of a finite sum to produce the theory of infinite series. We begin the study of infinite series in this section.

Definition 3.21 If $\{a_n\}_k^\infty$ is an infinite sequence of real numbers, the symbol

$$\sum_{n=k}^{\infty} a_n$$

is an *infinite series*, and a_n is the *nth term* of the series. We say that $\sum_{n=k}^{\infty} a_n$ *converges to the sum* A, and write

$$\sum_{n=k}^{\infty} a_n = A,$$

if the sequence $\{A_n\}_k^\infty$ defined by

$$A_n = a_k + a_{k+1} + \cdots + a_n, \quad n \geq k,$$

converges to A. The finite sum A_n is the *nth partial sum of* $\sum_{n=k}^\infty a_n$. If $\{A_n\}_k^\infty$ diverges, we say that $\sum_{n=k}^\infty a_n$ *diverges*; in particular, if $\lim_{n\to\infty} A_n = \infty$ or $-\infty$, we say that $\sum_{n=k}^\infty a_n$ *diverges to* ∞ *or* $-\infty$, and write

$$\sum_{n=k}^\infty a_n = \infty \quad \text{or} \quad \sum_{n=k}^\infty a_n = -\infty.$$

A divergent infinite series that does not diverge to $\pm\infty$ is said to *oscillate*, or *be oscillatory*. ■

We will usually refer to infinite series more briefly as *series*.

■ **Example 3.22** Consider the series

$$\sum_{n=0}^\infty r^n, \quad -1 < r < 1.$$

Here $a_n = r^n$ $(n \geq 0)$ and

$$A_n = 1 + r + r^2 + \cdots + r^n = \frac{1 - r^{n+1}}{1 - r}, \tag{3.10}$$

which converges to $1/(1-r)$ as $n \to \infty$; thus, we write

$$\sum_{n=0}^\infty r^n = \frac{1}{1-r}, \quad -1 < r < 1.$$

If $|r| > 1$, then (3.10) is still valid, but $\sum_{n=0}^\infty r^n$ diverges; if $r > 1$, then

$$\sum_{n=0}^\infty r^n = \infty, \tag{3.11}$$

while if $r < -1$, $\sum_{n=0}^\infty r^n$ oscillates, since its partial sums alternate in sign and their magnitudes become arbitrarily large for large n. If $r = -1$, then $A_{2m+1} = 0$ and $A_{2m} = 1$ for $m \geq 0$, while if $r = 1$, $A_n = n + 1$; in both cases the series diverges, and (3.11) holds if $r = 1$. ■

The series $\sum_{n=0}^\infty r^n$ is called the *geometric series with ratio* r. It occurs in many applications.

An infinite series can be viewed as a generalization of a finite sum

$$A = \sum_{n=k}^N a_n = a_k + a_{k+1} + \cdots + a_N$$

by thinking of the finite sequence $\{a_k, a_{k+1}, \ldots, a_N\}$ as being extended to an infinite sequence $\{a_n\}_k^\infty$ with $a_n = 0$ for $n > N$. Then the partial sums of $\sum_{n=k}^\infty a_n$ are

$$A_n = a_k + a_{k+1} + \cdots + a_n, \quad k \leq n < N,$$

and

$$A_n = A, \quad n \geq N;$$

that is, the terms of $\{A_n\}_k^\infty$ equal the finite sum A for $n \geq k$. Therefore, $\lim_{n\to\infty} A_n = A$.

Cauchy's Convergence Criterion for Series

The Cauchy convergence criterion for sequences yields a useful criterion for convergence of series.

Theorem 3.23 — Cauchy's Convergence Criterion for Series. A series $\sum a_n$ converges if and only if for every $\epsilon > 0$ there is an integer N such that

$$|a_n + a_{n+1} + \cdots + a_m| < \epsilon \quad \text{if} \quad m \geq n \geq N. \tag{3.12}$$

Proof. In terms of the partial sums $\{A_n\}$ of $\sum a_n$,

$$a_n + a_{n+1} + \cdots + a_m = A_m - A_{n-1}.$$

Therefore, (3.12) can be written as

$$|A_m - A_{n-1}| < \epsilon \quad \text{if} \quad m \geq n \geq N.$$

Since $\sum a_n$ converges if and only if $\{A_n\}$ converges, Lemma 3.17 and Theorem 3.18 implies the conclusion. ■

Intuitively, Theorem 3.23 means that $\sum a_n$ converges if and only if arbitrarily long sums

$$a_n + a_{n+1} + \cdots + a_m, \quad m \geq n,$$

can be made as small as we please by picking n large enough.

■ **Example 3.24** Consider the geometric series $\sum r^n$ of Example 3.22. If $|r| \geq 1$, then $\{r^n\}$ does not converge to zero. Therefore $\sum r^n$ diverges, as we saw in Example 3.22. If $|r| < 1$ and $m \geq n$, then

$$\begin{aligned} |A_m - A_n| &= |r^{n+1} + r^{n+2} + \cdots + r^m| \\ &\leq |r|^{n+1}(1 + |r| + \cdots + |r|^{m-n-1}) \\ &= |r|^{n+1}\frac{1 - |r|^{m-n}}{1 - |r|} < \frac{|r|^{n+1}}{1 - |r|}. \end{aligned} \tag{3.13}$$

If $\epsilon > 0$, choose N so that

$$\frac{|r|^{N+1}}{1 - |r|} < \epsilon.$$

Then (3.13) implies that

$$|A_m - A_n| < \epsilon \quad \text{if} \quad m \geq n \geq N.$$

Now Theorem 3.23 implies that $\sum r^n$ converges if $|r| < 1$, as in Example 3.22. ■

Letting $m = n$ in (3.12) yields the following important corollary of Theorem 3.23.

Corollary 3.25 If $\sum a_n$ converges, then $\lim_{n\to\infty} a_n = 0$.

It must be emphasized that Corollary 3.25 gives a *necessary* condition for convergence; that is, $\sum a_n$ cannot converge unless $\lim_{n\to\infty} a_n = 0$. The condition is *not sufficient*; $\sum a_n$ may diverge even if $\lim_{n\to\infty} a_n = 0$. We will see examples below.

We leave the proof of the following corollary of Theorem 3.23 as an exercise for the reader.

Corollary 3.26 If $\sum a_n$ converges, then for each $\epsilon > 0$ there is an integer K such that

$$\left|\sum_{n=k}^{\infty} a_n\right| < \epsilon \quad \text{if} \quad k \geq K;$$

that is,

$$\lim_{k\to\infty} \sum_{n=k}^{\infty} a_n = 0.$$

■ **Example 3.27** If $|r| < 1$, then

$$\left|\sum_{n=k}^{\infty} r^n\right| = \left|r^k \sum_{n=k}^{\infty} r^{n-k}\right| = \left|r^k \sum_{n=0}^{\infty} r^n\right| = \frac{|r|^k}{1-r}.$$

Therefore, if

$$\frac{|r|^K}{1-r} < \epsilon,$$

then

$$\left|\sum_{n=k}^{\infty} r^n\right| < \epsilon \quad \text{if} \quad k \geq K,$$

which implies that $\lim_{k\to\infty} \sum_{n=k}^{\infty} r^n = 0$.

3.4 Sequences and Series of Functions

Until now we have considered sequences and series of constants. Now we turn our attention to sequences and series of real-valued functions defined on subsets of the reals. Throughout this section, "subset" means "nonempty subset."

If $F_k, F_{k+1}, \ldots, F_n, \ldots$ are real-valued functions defined on a subset D of the reals, we say that $\{F_n\}$ is an *infinite sequence* or (simply a *sequence*) *of functions on* D. If the sequence of values $\{F_n(x)\}$ converges for each x in some subset S of D, then $\{F_n\}$ defines a limit function on S. The formal definition is as follows.

Definition 3.28 Suppose that $\{F_n\}$ is a sequence of functions on D and the sequence of values $\{F_n(x)\}$ converges for each x in some subset S of D. Then we say that $\{F_n\}$ *converges pointwise on* S *to the limit function* F, defined by

$$F(x) = \lim_{n\to\infty} F_n(x), \quad x \in S.$$

■ **Example 3.29** The functions

$$F_n(x) = \left(1 - \frac{nx}{n+1}\right)^{n/2}, \quad n \geq 1,$$

define a sequence on $D = (-\infty, 1]$, and

$$\lim_{n\to\infty} F_n(x) = \begin{cases} \infty, & x < 0, \\ 1, & x = 0, \\ 0, & 0 < x \leq 1. \end{cases}$$

Therefore, $\{F_n\}$ converges pointwise on $S = [0, 1]$ to the limit function F defined by

$$F(x) = \begin{cases} 1, & x = 0, \\ 0, & 0 < x \leq 1. \end{cases}$$

■ **Example 3.30** Consider the functions

$$F_n(x) = x^n e^{-nx}, \quad x \geq 0, \quad n \geq 1,$$

see Figure 3.1.

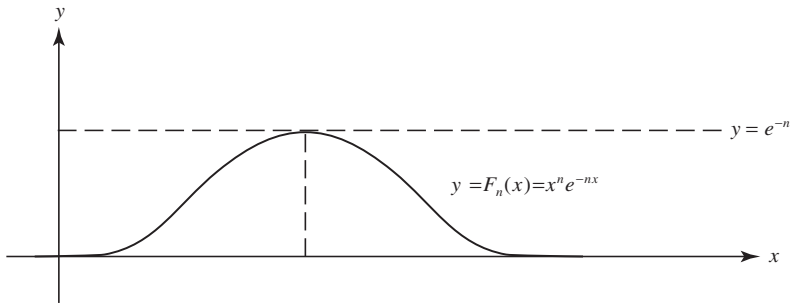


Figure 3.1

Equating the derivative

$$F_n'(x) = nx^{n-1}e^{-nx}(1-x)$$

to zero shows that the maximum value of $F_n(x)$ on $[0,\infty)$ is e^{-n}, attained at $x = 1$. Therefore,

$$|F_n(x)| \leq e^{-n}, \quad x \geq 0,$$

so $\lim_{n\to\infty} F_n(x) = 0$ for all $x \geq 0$. The limit function in this case is identically zero on $[0,\infty)$.

■ **Example 3.31** For $n \geq 1$, let F_n be defined on $(-\infty,\infty)$ by

$$F_n(x) = \begin{cases} 0, & x < -\frac{2}{n}, \\ -n(2+nx), & -\frac{2}{n} \leq x < -\frac{1}{n}, \\ n^2 x, & -\frac{1}{n} \leq x < \frac{1}{n}, \\ n(2-nx), & \frac{1}{n} \leq x < \frac{2}{n}, \\ 0, & x \geq \frac{2}{n} \end{cases}$$

see Figure 3.2, page 68).

Since $F_n(0) = 0$ for all n, $\lim_{n\to\infty} F_n(0) = 0$. If $x \neq 0$, then $F_n(x) = 0$ if $n \geq 2/|x|$. Therefore,

$$\lim_{n\to\infty} F_n(x) = 0, \quad -\infty < x < \infty,$$

so the limit function is identically zero on $(-\infty,\infty)$.

■ **Example 3.32** For each positive integer n, let S_n be the set of numbers of the form $x = p/q$, where p and q are integers with no common factors and $1 \leq q \leq n$. Define

$$F_n(x) = \begin{cases} 1, & x \in S_n, \\ 0, & x \notin S_n. \end{cases}$$

If x is irrational, then $x \notin S_n$ for any n, so $F_n(x) = 0$, $n \geq 1$. If x is rational, then $x \in S_n$ and $F_n(x) = 1$ for all sufficiently large n. Therefore,

$$\lim_{n\to\infty} F_n(x) = F(x) = \begin{cases} 1 & \text{if } x \text{ is rational,} \\ 0 & \text{if } x \text{ is irrational.} \end{cases}$$

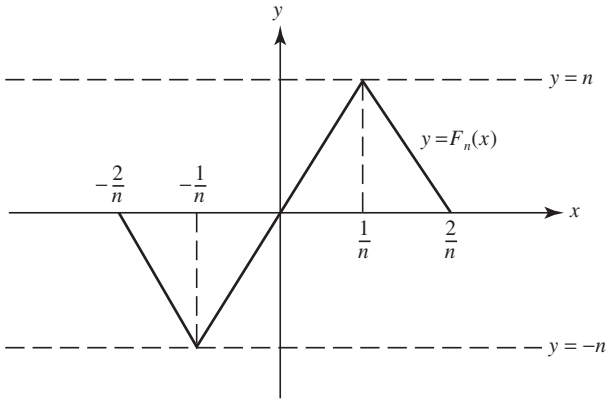


Figure 3.2

Uniform Convergence

The pointwise limit of a sequence of functions may differ radically from the functions in the sequence. In Example 3.29, each F_n is continuous on $(-\infty, 1]$, but F is not. In Example 3.31, the graph of each F_n has two triangular spikes with heights that tend to ∞ as $n \to \infty$, while the graph of F (the x-axis) has none. There is nothing in Definition 3.28 to preclude these apparent anomalies; although the definition implies that for each x_0 in S, $F_n(x_0)$ approximates $F(x_0)$ if n is sufficiently large, it does not imply that any particular F_n approximates F well over *all* of S. To formulate a definition that does, it is convenient to introduce the notation

$$\|g\|_S = \sup_{x \in S} |g(x)|$$

and to state the following lemma. We leave the proof as an exercise.

Lemma 3.33 If g and h are defined on S, then

$$\|g + h\|_S \le \|g\|_S + \|h\|_S \quad \text{and} \quad \|gh\|_S \le \|g\|_S \|h\|_S.$$

Moroever, if either g or h is bounded on S, then

$$\|g - h\|_S \ge \left| \|g\|_S - \|h\|_S \right|.$$

Definition 3.34 A sequence $\{F_n\}$ of functions defined on a set S *converges uniformly to the limit function F on S* if

$$\lim_{n \to \infty} ||F_n - F\|_S = 0.$$

Thus, $\{F_n\}$ converges uniformly to F on S if for each $\epsilon > 0$ there is an integer N such that

$$\|F_n - F\|_S < \epsilon \quad \text{if} \quad n \ge N. \tag{3.14}$$

If $S = [a, b]$ and F is the function with graph shown in Figure 3.3, then (3.14) implies that the graph of

$$y = F_n(x), \quad a \le x \le b,$$

lies in the shaded band

$$F(x) - \epsilon < y < F(x) + \epsilon, \quad a \le x \le b,$$

if $n \geq N$.

From Definition 3.34, if $\{F_n\}$ converges uniformly on S, then $\{F_n\}$ converges uniformly on any subset of S.

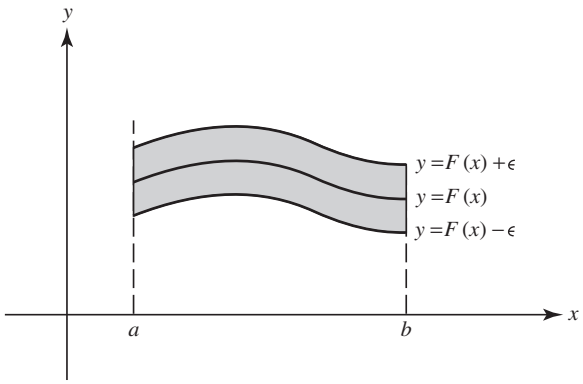


Figure 3.3

■ **Example 3.35** The sequence $\{F_n\}$ defined by

$$F_n(x) = x^n e^{-nx}, \quad n \geq 1,$$

converges uniformly to $F \equiv 0$ (that is, to the identically zero function) on $S = [0, \infty)$, since we saw in Example 3.30 that

$$\|F_n - F\|_S = \|F_n\|_S = e^{-n},$$

so

$$\|F_n - F\|_S < \epsilon$$

if $n > -\log \epsilon$. For these values of n, the graph of

$$y = F_n(x), \quad 0 \leq x < \infty,$$

lies in the strip $-\epsilon \leq y \leq \epsilon, \quad x \geq 0$, see Figure 3.4. ■

The next theorem provides alternative definitions of pointwise and uniform convergence. It follows immediately from Definitions 3.28 and 3.34.

Theorem 3.36 Let $\{F_n\}$ be defined on S. Then

(a) $\{F_n\}$ converges pointwise to F on S if and only if there is, for each $\epsilon > 0$ and $x \in S$, an integer N (which may depend on x as well as ϵ) such that

$$|F_n(x) - F(x)| < \epsilon \quad \text{if} \quad n \geq N.$$

(b) $\{F_n\}$ converges uniformly to F on S if and only if there is for each $\epsilon > 0$ an integer N (which depends only on ϵ and not on any particular x in S) such that

$$|F_n(x) - F(x)| < \epsilon \quad \text{for all } x \text{ in } S \text{ if } n \geq N.$$

The next theorem follows immediately from Theorem 3.36 and Example 3.38.

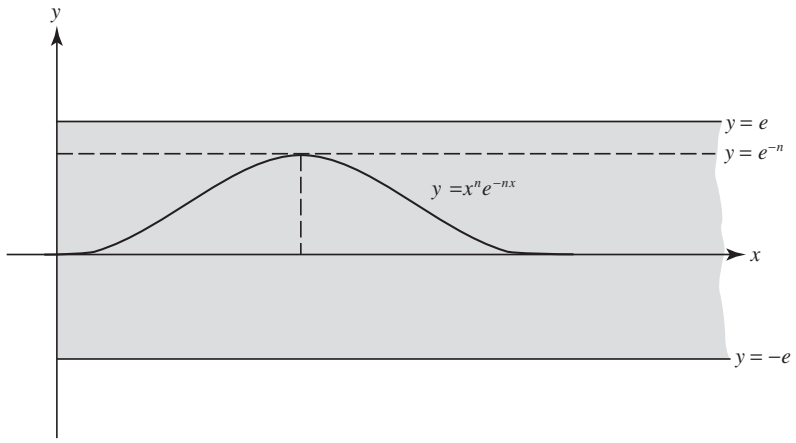


Figure 3.4

Theorem 3.37 If $\{F_n\}$ converges uniformly to F on S, then $\{F_n\}$ converges pointwise to F on S. The converse is false; that is, pointwise convergence does not imply uniform convergence.

■ **Example 3.38** The sequence $\{F_n\}$ of Example 3.31 converges pointwise to $F \equiv 0$ on $(-\infty, \infty)$, but not uniformly, since

$$\|F_n - F\|_{(-\infty,\infty)} = F_n\left(\frac{1}{n}\right) = \left|F_n\left(\frac{-1}{n}\right)\right| = n,$$

so

$$\lim_{n\to\infty} \|F_n - F\|_{(-\infty,\infty)} = \infty.$$

However, the convergence is uniform on

$$S_\rho = (-\infty, \rho] \cup [\rho, \infty)$$

for any $\rho > 0$, since

$$\|F_n - F\|_{S_\rho} = 0 \quad \text{if} \quad n > \frac{2}{\rho}.$$

■ **Example 3.39** If $F_n(x) = x^n$, $n \geq 1$, then $\{F_n\}$ converges pointwise on $S = [0, 1]$ to

$$F(x) = \begin{cases} 1, & x = 1, \\ 0, & 0 \leq x < 1. \end{cases}$$

The convergence is not uniform on S. To see this, suppose that $0 < \epsilon < 1$. Then

$$|F_n(x) - F(x)| > 1 - \epsilon \quad \text{if} \quad (1-\epsilon)^{1/n} < x < 1.$$

Therefore,

$$1 - \epsilon \leq \|F_n - F\|_S \leq 1$$

for all $n \geq 1$. Since ϵ can be arbitrarily small, it follows that

$$\|F_n - F\|_S = 1$$

for all $n \geq 1$.

However, the convergence is uniform on $[0, \rho]$ if $0 < \rho < 1$, since then

$$\|F_n - F\|_{[0,\rho]} = \rho^n$$

and $\lim_{n\to\infty} \rho^n = 0$. Another way to say the same thing: $\{F_n\}$ converges uniformly on every closed subset of $[0, 1)$. ■

The next theorem enables us to test a sequence for uniform convergence without guessing what the limit function might be. It is analogous to Cauchy's convergence criterion for sequences of constants.

Theorem 3.40 — Cauchy's Uniform Convergence Criterion. A sequence of functions $\{F_n\}$ converges uniformly on a set S if and only if for each $\epsilon > 0$ there is an integer N such that

$$\|F_n - F_m\|_S < \epsilon \quad \text{if} \quad n, m \geq N. \tag{3.15}$$

Proof. For necessity, suppose that $\{F_n\}$ converges uniformly to F on S. Then, if $\epsilon > 0$, there is an integer N such that

$$\|F_k - F\|_S < \frac{\epsilon}{2} \quad \text{if} \quad k \geq N.$$

Therefore,

$$\begin{aligned} \|F_n - F_m\|_S &= \|(F_n - F) + (F - F_m)\|_S \\ &\leq \|F_n - F\|_S + \|F - F_m\|_S \quad \text{(Lemma 3.33)} \\ &< \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon \quad \text{if} \quad m, n \geq N. \end{aligned}$$

For sufficiency, we first observe that (3.15) implies that

$$|F_n(x) - F_m(x)| < \epsilon \quad \text{if} \quad n, m \geq N,$$

for any fixed x in S. Therefore, Cauchy's convergence criterion for sequences of constants (Theorem 3.18) implies that $\{F_n(x)\}$ converges for each x in S; that is, $\{F_n\}$ converges pointwise to a limit function F on S. To see that the convergence is uniform, we write

$$\begin{aligned} |F_m(x) - F(x)| &= |[F_m(x) - F_n(x)] + [F_n(x) - F(x)]| \\ &\leq |F_m(x) - F_n(x)| + |F_n(x) - F(x)| \\ &\leq \|F_m - F_n\|_S + |F_n(x) - F(x)|. \end{aligned}$$

This and (3.15) imply that

$$|F_m(x) - F(x)| < \epsilon + |F_n(x) - F(x)| \quad \text{if} \quad n, m \geq N. \tag{3.16}$$

Since $\lim_{n\to\infty} F_n(x) = F(x)$,

$$|F_n(x) - F(x)| < \epsilon$$

for some $n \geq N$, so (3.16) implies that

$$|F_m(x) - F(x)| < 2\epsilon \quad \text{if} \quad m \geq N.$$

But this inequality holds for all x in S, so

$$\|F_m - F\|_S \leq 2\epsilon \quad \text{if} \quad m \geq N.$$

Since ϵ is an arbitrary positive number, this implies that $\{F_n\}$ converges uniformly to F on S. ∎

■ **Example 3.41** Suppose that g is differentiable on $S = (-\infty, \infty)$ and

$$|g'(x)| \leq r < 1, \quad -\infty < x < \infty. \tag{3.17}$$

Let F_0 be bounded on S and define

$$F_n(x) = g(F_{n-1}(x)), \quad n \geq 1. \tag{3.18}$$

We will show that $\{F_n\}$ converges uniformly on S. We first note that if u and v are any two real numbers, then (3.17) and the mean value theorem imply that

$$|g(u) - g(v)| \leq r|u - v|. \tag{3.19}$$

Recalling (3.18) and applying this inequality with $u = F_{n-1}(x)$ and $v = 0$ shows that

$$\begin{aligned} |F_n(x)| &= |g(0) + (g(F_{n-1}(x)) - g(0))| \leq |g(0)| + |g(F_{n-1}(x)) - g(0)| \\ &\leq |g(0)| + r|F_{n-1}(x)|; \end{aligned}$$

therefore, since F_0 is bounded on S, it follows by induction that F_n is bounded on S for $n \geq 1$. Moreover, if $n \geq 1$, then (3.18) and (3.19) with $u = F_n(x)$ and $v = F_{n-1}(x)$ imply that

$$|F_{n+1}(x) - F_n(x)| = |g(F_n(x)) - g(F_{n-1}(x))| \leq r|F_n(x) - F_{n-1}(x)|,$$

with $-\infty < x < \infty$, so

$$\|F_{n+1} - F_n\|_S \leq r\|F_n - F_{n-1}\|_S.$$

By induction, this implies that

$$\|F_{n+1} - F_n\|_S \leq r^n\|F_1 - F_0\|_S. \tag{3.20}$$

If $n > m$, then

$$\begin{aligned} \|F_n - F_m\|_S &= \|(F_n - F_{n-1}) + (F_{n-1} - F_{n-2}) + \cdots + (F_{m+1} - F_m)\|_S \\ &\leq \|F_n - F_{n-1}\|_S + \|F_{n-1} - F_{n-2}\|_S + \cdots + \|F_{m+1} - F_m\|_S, \end{aligned}$$

from Lemma 3.33. Now (3.20) implies that

$$\begin{aligned} \|F_n - F_m\|_S &\leq \|F_1 - F_0\|_S(1 + r + r^2 + \cdots + r^{n-m-1})r^m \\ &< \|F_1 - F_0\|_S \frac{r^m}{1-r}. \end{aligned}$$

Therefore, if

$$\|F_1 - F_0\|_S \frac{r^N}{1-r} < \epsilon,$$

then $\|F_n - F_m\|_S < \epsilon$ if $n, m \geq N$. Therefore, $\{F_n\}$ converges uniformly on S, by Theorem 3.40.

Properties Preserved by Uniform Convergence

We now study properties of the functions of a uniformly convergent sequence that are inherited by the limit function. We first consider continuity.

Theorem 3.42 If $\{F_n\}$ converges uniformly to F on S and each F_n is continuous at a point x_0 in S, then so is F. Similar statements hold for continuity from the right and left.

Proof. Suppose that each F_n is continuous at x_0. If $x \in S$ and $n \geq 1$, then

$$\begin{aligned} |F(x) - F(x_0)| &\leq |F(x) - F_n(x)| + |F_n(x) - F_n(x_0)| + |F_n(x_0) - F(x_0)| \\ &\leq |F_n(x) - F_n(x_0)| + 2\|F_n - F\|_S. \end{aligned} \tag{3.21}$$

Suppose that $\epsilon > 0$. Since $\{F_n\}$ converges uniformly to F on S, we can choose n so that $\|F_n - F\|_S < \epsilon$. For this fixed n, (3.21) implies that

$$|F(x) - F(x_0)| < |F_n(x) - F_n(x_0)| + 2\epsilon, \quad x \in S. \tag{3.22}$$

Since F_n is continuous at x_0, there is a $\delta > 0$ such that

$$|F_n(x) - F_n(x_0)| < \epsilon \quad \text{if} \quad |x - x_0| < \delta,$$

so, from (3.22),

$$|F(x) - F(x_0)| < 3\epsilon, \quad \text{if} \quad |x - x_0| < \delta.$$

Therefore, F is continuous at x_0. Similar arguments apply to the assertions on continuity from the right and left. ■

Corollary 3.43 If $\{F_n\}$ converges uniformly to F on S and each F_n is continuous on S, then so is F; that is, a uniform limit of continuous functions is continuous.

Infinite Series of Functions

In Section 3.3 we defined the sum of an infinite series of constants as the limit of the sequence of partial sums. The same definition can be applied to series of functions, as follows.

Definition 3.44 If $\{f_j\}_k^\infty$ is a sequence of real-valued functions defined on a set D of reals, then $\sum_{j=k}^\infty f_j$ is an *infinite series* (or simply a *series*) of functions on D. The *partial sums of*, $\sum_{j=k}^\infty f_j$ are defined by

$$F_n = \sum_{j=k}^{n} f_j, \quad n \geq k.$$

If $\{F_n\}_k^\infty$ converges pointwise to a function F on a subset S of D, we say that $\sum_{j=k}^\infty f_j$ *converges pointwise to the sum F on S*, and write

$$F = \sum_{j=k}^{\infty} f_j, \quad x \in S.$$

If $\{F_n\}$ converges uniformly to F on S, we say that $\sum_{j=k}^\infty f_j$ *converges uniformly to F on S*.

■ **Example 3.45** The functions

$$f_j(x) = x^j, \quad j \geq 0,$$

define the infinite series

$$\sum_{j=0}^{\infty} x^j$$

on $D = (-\infty, \infty)$. The nth partial sum of the series is

$$F_n(x) = 1 + x + x^2 + \cdots + x^n,$$

or, in closed form,

$$F_n(x) = \begin{cases} \dfrac{1 - x^{n+1}}{1 - x}, & x \neq 1, \\ n + 1, & x = 1 \end{cases}$$

We have seen earlier that $\{F_n\}$ converges pointwise to

$$F(x) = \frac{1}{1 - x}$$

if $|x| < 1$ and diverges if $|x| \geq 1$; hence, we write

$$\sum_{j=0}^{\infty} x^j = \frac{1}{1 - x}, \quad -1 < x < 1.$$

Since the difference

$$F(x) - F_n(x) = \frac{x^{n+1}}{1 - x}$$

can be made arbitrarily large by taking x close to 1,

$$\|F - F_n\|_{(-1,1)} = \infty,$$

so the convergence is not uniform on $(-1, 1)$. Neither is it uniform on any interval $(-1, r]$ with $-1 < r < 1$, since

$$\|F - F_n\|_{(-1,r)} \geq \frac{1}{2}$$

for every n on every such interval. (Why?) The series does converge uniformly on any interval $[-r, r]$ with $0 < r < 1$, since

$$\|F - F_n\|_{[-r,r]} = \frac{r^{n+1}}{1 - r}$$

and $\lim_{n\to\infty} r^n = 0$. Put another way, the series converges uniformly on closed subsets of $(-1, 1)$. ■

As for series of constants, the convergence, pointwise or uniform, of a series of functions is not changed by altering or omitting finitely many terms. This justifies adopting the convention that we used for series of constants: when we are interested only in whether a series of functions converges, and not in its sum, we will omit the limits on the summation sign and write simply $\sum f_n$.

Tests for Uniform Convergence of Series

Theorem 3.40 is easily converted to a theorem on uniform convergence of series, as follows.

Theorem 3.46 — Cauchy's Uniform Convergence Criterion. A series $\sum f_n$ converges uniformly on a set S if and only if for each $\epsilon > 0$ there is an integer N such that

$$\|f_n + f_{n+1} + \cdots + f_m\|_S < \epsilon \quad \text{if} \quad m \geq n \geq N. \tag{3.23}$$

Proof. Apply Theorem 3.40 to the partial sums of $\sum f_n$, observing that

$$f_n + f_{n+1} + \cdots + f_m = F_m - F_{n-1}.$$

Setting $m = n$ in (3.23) yields the following necessary, but not sufficient, condition for uniform convergence of series. It is analogous to Corollary 3.25.

Corollary 3.47 If $\sum f_n$ converges uniformly on S, then $\lim_{n\to\infty} \|f_n\|_S = 0$.

Theorem 3.46 leads immediately to the following important test for uniform convergence of series.

Theorem 3.48 — Weierstrass's Test. The series $\sum f_n$ converges uniformly on S if

$$\|f_n\|_S \leq M_n, \quad n \geq k, \tag{3.24}$$

where $\sum M_n < \infty$.

Proof. From Cauchy's convergence criterion for series of constants, there is for each $\epsilon > 0$ an integer N such that

$$M_n + M_{n+1} + \cdots + M_m < \epsilon \quad \text{if} \quad m \geq n \geq N,$$

which, because of (3.24), implies that

$$\|f_n\|_S + \|f_{n+1}\|_S + \cdots + \|f_m\|_S < \epsilon \quad \text{if} \quad m, n \geq N.$$

Lemma 3.33 and Theorem 3.46 imply that $\sum f_n$ converges uniformly on S. ■

■ **Example 3.49** Taking $M_n = 1/n^2$ and recalling that

$$\sum \frac{1}{n^2} < \infty,$$

we see that

$$\sum \frac{1}{x^2 + n^2} \quad \text{and} \quad \sum \frac{\sin nx}{n^2}$$

converge uniformly on $(-\infty, \infty)$.

■ **Example 3.50** The series

$$\sum f_n(x) = \sum \left(\frac{x}{1+x}\right)^n$$

converges uniformly on any set S such that

$$\left|\frac{x}{1+x}\right| \leq r < 1, \quad x \in S, \tag{3.25}$$

because if S is such a set, then

$$\|f_n\|_S \leq r^n$$

and Weierstrass's test applies, with

$$\sum M_n = \sum r^n < \infty.$$

Since (3.25) is equivalent to

$$\frac{-r}{1+r} \leq x \leq \frac{r}{1-r}, \quad x \in S,$$

this means that the series converges uniformly on any compact subset of $(-1/2, \infty)$. (Why?) From Corollary 3.47, the series does not converge uniformly on $S = (-1/2, b)$ with $b < \infty$ or on $S = [a, \infty)$ with $a > -1/2$, because in these cases $\|f_n\|_S = 1$ for all n. ■

Weierstrass's test is very important, but applicable only to series that actually exhibit a stronger kind of convergence than we have considered so far. We say that $\sum f_n$ *converges absolutely on* S if $\sum |f_n|$ converges pointwise on S, and *absolutely uniformly* on S if $\sum |f_n|$ converges uniformly on S. We leave it as a exercise to the reader to verify that our proof of Weierstrass's test actually shows that $\sum f_n$ converges absolutely uniformly on S. We also leave it to you to show that if a series converges absolutely uniformly on S, then it converges uniformly on S.

4 A Little More on Continuity and Differentiability

4.1 Function Classes

First, we will introduce some notation. We will use C^0 to denote the class of all continuous functions. In this section, we introduce the class of functions on $[a,b]$ that have derivatives.

Definition 4.1 Function $f : [a,b] \to \mathbb{R}$ is *continuously differentiable* if it is differentiable on $[a,b]$ and it's derivative f' is continuous on $[a,b]$. We say "f is C^1 on $[a,b]$", we can also say "f is of class C^1".

Remark 4.2

The set of *all continuously differentiable functions on* $[a,b]$ is denoted by $C^1([a,b];\mathbb{R})$ and written as $C^1([a,b])$ for short. Note that if f is of class C^1 then f' is of class C^0.

Similarly, we can define the derivative of the derivative of a function (the *second order derivative*). If the derivative f' of a function f is differentiable at x_0, the second order derivative of f at x_0 is given by

$$f''(x_0) = (f')'(x_0).$$

Definition 4.3 The function $f : [a,b] \to \mathbb{R}$ is *twice continuously differentiable* if f' is continuously differentiable (i.e. f' is of class C^1). We say f is of class C^2.

We can also define what it means for a function to be third continuously differentiable, and fourth continuously differentiable, and so on. The kth derivative of f at x_0 is

$$f^{(k)}(x_0) = \left(f^{(k-1)}\right)'(x_0).$$

To that end, we have the following general definition.

Definition 4.4 The function $f : [a,b] \to \mathbb{R}$ is k times continuously differentiable if $f^{(k-1)}$ is continuously differentiable. We say f is C^k on $[a,b]$.

Remark 4.5

f being k times continuously differentiable means that all the lower order derivatives, as well as f itself, are continuous.

Definition 4.6 The function $f : [a,b] \to \mathbb{R}$ is *infinitely many times continuously differentiable* if it is k times continuously differentiable for all $k \in \mathbb{N}$. We say f is C^∞ on $[a,b]$. We also say that f is *smooth*.

Remark 4.7

We have the following relation (why?):

$$C^\infty([a,b]) \subset \cdots \subset C^k([a,b]) \subset \cdots \subset C^1([a,b]) \subset C^0([a,b]).$$

The following exercise demonstrates (at least for the first few k) that these classes form proper subsets.

Exercise 4.8 Find examples of functions $f : [a,b] \to \mathbb{R}$ such that:

(a) f is C^0, but not C^1,

(b) f is C^1, but not C^2,

(c) f is C^2, but not C^3,

(d) f is C^∞.

We could have similarly given the above definitions on the open interval (a,b), and denoted the class by $C^n(a,b)$.

4.2 Lipschitz Continuity

Lipschitz [1] continuity provides a powerful form of continuity. We will start by giving the definition.

Definition 4.9 A function $f : [a,b] \to \mathbb{R}$ is *Lipschitz continuous* if there exists $L > 0$, such that, for all $x, y \in [a,b]$,

$$|f(x) - f(y)| \leq L|x - y|.$$

The constant L is called the *Lipschitz constant* for f.

From Definition 4.9, $|f(x) - f(y)| \leq L|x - y|$ can be rewritten as

$$\frac{|f(x) - f(y)|}{|x - y|} \leq L.$$

However, we know that the standard definition of the derivative of $f(x)$ is

$$\begin{aligned} f'(x) &= \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{y \to x} \frac{f(y) - f(x)}{y - x} \qquad \text{(by using a change of variable } y = x + h.) \end{aligned}$$

Hence, if $f'(x)$ exists,

$$|f'(x)| = \lim_{y \to x} \left| \frac{f(y) - f(x)}{y - x} \right|.$$

This is true because of the result of the Lemma: if $|z_n| \leq L$ and $|z_n|$ converges, then the limit $\lim_{n\to\infty} |z_n| \leq L$.

[1] Lipschitz continuity was named after Rudolf Lipschitz. He was a German mathematician, born in 1832 and spent most of his life working at the University of Bonn. Lipschitz's most notable discovery is the Lipschitz continuity condition, but he also carried out significant research in many other areas of pure and applied maths, particularly in number theory and potential theory. He was also the doctoral examiner of famous mathematician Felix Christian Klein.

How do we calculate L?

Definition 4.10 (Reminder)
Let $f : S \to \mathbb{R}$ be a bounded function on S. We define

$$\|f\|_S = \sup_{x \in S} |f(x)|.$$

We will refer to this as the *sup-norm* (it's also known as the *uniform norm*).

■ **Example 4.11** Prove the following Lemma which states the properties of a sup-norm. These properties hold for other norms too, and the proofs are similar.

Lemma 4.12 For all $f, g \in C^0([a,b])$, we have

(a) $\|f\|_S \geq 0$ with $\|f\|_S = 0$ if and only if $f \equiv 0$.

(b) If $a \in \mathbb{R}$, then $\|af\|_S = |a|\|f\|_S$.

(c) The norm satisfies the triangle inequality

$$\|f + g\|_S \leq \|f\|_S + \|g\|_S.$$

Proof. We will prove the above properties one by one

(a) By the definition of the sup-norm, $\|f\|_S$ is the supremum of a set of non-negative numbers, so clearly, this norm must be non-negative. For the case when $\|f\|_S = 0$, we have

$$0 \leq |f(x)| \leq \sup_{x \in [a,b]} |f(x)| = 0$$

so that $f(x) = 0$ for all $x \in [a,b]$.

(b) If $a \in \mathbb{R}$, then

$$\begin{aligned} \|af\|_S &= \sup_{x \in [a,b]} |af(x)| && (4.1)\\ &= |a| \sup_{x \in [a,b]} |f(x)| && (4.2)\\ &= |a|\|f\|_S && (4.3) \end{aligned}$$

(c) We will estimate the norm of the sum of functions f and g as follows

$$\begin{aligned} \|f + g\|_S &= \sup_{x \in [a,b]} |f(x) + g(x)| \\ &\leq \sup_{x \in [a,b]} (|f(x)| + |g(x)|) \\ &\leq \sup_{x \in [a,b]} |f(x)| + \sup_{x \in [a,b]} |g(x)| \end{aligned}$$

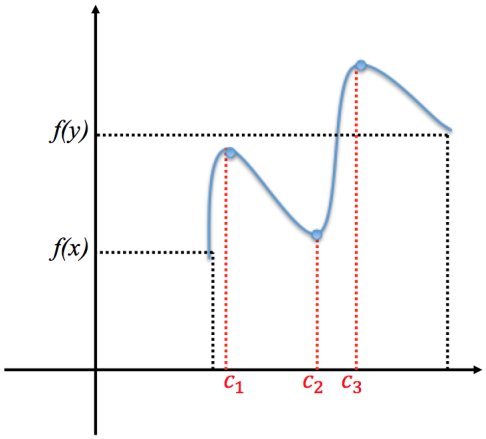
where in the first inequality we have used the triangle inequality for $|\cdot|$ and where in the second inequality we have used the fact that $\sup(a + b) \leq \sup(a) + \sup(b)$. ■

Proposition 4.13 Let $f : [a,b] \to \mathbb{R}$ be continuously differentiable. Then the smallest Lipschitz constant of f is

$$\|f'(x)\|_S = \sup_{x\in[a,b]} |f'(x)|.$$

Proof. Firstly, note that, if L_1 is a Lipschitz constant, then every number $k \geq L_1$ is a Lipschitz constant since

$$|f(x) - f(y)| \leq L_1|x-y| \leq k|x-y|.$$



Secondly, from the Mean Value Theorem, we know that

$$f'(c) = \frac{f(y)-f(x)}{y-x}, \qquad \text{where} \quad c \in [x,y].$$

Hence,

$$|f(y) - f(x)| = |f'(c)| \cdot |y-x| \leq \|f'\|_S |y-x|.$$

Comparing this inequality with the definition of Lipschitz continuity, we can conclude that $\|f'\|_S$ is certainly a Lipschitz constant.

Finally, we know that the supremum is the lowest upper bound, so $\|f'\|_S$ is the smallest Lipschitz constant. ■

Remark 4.14

If $f \in C^1([a,b])$ then f is Lipschitz continuous.

▪ **Example 4.15** Show that the function $f : [-1,1] \to \mathbb{R}$ is Lipschitz on $[-1,1]$ and give the smallest Lipschitz constant

$$f(x) = \begin{cases} x^2 & \text{if } 0 \leq x \leq 1 \\ -3x & \text{if } -1 \leq x \leq 0 \end{cases}$$

We will approach this problem by splitting the interval of x into two parts, $[-1,0)$ and $[0,1]$. If $x,y \in [-1,0)$, then

$$|f(x) - f(y)| = 3|x-y|.$$

For $x > 0$, $f'(x) = 2x$ and thus if $x, y \in [0,1]$ then

$$|f(x) - f(y)| \leq f'(1)|x - y| = 2|x - y|.$$

Now if $x > 0$, $y < 0$ then $x - y = |x| + |y|$ and

$$\begin{aligned}
&|f(x) - f(0)| \leq 2|x| \\
&|f(0) - f(y)| \leq 3|y| \\
&|f(x) - f(y)| \leq 2|x| + 3|y| < 3(|x| + |y|) = 3|x - y|.
\end{aligned}$$

So, 3 is the smallest Lipschitz constant.

Exercise 4.16 Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = \ln(x^2 + k^2)$ with $k > 0$. Show that the smallest Lipschitz constant is $1/k$.

5 Integrability

As you very well know, the geometrical interpretation of the integral of a function $f(x)$ is the area under the graph of $f(x)$. If the area is for a specified range of x values, say $x \in [a, b]$, then the corresponding integral over that range is called the *definite integral*, and when the integral is over a varying end-point, x, it is an *indefinite integral*, the indefinite integral is sometimes called the antiderivative or primitive.

Integrability (meaning the integral exists) is a powerful quality. In contrast to differentiability, the indefinite integral of every continuous function exists and is differentiable, whilst we know there are cases when the derivative of a continuous function does not exist.

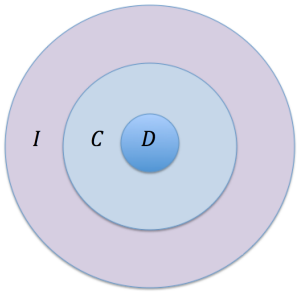


Figure 5.1 *Relationship between Differentiability, Continuity, and Integrability. All differentiable functions are continuous, but not all continuous functions are differentiable. While all continuous functions are integrable, not all integrable functions are continuous. One way to remember the order of integrability, continuity, and differentiability is using Ice-Cream-Doughnuts as suggested by Damian McCrea - class of 2020*

For many functions, as well as the composition of functions, you have learnt techniques to obtain the antiderivative and you are also aware of the list of antiderivatives for various functions. However, sometimes we are interested in knowing the antiderivative of a general function f, particularly for the cases where we are unable to find the antiderivative of a function. The approach used that gives us the definition of the integral of a general function is known as the *Riemann integral*.

In this chapter, we will introduce what it means for a function to be Riemann integrable, and when it is, we define its Riemann integral.

5.1 Riemann Integration

There are many techniques used to find definite integrals, but for cases when a function does not have an easily computable anti-derivative, we use finite sums to approximate the area under the curve. Using sums is the discrete analog of the definite integral.[2]

Riemann Sums and the Criterion for Integrability

Now, we will define *partitions* then *Riemann sums*, then the *Riemann integral*.

Definition 5.1 — Partition. Let $[a, b]$ be a closed set. A *partition* P of $[a, b]$ is a finite ordered sequence of real numbers

$$P = (x_0, x_1, x_2, \ldots, x_n)$$

with

$$a = x_0 < x_1 < x_2 < \cdots < x_n = b.$$

The points $x_0, \ldots, x_n$ are called the partition points of P. So, P divides the interval $[a, b]$ into n subintervals, where n is a positive integer.

[2] Riemann integrals were named after German mathematician Bernhard Riemann, who lived between 1826 and 1866. Despite his intense introvert personality, his advanced mathematical abilities were very strong and easily recognised. He was always a devout Christian and went to the University of Gottingen in Germany to study theology. Whilst there, he continued studying mathematics too and was lectured by many notable mathematicians such as Gauss, Jacobi, Dirichlet, Steiner, and Eisenstein. Riemann's work on Real Analysis and Analytic Number Theory was influential, and he was the founder of Riemannian geometry. It was Riemann's findings that formed the structure on which Einstein's work on the General Theory of Relativity is based.

Sometimes, the partition P is written as in Definition 5.1, i.e. with the discrete points x_i ordered within the sequence, and sometimes P is represented as the set of subintervals I_k where $I_k = [x_k, x_{k-1}]$, i.e. $P = \{[x_0, x_1], [x_1, x_2], \ldots, [x_{n-1}, x_n]\}$. In what follows, we will use the notation $P = (x_0, x_1, \ldots, x_n)$ to denote the partition P and points that lie within.

When the points in $[a, b]$ are equally spaced, the partition is called a *regular partition* (or *uniform partition*), and when they are not equidistant, it's an *irregular partition*.

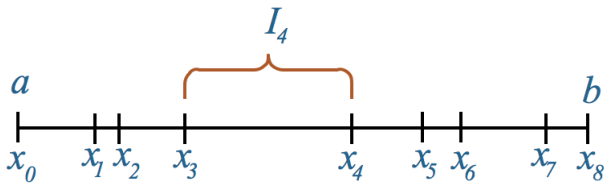


Figure 5.2 *An irregular partition with 8 subintervals*

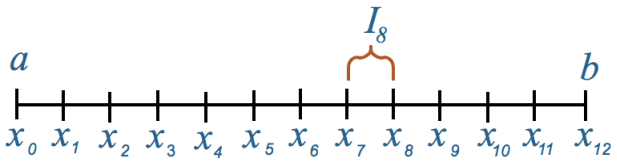


Figure 5.3 *A regular partition with 12 subintervals*

The discrete x values start from x_0, so we will always have $n+1$ values for x, but only n subintervals/rectangles. We will denote the width of each interval by

$$\Delta x_i = x_i - x_{i-1}$$

where $i = 1, \ldots, n$ and Δ (Greek uppercase delta) is most commonly used to represent the difference in a certain quantity. Here we use it to represent the difference in subsequent x values. We say "delta x_i". We will be interested in taking finer and finer partitions in order to better approximate the area. We define

$$\|P\| = \max_i \Delta x_i,$$

the width of the largest interval in the partition. Then we will be interested in sequences of partitions P_n such that

$$\lim_{n\to\infty} ||P_n|| = 0.$$

There are many ways we can place the rectangles leading to the different "types" of Riemann sums.

The *left Riemann sum* places the rectangles such that the top (or bottom of the curve is below the horizontal axis) left endpoint of each rectangle is touching the curve, while the *right Riemann sum* places the rectangles such that the top (or bottom) right endpoint of each rectangle is touching the curve. The middle Riemann sum is placing the midpoint of each rectangle on the curve.

For each of these finite sums it matters how we place the rectangles: the point on the rectangle that touches the curve will be the point where we evaluate the function for computing the Riemann sum, and this will generally

give a different total area. In the more general formulation of Riemann sums, we consider a *tagged partition*. This is a partition P together with another finite ordered sequence $T = (t_1, t_2, \ldots, t_n)$ such that $t_k \in I_k$ for each k, and we then evaluate the function at the point t_k. We can denote such a sum as

$$S(f; P; T) = \sum_{i=1}^{n} f(t_i)\Delta x_i.$$

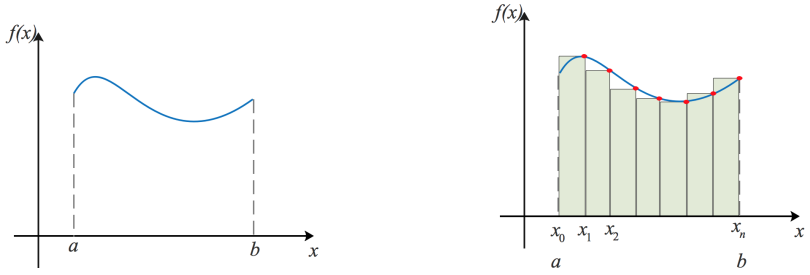


Figure 5.4 *Rectangles for the right Riemann sum*

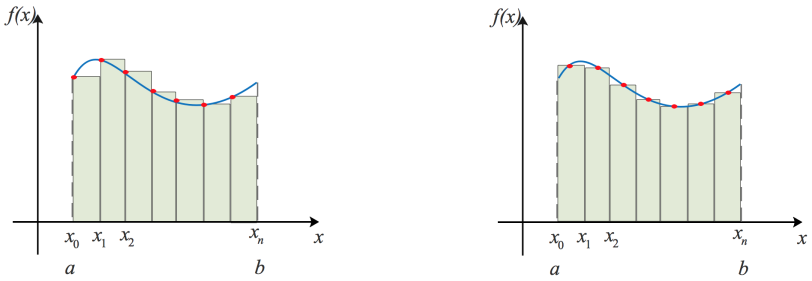


Figure 5.5 *Rectangles for the left and middle Riemann sums*

A more elegant type of Riemann sum is using lower Riemann sums and upper Riemann sums. This particular method should be attributed to Darboux [3], but is now more casually referred to as a Riemann sum (we will see why soon). The *lower Riemann sum* is the sum of the areas of rectangles which are always just below the graph of the function $f(x)$. The *upper Riemann sum* is the sum of the areas of rectangles which are always just above the graph of the function $f(x)$. The point where the rectangle hits the curve could be on a left corner, right corner, or somewhere in between. So, for a function which is monotonically decreasing, the lower sum and the upper sum are essentially the right and left sums respectively. Similarly, if the function is increasing, the lower sum and the upper sum are the left and right sum respectively.

[3] Jean-Gaston Darboux (1842-1917) was a French mathematician. He is famous for his exceptional geometrical proofs and substantial work on mathematical analysis. The Darboux Integral as well as many other findings bear his name.

Now we are ready to introduce the formal definition of upper and lower Riemann sums. We will use the following notation, needed for our definition and thereafter:

$$M_i = \sup_{x \in [x_{i-1}, x_i]} f(x)$$

$$m_i = \inf_{x \in [x_{i-1}, x_i]} f(x)$$

where $i = 1, \ldots, n$. Essentially, M_i and m_i are the values of the function evaluated at the point where the rectangle touches the curve in Figs. 5.7 and 5.6 respectively.

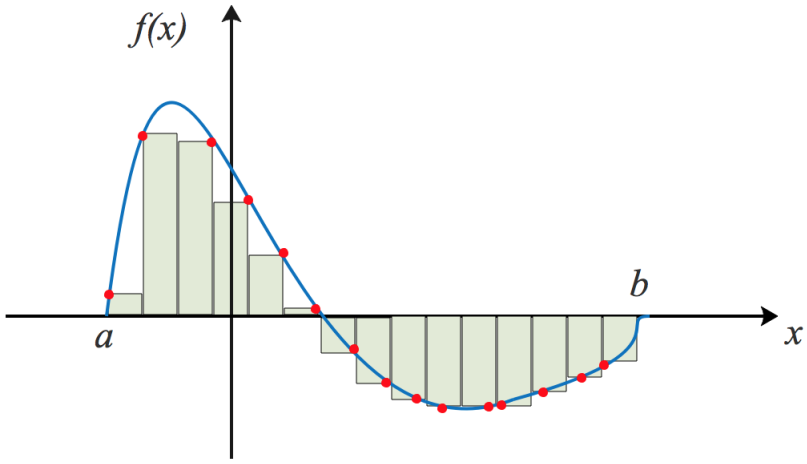


Figure 5.6*Rectangles for the lower Riemann sum*

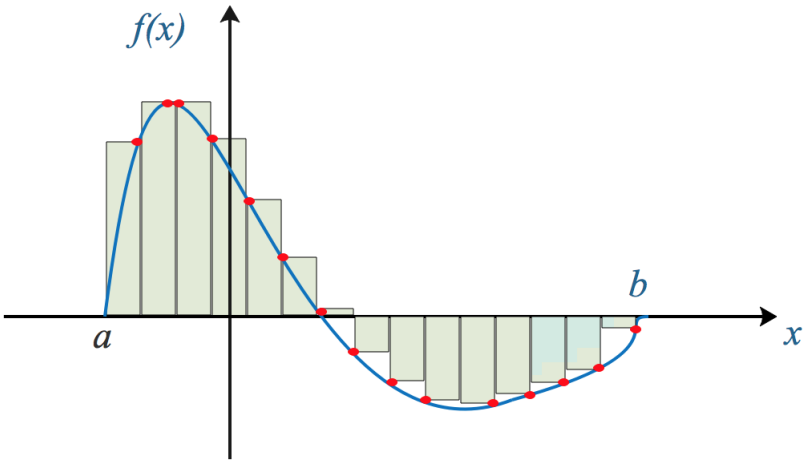


Figure 5.7*Rectangles for the upper Riemann sum*

Definition 5.2 Let $f : [a,b] \to \mathbb{R}$ be a bounded function on the interval $[a,b]$, and $P = (x_0, x_1, \ldots, x_n)$ be a partition of $[a,b]$.

(a) The lower Riemann sum of f with respect to partition P is

$$L(f;P) = \sum_{i=1}^{n} m_i \Delta x_i.$$

(b) The upper Riemann sum of f with respect to partition P is

$$U(f;P) = \sum_{i=1}^{n} M_i \Delta x_i.$$

So, geometrically, these Riemann sums represent the sum of the areas of the rectangles in Figs. 5.6 and 5.7 respectively. Since $m_i \leq M_i$ for all $i \in \{1,2,\ldots,n\}$, we have

$$L(f;P) \leq U(f;P).$$

More generally, letting $m = \inf_{x\in[a,b]} f(x)$ and $M = \sup_{x\in[a,b]} f(x)$, the set of all upper Riemann sums is bounded from below by $m(b-a)$ and all lower Riemann sums is bounded from above by $M(b-a)$, that is

$$m(b-a) \leq L(f;P) \leq U(f;P) \leq M(b-a)$$

for any partition P. Clearly, the more subintervals we have, the closer our Riemann sum will be to the exact area under the curve. Therefore, if we want the Riemann sum to be more accurate, we take a larger number of subintervals. Taking more subintervals in a partition is called making a *refinement*.

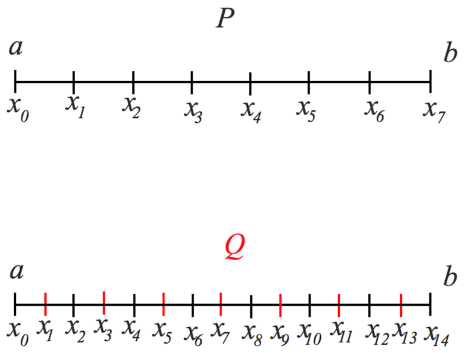


Figure 5.8 *An example of a refinement: Partition Q is a refinement of partition P. The refinement was made by splitting each subinterval in half and adding another point in the middle.*

The following theorem is the most fundamental result of Riemann sums. It tells us that by taking refinements of a partition, our Riemann sum approaches the actual area under the graph. Refining our partitions decreases the upper Riemann sums and increases the lower Riemann sums.

Theorem 5.3 — Convergence of Riemann sums. Let $f : [a,b] \to \mathbb{R}$, and let P be a partition of the interval $[a,b]$ and let Q be a refinement of P. Then

$$L(f;P) \leq L(f;Q) \leq U(f;Q) \leq U(f;P).$$

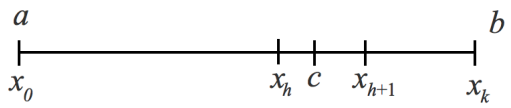


Figure 5.9 *An extra single point to refine a partition.*

Proof. As you read along, try to imagine the geometrical meaning of it as it makes the proof much simpler.

Let's introduce the partition $P = (x_0, x_1, \ldots, x_k)$. Let x_h lie somewhere between x_0 and x_k. To make a refinement, we will split the subinterval $[x_h, x_{h+1}]$ into two subintervals, so our refined partition Q contains one more point than partition P and Q is given by

$$Q = (x_0, x_1, \ldots, x_h, c, x_{h+1}, \ldots, x_k).$$

Using properties of the infimum, clearly the infimum of a function over a smaller subinterval is larger than (or equal to) the infimum over the larger subinterval. We have

$$\inf_{x\in[x_h,c]} f(x) \geq \inf_{x\in[x_h,x_{h+1}]} f(x)$$

and

$$\inf_{x\in[c,x_{h+1}]} f(x) \geq \inf_{x\in[x_h,x_{h+1}]} f(x).$$

Now, we will take the difference of the lower sums associated to P and Q. Since most subintervals of Q are also the same subintervals in P, most of these differences will cancel out. The only differences that will remain are those in the subinterval $[x_h, x_{h+1}]$. We have

$$\begin{aligned} L(f;Q) - L(f;P) &= (c - x_h) \inf_{x\in[x_h,c]} f(x) + (x_{h+1} - c) \inf_{x\in[c,x_{h+1}]} f(x) \\ &\quad - (x_{h+1} - x_h) \inf_{x\in[x_h,x_{h+1}]} f(x) \\ &\geq [(c - x_h) + (x_{h+1} - c) - (x_{h+1} - x_h)] \inf_{x\in[x_h,x_{h+1}]} f(x) \\ &\geq 0 \end{aligned}$$

Consequently, this gives $L(f;P) \leq L(f;Q)$.

Exercise 5.4 The approach for the upper Riemann sum $U(f;Q) \leq U(f;P)$ is very similar and left as an exercise.

This leads to the result

$$L(f;P) \leq L(f;Q) \leq U(f;Q) \leq U(f;P).$$

This was just for the case of inserting one extra point in Q. Now, for the more general result, of having i more points in Q, we simply apply the above process i times. ■

The following corollary tells us that the lower Riemann sum of one partition is less than the upper Riemann sum of another.

Corollary 5.5 Let $f : [a,b] \to \mathbb{R}$ be bounded. Let P_1 and P_2 be two partitions of $[a,b]$. Then

$$L(f;P_1) \leq U(f;P_2).$$

Proof. Let Q be a common refinement of P_1 and P_2. Then, by Theorem 5.3, we have

$$L(f;P_1) \leq L(f;Q) \leq U(f;Q) \leq U(f;P_2).$$

and we arrived at our result. ■

We could have partitioned our interval $[a,b]$ in many ways. Our choice of how we split the interval determines the size of n, effects the length of each subinterval, and effects where the rectangles touch the curve. So, our Riemann sum is highly influenced by our choice of P.

Exercise 5.6 Let $[a,b] \subset \mathbb{R}$ be any finite interval and P be any partition of $[a,b]$. Suppose $f,g : [a,b] \to \mathbb{R}$ are bounded functions, and that $c \geq 0$. Show that

(a) $U(-f;P) = -L(f;P)$

(b) $L(-f;P) = -U(f;P)$

(c) $U(cf;P) = cU(f;P)$

(d) $L(cf;P) = cL(f;P)$

(e) $U(f+g;P) \leq U(f;P) + U(g;P)$

(f) $L(f+g;P) \geq L(f;P) + L(g;P)$

We have the following key definition.

Definition 5.7 Let $f : [a,b] \to \mathbb{R}$ be a bounded function on the interval $[a,b]$, and P ranges over *all partitions* of $[a,b]$. Then

(a) The lower Riemann integral of f over $[a,b]$ is

$$\underline{\int_a^b} f(x)\,\mathrm{d}x = \sup_P L(f;P).$$

(b) The upper Riemann integral of f over $[a,b]$ is

$$\overline{\int_a^b} f(x)\,\mathrm{d}x = \inf_P U(f;P).$$

Remark 5.8

Notice in Definition 5.7, we take the *supremum* of all the *lower* Riemann sums and the *infimum* of all the *upper* Riemann sums.

A consequence of Corollary 5.5 is the following proposition.

Proposition 5.9 If $f : [a,b] \to \mathbb{R}$ is bounded, then

$$\underline{\int_a^b} f(x)\,\mathrm{d}x \leq \overline{\int_a^b} f(x)\,\mathrm{d}x$$

Proof. Let A and B be the sets of all lower Riemann sums and upper Riemann sums respectively taken over all partitions of $[a,b]$. Then by Corollary 5.5, $\sup A \leq \inf B$, so

$$\underline{\int_a^b} f(x)\,\mathrm{d}x \leq \overline{\int_a^b} f(x)\,\mathrm{d}x.$$

■

Finally, we come to the definition of the Riemann integral and what it means to say a function is integrable.

Definition 5.10 A bounded function f is said to be *Riemann integrable* on the interval $[a,b]$ if the lower and upper Riemann integrals are equal. When this common value exists, we call it the *Riemann integral* and denote it by

$$\int_a^b f(x)\,\mathrm{d}x.$$

If they are not equal, the function is *not Riemann integrable*.

Exercise 5.11 Let $f : [a,b] \to \mathbb{R}$ be a Riemann integrable function, and let P_n be a sequence of partitions such that

$$\lim_{n\to\infty} \|P_n\| = 0.$$

Show that

$$\int_a^b f(x)\,\mathrm{d}x = \lim_{n\to\infty} L(f;P_n) = \lim_{n\to\infty} U(f;P_n).$$

What we are calling the Riemann integral here is really Darboux's formulation of the integral, whereas Riemann's formulation used tagged partitions. But, there is a theorem that states that a function is Darboux integrable (what we have called Riemann integrable above) if and only if $\lim_{n\to\infty} S(f;P_n;T_n)$ converges to the same value for any sequence of tagged partitions (P_n,T_n) such that $\lim_{n\to\infty}\|P_n\| = 0$. For that reason, we may ignore the minor differences in forming the integral (the exact nature of the partitions and the point in each interval at which the function is evaluated) and we simply refer to the Riemann integral.

▪ **Example 5.12** We will start with a basic example: the constant function. Let $f : [a,b] \to \mathbb{R}$ be the constant function defined by $f(x) = c$. Is f Riemann integrable? If it is, what is the integral? For any partition P, $m_i = M_i = c$,

and so

$$\begin{aligned} L(f;P) = U(f;P) &= \sum_{i=1}^{n} c\Delta x_i \\ &= c\sum_{i=1}^{n}(x_i - x_{i-1}) \\ &= c\left[(x_1 - x_0) + (x_2 - x_1) + \cdots + (x_n - x_{n-1})\right] \\ &= c(a-b), \end{aligned}$$

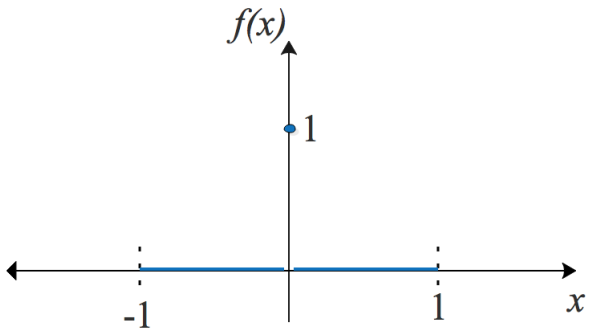
since the sum is telescoping and $a = x_0$, $b = x_n$ are the endpoints of the interval. Thus, f is integrable and

$$\int_a^b f(x)\mathrm{d}x = c(a-b).$$

■ **Example 5.13** Now we will look at a piecewise function. Let $f : [-1,1] \to \mathbb{R}$ be defined by

$$f(x) = \begin{cases} 1 & \text{if } x = 0 \\ 0 & \text{if } x \neq 0 \end{cases}$$

Is f Riemann integrable? If it is, what is the integral?



There are many (in fact infinitely many) ways we can create partitions of $[-1,1]$, but let's consider the partitions given by

$$P_h = \left(-1, -\frac{1}{2^h}, \frac{1}{2^h}, 1\right),$$

so the first partition would be $P_1 = \left(-1, -\frac{1}{2}, \frac{1}{2}, 1\right)$. We can see that we have three subintervals for any h. Now, for each of the subintervals $I = \left[-1, -\frac{1}{2^h}\right], \left[\frac{1}{2^h}, 1\right]$, we have $\sup_{x\in I} f(x) = \inf_{x\in I} f(x) = 0$ so these subintervals do not contribute to the upper or lower Riemann sums. Now let's look at the middle subinterval, $I = \left[-\frac{1}{2^h}, \frac{1}{2^h}\right]$. This is the subinterval that contains the point of discontinuity, at $x = 0$. We have

$$\inf_{x\in I} f(x) = 0$$

so

$$L(f;P_h) = 0$$

and

$$\sup_{x\in I} f(x) = 1 \quad \text{and} \quad \Delta x = \frac{1}{2^h} + \frac{1}{2^h} = \frac{1}{2^{h-1}}$$

so that

$$U(f;P_h) = \frac{1}{2^{h-1}}.$$

Now we can look at the upper and lower Riemann integrals. We have

$$\overline{\int_{-1}^{1}} f(x)\,\mathrm{d}x = \inf_P U(f;P) \leq \inf_h U(f;P_h) = 0$$

and

$$\underline{\int_{-1}^{1}} f(x)\,\mathrm{d}x = \sup_P L(f;P) \geq \sup_h L(f;P_h) = 0$$

Now from Proposition 5.9

$$\underline{\int_{-1}^{1}} f(x)\,\mathrm{d}x \leq \overline{\int_{-1}^{1}} f(x)\,\mathrm{d}x$$

hence f is integrable and

$$\int_{-1}^{1} f(x)\,\mathrm{d}x = 0.$$

■ **Example 5.14** Let $n \in \mathbb{N}$ with $n \geq 2$. By choosing a suitable partition to approximate the integral

$$\int_1^n \log x\,\mathrm{d}x$$

using upper and lower Riemann sums, show that

$$\sum_{k=1}^{n-1} \log k \leq n\log n - n + 1 \leq \sum_{k=2}^{n} \log k.$$

Let $f(x) = \log x$. We take the following, as our partition of the interval $[1,n]$

$$P = (1,2,3,\ldots,n-1,n).$$

We have $\Delta x = 1$ for each subinterval. Now since f is monotonically increasing, we have

$$f(k-1) \leq f(x) \leq f(k) \quad \text{for } x \in [k-1,k].$$

So we can say

$$\sup_{x\in[k-1,k]} f(x) = f(k) \quad \text{and} \quad \inf_{x\in[k-1,k]} f(x) = f(k-1).$$

We deduce that

$$L(f;P) = \sum_{k=2}^{n} \log(k-1) = \sum_{k=1}^{n-1} \log k$$

$$U(f;P) = \sum_{k=2}^{n} f(k).$$

Therefore, we have

$$\sum_{k=1}^{n-1} \log k \leq \int_1^n \log x\,\mathrm{d}x \leq \sum_{k=2}^{n} \log k.$$

Next, note that

$$\int_1^n \log x\,\mathrm{d}x = \int_1^n \frac{\mathrm{d}}{\mathrm{d}x}(x\log(x) - x)\mathrm{d}x = n\log(n) - n + 1,$$

which yields the result

$$\sum_{k=1}^{n-1} \log k \leq n\log n - n + 1 \leq \sum_{k=2}^{n} \log k.$$

Exercise 5.15 As a follow up to Example 5.14, show that

$$n^n e^{-n+1} \leq n! \leq (n+1)^{n+1} e^{-n}.$$

The following theorem is known as Riemann's criterion for integrability. It is analogous to Cauchy's criterion for converging sequences and therefore sometimes referred to as Cauchy's criterion for integrability.

Theorem 5.16 – Riemann's criterion for integrability
A function $f : [a,b] \to \mathbb{R}$ is Riemann integrable if and only if, for all $\varepsilon > 0$, there exists a partition P such that

$$U(f;P) - L(f;P) < \varepsilon.$$

Proof. This is an if and only if ($\Leftrightarrow$) statement. So, we should prove the statement in both ways $\Rightarrow$ and $\Leftarrow$. We will first prove the "$\Leftarrow$" statement: that for all $\varepsilon > 0$, there exists a partition P such that $U(f;P) - L(f;P) < \varepsilon$ implies that f is integrable. If P is any partition, from the definition of supremum and infimum, we have

$$0 \leq L(P;f) \leq \underline{\int_a^b} f(x)\,\mathrm{d}x \leq \overline{\int_a^b} f(x)\,\mathrm{d}x \leq U(P;f).$$

This inequality and our statement that $U(f;P) - L(f;P) < \varepsilon$ implies that for all $\varepsilon > 0$,

$$0 \leq \overline{\int_a^b} f(x)\,\mathrm{d}x - \underline{\int_a^b} f(x)\,\mathrm{d}x < \varepsilon.$$

Since ε is arbitrary, this implies

$$0 = \overline{\int_a^b} f(x)\,\mathrm{d}x - \underline{\int_a^b} f(x)\,\mathrm{d}x$$

which means that $\overline{\int_a^b} f(x)\,\mathrm{d}x = \underline{\int_a^b} f(x)\,\mathrm{d}x$, so f is integrable. Now we prove the "$\Rightarrow$" statement. Suppose f is integrable and $\varepsilon > 0$. Then there exist partitions P_1 and P_2 such that

$$U(f;P_1) - \int_a^b f(x)\,\mathrm{d}x < \frac{\varepsilon}{2} \quad \text{and} \quad \int_a^b f(x)\,\mathrm{d}x - L(f;P_2) < \frac{\varepsilon}{2}.$$

Now, taking Q to be the common refinement of P_1 and P_2, we have

$$U(f;Q) < \int_a^b f(x)\,\mathrm{d}x + \frac{\varepsilon}{2} \quad \text{and} \quad L(f;Q) > \int_a^b f(x)\,\mathrm{d}x - \frac{\varepsilon}{2}$$

which implies $U(f;Q) - L(f;Q) < \varepsilon$. ∎

Properties of Riemann Integration

Now we will give two fundamental properties of the Riemann integral, namely the properties of linearity, and additivity.

Theorem 5.17 — Linearity of the Riemann integral. Let $f : [a,b] \to \mathbb{R}$ and $g : [a,b] \to \mathbb{R}$ be Riemann integrable, and let $c \in \mathbb{R}$. Then $f + cg$ is integrable and

$$\int_a^b f(x) + cg(x)\,\mathrm{d}x = \int_a^b f(x)\,\mathrm{d}x + c\int_a^b g(x)\,\mathrm{d}x$$

Proof. We will prove this in two stages. First we will show that g is integrable, then so is $-g$. Then we will show that $f + cg$ is integrable for $c \geq 0$. Combining these two facts yields the result.

Let P be any partition of the interval $[a,b]$. Then $L(-g;P) = -U(g;P)$ which implies that

$$\begin{aligned}\underline{\int_a^b} -g(x)\,\mathrm{d}x &= \sup_P L(-g;P) = \sup_P -U(g;P) = -\inf_P U(g;P)\\ &= -\int_a^b g(x)\mathrm{d}x\end{aligned}$$

Similarly,

$$\begin{aligned}\overline{\int_a^b} -g(x)\,\mathrm{d}x &= \inf_P U(-g;P) = \inf_P -L(g;P) = -\sup_P U(g;P)\\ &= -\int_a^b g(x)\mathrm{d}x.\end{aligned}$$

Therefore, $-g$ is integrable and

$$\int_a^b -g(x)\mathrm{d}x = -\int_a^b g(x)\mathrm{d}x.$$

Now for the second part of the proof, we will consider functions f and g and suppose $c \geq 0$. From a property given in Exercise 5.6, we have

$$L(f;P) + cL(g;P) \leq L(f+cg;P). \tag{5.1}$$

Also, if Q is a refined partition of P, from the definition of the supremum, we have

$$L(f+cg;P) \leq \sup_Q L(f+cg;Q),$$

so this with inequality (5.1) gives

$$L(f;P) + cL(g;P) \leq \sup_Q L(f+cg;Q). \tag{5.2}$$

Now, suppose we have two partitions P_1 and P_2 with R being their common refinement. Then

$$L(f;P_1) + cL(g;P_2) \leq L(f;R) + cL(g;R).$$

Combining this with the inequality in (5.2) yields

$$L(f;P_1) + cL(g;P_2) \leq \sup_Q L(f+cg;Q)$$

for any pair of partitions P_1 and P_2. Now, taking the supremum over P_1 and P_2 we have

$$\sup_{P_1} L(f;P_1) + c\sup_{P_2} L(g;P_2) \leq \sup_Q L(f+cg;Q)$$

or equivalently

$$\int_a^b f(x)\mathrm{d}x + c\int_a^b g(x)\mathrm{d}x \leq \underline{\int_a^b} f(x) + cg(x)\,\mathrm{d}x$$

Now, we consider the same approach for functions $-f$ and $-g$. Very similarly to the case with f and g, (the intermediate steps are left to the reader as an exercise), we arrive at the following result

$$\overline{\int_a^b} f(x) + cg(x)\,\mathrm{d}x \leq \int_a^b f(x)\mathrm{d}x + c\int_a^b g(x)\mathrm{d}x.$$

So, by substitution, we have

$$\underline{\int_a^b} f(x) + cg(x)\,\mathrm{d}x \leq \overline{\int_a^b} f(x) + cg(x)\,\mathrm{d}x$$

so

$$\underline{\int_a^b} f(x) + cg(x)\,\mathrm{d}x = \overline{\int_a^b} f(x) + cg(x)\,\mathrm{d}x = \int_a^b f(x)\,\mathrm{d}x + c\int_a^b g(x)\,\mathrm{d}x$$

and by definition, this implies that $f + cg$ is integrable. Combining the two results gives us the following result we require

$$\int_a^b f(x) + cg(x)\,\mathrm{d}x = \int_a^b f(x)\,\mathrm{d}x + c\int_a^b g(x)\,\mathrm{d}x.$$

■

Theorem 5.18 — Additivity of the Riemann integral. Let $f : [a,b] \to \mathbb{R}$ and let $c \in (a,b)$. Then f is integrable if and only if $f|_{[a,c]}$ and $f|_{[c,b]}$ are integrable and

$$\int_a^b f(x)\,\mathrm{d}x = \int_a^c f(x)\,\mathrm{d}x + \int_c^b f(x)\,\mathrm{d}x.$$

Proof. Since this is an if and inly if statement, we need to prove it in both ways $\Rightarrow$ and $\Leftarrow$. We will first prove the $\Rightarrow$ statement.

Suppose $f : [a,b] \to \mathbb{R}$ is integrable. Fix $\varepsilon > 0$. From the Riemann Criterion for Integrability, Theorem 5.16, there exists a partition of $[a,b]$, P such that

$$U(f;P) - L(f;P) < \varepsilon. \tag{5.3}$$

Now, let's take this partition P and refine it by including a point c between points x_h and x_{h+1} so

$$P = (x_0, x_1, \ldots, x_h, c, x_{h+1}, \ldots, x_k).$$

Consider the partitions P_{c_1} and P_{c_2} where $P_{c_1} = (x_0, x_1, \ldots, x_h, c)$ and $P_{c_2} = (c, x_{h+1}, \ldots, x_k)$.

We have

$$L(f;P) = L(f|_{[a,c]}, P_{c_1}) + L(f|_{[c,b]}, P_{c_2}),$$
$$U(f;P) = U(f|_{[a,c]}, P_{c_1}) + U(f|_{[c,b]}, P_{c_2}).$$

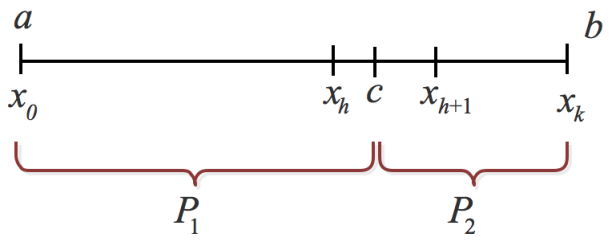


Figure 5.10*Partition split into two*

Substituting these expressions in (5.3) and combining the terms for each partition, gives

$$\Big[U(f|_{[a,c]}, P_{c_1}) - L(f|_{[a,c]}, P_{c_1})\Big] + \Big[U(f|_{[c,b]}, P_{c_2}) - L(f|_{[c,b]}, P_{c_2})\Big] < \varepsilon.$$

Since both terms are non-negative, we must have that

$$U(f|_{[a,c]}, P_{c_1}) - L(f|_{[a,c]}, P_{c_1}) < \varepsilon$$

and

$$U(f|_{[c,b]}, P_{c_2}) - L(f|_{[c,b]}, P_{c_2}) < \varepsilon.$$

This implies, using the Riemann Criterion for Integrability, that both $f|_{[a,c]}$ and $f|_{[c,b]}$ are integrable.

Now, we will show the $\Leftarrow$ statement. Here, this is simply just working backwards. To that end, suppose $f|_{[a,c]}$ and $f|_{[c,b]}$ are integrable. So, for a fixed $\varepsilon > 0$ there exist partitions P_{c_1} and P_{c_2} of $[a,c]$ and $[c,b]$ respectively such that

$$U(f|_{[a,c]}, P_{c_1}) - L(f|_{[a,c]}, P_{c_1}) < \frac{\varepsilon}{2}$$

and

$$U(f|_{[c,b]}, P_{c_2}) - L(f|_{[c,b]}, P_{c_2}) < \frac{\varepsilon}{2}.$$

Now, we will form a partition P which is formed by concatenating the two partitions P_{c_1} and P_{c_2}. We have

$$L(f;P) = L(f|_{[a,c]}, P_{c_1}) + L(f|_{[c,b]}, P_{c_2}),$$
$$U(f;P) = U(f|_{[a,c]}, P_{c_1}) + U(f|_{[c,b]}, P_{c_2}).$$

so

$$\begin{aligned} U(f;P) - L(f;P) = &\Big[U(f|_{[a,c]}, P_{c_1}) - L(f|_{[a,c]}, P_{c_1})\Big] \\ &+ \Big[U(f|_{[c,b]}, P_{c_2}) - L(f|_{[c,b]}, P_{c_2})\Big] < \varepsilon. \end{aligned}$$

This implies that f is integrable. Furthermore, we have that

$$\begin{aligned} \int_a^b f(x)\,\mathrm{d}x \le U(f;P) &= U(f|_{[a,c]}, P_{c_1}) + U(f|_{[c,b]}, P_{c_2}) \\ &< \int_a^c f(x)\,\mathrm{d}x + \int_c^b f(x)\,\mathrm{d}x + \varepsilon. \end{aligned}$$

Similarly, for the lower Riemann sums, we have

$$\begin{aligned} \int_a^b f(x)\mathrm{d}x \ge L(f;P) &= L(f|_{[a,c]}, P_{c_1}) + L(f|_{[c,b]}, P_{c_2}) \\ &> \int_a^c f(x)\,\mathrm{d}x + \int_c^b f(x)\,\mathrm{d}x - \varepsilon \end{aligned}$$

Since ε is arbitrary, we arrive at the required result that

$$\int_a^b f(x)\,\mathrm{d}x = \int_a^c f(x)\,\mathrm{d}x + \int_c^b f(x)\,\mathrm{d}x.$$

■

Now, we will state a powerful theorem that tells us that all continuous functions are integrable.

Theorem 5.19 – Continuous functions are integrable
Let $f : [a,b] \to \mathbb{R}$ be continuous on $[a,b]$. Then f is integrable.

Proof. We will show that by f being continuous, the Riemann Criterion for Integrability, Theorem 5.16, is satisfied.

Since f is continuous on $[a,b]$, it is uniformly continuous. So, by definition of uniform continuity, we can say that fro all $\varepsilon > 0$, there exists $\delta > 0$ such that for all $x, y \in [a,b]$ with $|x-y| < \delta$ we have

$$|f(x) - f(y)| < \frac{1}{2(b-a)}\varepsilon.$$

Now, let P be a partition of $[a,b]$ such that the distance of each subinterval is less than δ. So, for $P = (x_0, \ldots, x_k)$ we have $\Delta x_i < \delta$ for all $i \in \{1, \ldots k\}$. One way of making this happen is to pick x_i to be

$$x_i = a + \frac{b-a}{k}i, \quad i = 0, \ldots, k$$

where the number of subintervals is chosen to be large enough for $(b-a) < k\delta$ to hold.

Now, for $x, y \in [x_{i-1}, x_i]$ we have

$$|f(x) - f(y)| < \frac{1}{2(b-a)}\varepsilon$$

implying that

$$\sup_{x\in[x_{i-1},x_i]} f(x) - \inf_{x\in[x_{i-1},x_i]} f(x) \leq \frac{1}{2(b-a)}\varepsilon < \frac{1}{b-a}\varepsilon$$

Now, using this inequality, we can estimate the difference of the upper and lower Riemann sums as follows

$$\begin{aligned} U(f;P) - L(f;P) &= \sum_{i=1}^{k} \left[\sup_{x\in[x_{i-1},x_i]} f(x) - \inf_{x\in[x_{i-1},x_i]} f(x) \right] \Delta x_i \\ &< \sum_{i=1}^{k} \frac{\varepsilon}{b-a}(x_i - x_{i-1}) \\ &= \frac{\varepsilon}{b-a}\left[(x_1 - x_0) + (x_2 - x_1) + \cdots + (x_k - x_{k-1})\right] = \varepsilon \end{aligned}$$

since the sum telescopes and the last and first terms are respectively $x_k = b$ and $x_0 = a$.

Therefore, for any $\varepsilon > 0$, we found a partition P such that

$$U(f;P) - L(f;P) < \varepsilon,$$

and by the Riemann Criterion for Integrability, this implies that f is integrable.

■

The above is a sufficient condition for integrability, but is not necessary (see Example 5.13). In fact we can integrate some fairly nasty looking functions, and yet we have the following pleasant result.

Theorem 5.20 – Integrals are continuous
Let $f : [a,b] \to \mathbb{R}$ be integrable on $[a,b]$ and F defined on $[a,b]$ as

$$F(x) = \int_a^x f(t)\,\mathrm{d}t.$$

Then F is continuous on $[a,b]$.

Proof. By definition (as an integrable function) f is bounded on $[a,b]$; that is, there exists a number $M > 0$ such that

$$|f(x)| < M$$

for all $x \in [a,b]$. Let $c \in [a,b]$ and $h > 0$, then

$$F(c+h) - F(c) = \int_a^{c+h} f(x)\,\mathrm{d}x - \int_a^c f(x)\,\mathrm{d}x = \int_c^{c+h} f(x)\,\mathrm{d}x$$

where we have used the additivity of the Riemann integral in the second equality. Since $-M \le f(x) \le M$, we have

$$-Mh \le \int_c^{c+h} f(x)\,\mathrm{d}x \le Mh$$

or

$$-Mh \le F(c+h) - F(c)\,\mathrm{d}x \le Mh.$$

If $h < 0$ we have

$$-M(-h) \le \int_{c+h}^c f(x)\,\mathrm{d}x \le M(-h)$$

and multiplying through by -1

$$-Mh \ge -\int_{c+h}^c f(x)\,\mathrm{d}x \ge Mh$$

or

$$Mh \le \int_c^{c+h} f(x)\,\mathrm{d}x \le -Mh$$

which is

$$Mh \le F(c+h) - F(c)\,\mathrm{d}x \le -Mh.$$

Combining these inequalities we have

$$|F(c+h) - F(c)| \le M|h|.$$

Hence, for any $\epsilon > 0$ we have

$$|F(c+h) - F(c)| < \epsilon$$

for all $|h| < \epsilon/M$, proving that

$$\lim_{h\to 0} F(c+h) = f(c)$$

so that F is continuous at any $c \in [a,b]$. ■

5.2 The Fundamental Theorem of Calculus

To evaluate integrals, you are required to find the anti-derivative of the integrand and, in the case of a definite integral, evaluate the result for the bounds of integration. Differentiation and integration being inverse processes is non-trivial and this result goes by the name of *The Fundamental Theorem of Calculus*, and is split into 2 parts.

Theorem 5.21 – The First Fundamental Theorem of Calculus
Let $f : [a,b] \to \mathbb{R}$ be a continuous function, and let's define the function $F(x) = \int_a^x f(t)\,\mathrm{d}t$. Then F is continuous on $[a,b]$, differentiable on (a,b), and we have $F'(x) = f(x)$ for all $x \in (a,b)$.

This part of the theorem is simply saying that the *derivative of the integral is the function itself*. It is worth comparing this result to Theorem 5.20: there we started with an integrable f and obtained that F was continuous, here we start with a continuous f and obtain that F is differentiable.

Proof. Let $x, y \in [a,b]$. Since f is continuous in $[a,b]$ it is uniformly continuous on $[a,b]$, hence for all $\varepsilon > 0$, there exists $\delta > 0$ such that for all x, y with $|y - x| < \delta$, we have $|f(x) - f(y)| < \varepsilon$.

Our goal is to show that F is differentiable and given by $F'(x) = f(x)$. Referring to the standard definition of a differentiable function, our goal is thus to show that

$$f(x) = F'(x) = \lim_{y \to x} \frac{F(y) - F(x)}{y - x}.$$

Let's estimate

$$\left| \frac{F(y) - F(x)}{y - x} - f(x) \right| = \left| \frac{1}{y - x} \int_x^y f(t)\,\mathrm{d}t - f(x) \right| \tag{5.4}$$

$$= \left| \frac{1}{y - x} \int_x^y f(t) - f(x)\,\mathrm{d}t \right| \tag{5.5}$$

where (5.4) follows from additivity of the integral. Now, since $x < t < y$, we have that $|x - t| < \delta$, and so $|f(t) - f(x)| < \varepsilon$. We can now bound (5.5) by

$$\left| \frac{1}{y - x} \int_x^y f(t) - f(x)\,\mathrm{d}t \right| = \frac{1}{|y - x|} \left| \int_x^y f(t) - f(x)\,\mathrm{d}t \right|$$

$$\leq \frac{1}{|y - x|} |y - x| \sup_{t \in [x,y]} |f(t) - f(x)| < \varepsilon \tag{5.6}$$

So, all in all, this tells us that

$$\left| \frac{F(y) - F(x)}{y - x} - f(x) \right| < \varepsilon$$

which is the precise definition of the limit. ∎

An immediate result of this theorem is the second part of the Fundamental Theorem of Calculus.

Theorem 5.22 – The Second Fundamental Theorem of Calculus
Let $f : [a,b] \to \mathbb{R}$ be continuous, and $F : [a,b] \to \mathbb{R}$ be continuous on $[a,b]$, and differentiable on (a,b) where $F'(x) = f(x)$. Then $\int_a^b f(t)\,\mathrm{d}t = F(b) - F(a)$.

Proof. Let's define a function $G : [a,b] \to \mathbb{R}$ by

$$G(x) = \int_a^x f(t)\,\mathrm{d}t.$$

Using the first Fundamental Theorem of Calculus, we have $G'(x) = f(x)$.

Now, consider a function $H(x)$ given by $H(x) = G(x) - F(x)$. The derivative of this function is $G'(x) - F'(x) = f(x) - f(x) = 0$. Thus, $H(x)$ is a constant function on $[a,b]$, namely $H(x) = A$ where A is a constant. However, $G(a) = \int_a^a f(t)\,\mathrm{d}t = 0$, so we have $G(a) - F(a) = -F(a) = A$. Finally,

$$G(x) = F(x) + A = F(x) - F(a),$$

for all $x \in [a,b]$. Taking $x = b$, we have

$$G(b) = \int_a^b f(t)\,\mathrm{d}t = F(b) - F(a).$$

■

Remark 5.23

Theorem 5.22 can actually be improved upon: we can replace continuity of f with the weaker hypothesis of integrability of f. However, this does not follow from Theorem 5.21 and requires a new proof starting from the definition of the integral. The interested reader can find this in any good calculus textbook.

The following theorem is a useful result which may be applied to many integrands we will be looking at where a change of variable is used to simplify the integral.

Theorem 5.24 – Substitution Formula
Let $I_1 = [\alpha,\beta]$ and $I_2 = [\gamma,\delta]$ be finite intervals. Let $g : I_1 \to (\gamma,\delta)$ be a differentiable function with g' continuous on (α,β). Suppose $f : I_2 \to \mathbb{R}$ is continuous. Then if $[a,b] \subset (\alpha,\beta)$,

$$\int_{g(a)}^{g(b)} f(t)\,\mathrm{d}t = \int_a^b f(g(t))g'(t)\,\mathrm{d}t.$$

Proof. By the First Fundamental Theorem of Calculus, Theorem 5.21, since $f(t)$ is continuous, there exists $F : [\gamma,\delta] \to \mathbb{R}$ with $F'(x) = f(x)$ for all $x \in (\gamma,\delta)$.

By the chain rule, the function $F \circ g : [\alpha,\beta] \to \mathbb{R}$ is differentiable on (α,β) with

$$(F \circ g)'(x) = F'(g(x))g'(x),$$

for all $x \in (\alpha, \beta)$. Now we use the Second Fundamental Theorem of Calculus, Theorem 5.22, to find that

$$\begin{aligned}\int_a^b f(g(t))g'(t)\,\mathrm{d}t &= \int_a^b (F\circ g)'(t)\,\mathrm{d}t \\ &= (F\circ g)(b) - (F\circ g)(a) = F(g(b)) - F(g(a)) \\ &= \int_{g(a)}^{g(b)} f(t)\,\mathrm{d}t.\end{aligned}$$

■

■ **Example 5.25** Given the change of variable $g(t) = \frac{1}{t}$, show that for $0 < a < b$

$$\int_a^b \frac{1}{1+t^2}\,\mathrm{d}t = \int_{1/b}^{1/a} \frac{1}{1+t^2}\,\mathrm{d}t.$$

We know that $g(t) = \frac{1}{t}$ and $f(t) = \frac{1}{1+t^2}$. If $0 < a < b$, then we can take

$$I_1 = \left[\frac{a}{2}, 2b\right], \quad I_2 = \left[\frac{1}{2b}, \frac{2}{a}\right].$$

Then we have $g : I_1 \to I_2$ with $g'(t) = -\frac{1}{t^2}$ being continuous on $\left(\frac{a}{2}, 2b\right)$. Now since $f : I_2 \to \mathbb{R}$ is continuous, we can use Theorem 5.24 to obtain

$$\int_{1/a}^{1/b} \frac{1}{1+t^2}\,\mathrm{d}t = \int_a^b \frac{1}{1+\left(\frac{1}{t}\right)^2}\frac{-1}{t^2}\,\mathrm{d}t.$$

After switching the integral bounds, we arrive at the result.

5.3 Interchanging Limits and Integrals

In this section, we look at evaluating the limit of the integral of a sequence of functions. Sometimes, we cannot evaluate integrals explicitly, so we ask ourselves: When can we interchange the limit with the integral so that we can find the limit of the integrand first, then integrate the result? In other words, if $f(x) = \lim_{n\to\infty} f_n(x)$, when can we say that

$$\lim_{n\to\infty} \int_a^b f_n(x)\,\mathrm{d}x = \int_a^b f(x)\,\mathrm{d}x\,?$$

Definition 5.26 (Reminder)
Let $f_n : A \subset \mathbb{R} \to \mathbb{R}$ be a countable set of functions defined on the same A. The sequence $\{f_n\}_{n=1}^{\infty}$ *converges pointwise* at $x_0 \in A$ if

$$\lim_{n\to\infty} f_n(x_0) = l < \infty.$$

If $\{f_n\}_{n=1}^{\infty}$ converges pointwise at every x in A, then we say that f is the pointwise limit of $\{f_n\}_{n=1}^{\infty}$ on A. We call the limit $f(x)$ because l might depend on x. So, we say $\lim_{n\to\infty}(f_n(x)) = l = f(x)$. Clearly, $f : A \to \mathbb{R}$ is a function on A because A is its domain, its value $f(x) = \lim_{n\to\infty} f_n(x)$ and its codomain is $\mathbb{R}$.

Definition 5.27 (Reminder)
A sequence $\{f_n\}_{n=1}^{\infty}$ of functions $f_n : S \to \mathbb{R}$ *converges uniformly* to f on S if

$$\lim_{n\to\infty} \|f_n - f\|_S = 0.$$

In other words, $\{f_n\}_{n=1}^{\infty}$ converges uniformly to f if for all $\varepsilon > 0$, there exists $N \in \mathbb{Z}$ such that

$$\|f_n - f\|_S < \varepsilon, \quad n \geq N.$$

Note that the $N(\epsilon)$ in Definition5.27 is independent of x.

Before we introduce the theorem of interchanging limits and integrals, we need the following lemma, which will aid with proving the theorem.

Lemma 5.28 Let $f \in C^0[a,b]$, $a < b$, then

$$\left|\int_a^b f(x)\,\mathrm{d}x\right| \leq \int_a^b |f(x)|\,\mathrm{d}x \leq \int_a^b \sup_{x\in[a,b]} |f(x)|\,\mathrm{d}x$$
$$\leq \|f\|_S \int_a^b \mathrm{d}x = (b-a)\|f\|_S$$

This inequality follows by considering the Riemann integral of $y = f(x)$, $y = |f(x)|$ and the constant function $y = \|f\|_S$.

Theorem 5.29 – Uniform convergence and Riemann integrability
Let $\{f_n\}_{n=1}^{\infty}$ be a sequence of Riemann integrable functions defined on a finite interval $[a,b]$. If f_n converges uniformly to f as $n \to \infty$, then f is Riemann integrable and

$$\lim_{n\to\infty} \int_a^b f_n(x)\,\mathrm{d}x = \int_a^b f(x)\,\mathrm{d}x. \tag{5.7}$$

Remark 5.30

This theorem tells us that uniform convergence commutes with integration: if our functions f_n are uniformly convergent then we can interchange the order of the limit and the integration sign

$$\lim_{n\to\infty} \int_a^b f_n(x)\,\mathrm{d}x = \int_a^b \lim_{n\to\infty} f_n(x)\,\mathrm{d}x.$$

Proof. This proof is split into 2 parts, first we prove that given the conditions, we have that f is Riemann integrable; then we prove the result given by equation (5.7)

By the uniform convergence of $\{f_n\}_{n=1}^{\infty}$ to f, and using Definition 5.27, let us fix $\varepsilon > 0$. In which case, there exists $N \in \mathbb{Z}$ such that for $n \geq N$, we have

$$\|f_n(x) - f(x)\| < \frac{\varepsilon}{2(b-a)},$$

for all $x \in [a,b]$. Therefore, assume $n > N$, if $I \subset [a,b]$ is any subinterval,

we have

$$\begin{aligned}\sup_{x\in I} f(x) &= \sup_{x\in I}\left(f(x) - f_n(x) + f_n(x)\right)\\ &\leq \sup_{x\in I}\left(f(x) - f_n(x)\right) + \sup_{x\in I} f_n(x)\\ &\leq \sup_{x\in I} f_n(x) + \frac{\varepsilon}{2(b-a)},\end{aligned}$$

where in the first inequality we have used the following proposition: If $f, g : A \to \mathbb{R}$ are bounded functions, then $\sup(f+g) \leq \sup f + \sup g$. This leads us to the following conclusion

$$\begin{aligned}U(f;P) &= \sum_{i=1}^{k} \sup_{x\in[x_{i-1},x_i]} f(x)\Delta x_i\\ &\leq \sum_{i=1}^{k}\left[\sup_{x\in[x_{i-1},x_i]} f_n(x) + \frac{\varepsilon}{2(b-a)}\right]\Delta x_i\\ &= U(f_n;P) + \frac{\varepsilon}{2(b-a)}\sum_{i=1}^{k}(x_i - x_{i-1})\\ &= U(f_n;P) + \frac{\varepsilon}{2} \qquad (5.8)\end{aligned}$$

Taking the infimum over P and that f_n is Riemann integrable (given in the theorem), we conclude that

$$\overline{\int_a^b} f(x)\,\mathrm{d}x \leq \int_a^b f_n(x)\,\mathrm{d}x + \frac{\varepsilon}{2} < \int_a^b f_n(x)\,\mathrm{d}x + \varepsilon. \qquad (5.9)$$

A similar approach is used for the infimum (while reversing the inequality), so that we have

$$L(f;P) \geq L(f_n,P) - \frac{\varepsilon}{2}$$

Taking the supremum yields the following

$$\underline{\int_a^b} f(x)\,\mathrm{d}x \geq \int_a^b f_n(x)\,\mathrm{d}x - \frac{\varepsilon}{2} > \int_a^b f_n(x)\,\mathrm{d}x - \varepsilon.. \qquad (5.10)$$

Now, combining (5.9) and (5.10), we have

$$\int_a^b f_n(x)\,\mathrm{d}x - \varepsilon < \underline{\int_a^b} f(x)\,\mathrm{d}x \leq \overline{\int_a^b} f(x)\,\mathrm{d}x < \int_a^b f_n(x)\,\mathrm{d}x + \varepsilon$$

From this, we have that

$$0 \leq \overline{\int_a^b} f(x)\,\mathrm{d}x - \underline{\int_a^b} f(x)\,\mathrm{d}x < 2\varepsilon.$$

Since ε is arbitrary, we must have that

$$\overline{\int_a^b} f(x)\,\mathrm{d}x = \underline{\int_a^b} f(x)\,\mathrm{d}x = \int_a^b f(x)\,\mathrm{d}x$$

and thus f is integrable.

The uniform convergence of $\{f_n\}_{n=1}^{\infty}$ implies that the limit function f is continuous on $[a,b]$. We have

$$\begin{aligned}\left|\int_a^b f_n(x)\,\mathrm{d}x - \int_a^b f(x)\,\mathrm{d}x\right| &= \left|\int_a^b f_n(x) - f(x)\,\mathrm{d}x\right|\\ &\leq \int_a^b |f_n(x) - f(x)|\,\mathrm{d}x\\ &\leq (b-a)\|f_n - f\|_S\end{aligned}$$

using Lemma 5.28. From the definition of uniform convergence, we know that $\|f_n - f\|_S \to 0$ as $n \to \infty$. So indeed this yields the result

$$\lim_{n\to\infty} \int_a^b f_n(x)\,\mathrm{d}x = \int_a^b f(x)\,\mathrm{d}x.$$

■

■ **Example 5.31** Let $f_n : [0, 1/2] \to \mathbb{R}$, be defined by $f_n(x) = \frac{e^{x+\sin^n(x)}}{\ln(\cos(x^n)+3)}$ for $n \in \mathbb{N}$. Find

$$\lim_{n\to\infty} \int_0^{\frac{1}{2}} f_n(x)\,\mathrm{d}x.$$

First we note that it is not possible to explicitly evaluate the integrals for the f_n's. Ideally, we would like to use Theorem 5.29.

For $x \in [0, 1/2]$, we notice that

$$\|x^n\|_S \leq (\|x\|_S)^n \leq \left(\frac{1}{2}\right)^n \quad \text{and} \quad \|\sin^n(x)\|_S \leq \left(\frac{1}{2}\right)^n,$$

therefore, both x^n and $\sin^n(x)$ tend uniformly to 0. Hence, $\{f_n\}_{n=1}^{\infty}$ tends uniformly to $f(x) = \frac{e^x}{\ln 4}$ on $[0, 1/2]$ (since $\cos(x^n) \to \cos(0) = 1$).

Hence, using Theorem 5.29, we can interchange the limit with the integral which yields

$$\begin{aligned}\lim_{n\to\infty} \int_0^{\frac{1}{2}} f_n(x)\,\mathrm{d}x &= \int_0^{\frac{1}{2}} \lim_{n\to\infty} f_n(x)\,\mathrm{d}x \\ &= \int_0^{\frac{1}{2}} f(x)\,\mathrm{d}x = \int_0^{\frac{1}{2}} \frac{e^x}{\ln 4}\,\mathrm{d}x = \frac{e^{\frac{1}{2}} - 1}{2\ln 2}\end{aligned}$$

■ **Example 5.32** For $a \geq 0$, show that for the sequence of functions

$$f_n(x) = \begin{cases} \frac{\sin(nx)}{nx} & x \neq 0 \\ 1 & x = 0 \end{cases}$$

we have

$$\lim_{n\to\infty} \int_a^{\pi} f_n(x)\,\mathrm{d}x = 0.$$

If $a > 0$, then we have uniform convergence on $[a, \pi]$ to $f(x) = 0$:

$$\|f_n - f\|_{[a,\pi]} = \sup_{[a,\pi]} \left|\frac{\sin(nx)}{nx}\right| \leq \sup_{[a,\pi]} \left|\frac{1}{nx}\right| = \frac{1}{na} \leq \frac{1}{Na} < \epsilon$$

for all $n \geq N$ with $N \geq \dfrac{1}{a\epsilon}$. This implies that

$$\lim_{n\to\infty} \int_a^{\pi} f_n(x)\,\mathrm{d}x = \int_a^{\pi} \lim_{n\to\infty} f_n(x)\,\mathrm{d}x = 0.$$

Alternatively, we can find estimates on the integral of f_n.

$$\left|\int_a^{\pi} \frac{\sin(nx)}{nx}\,\mathrm{d}x\right| \leq \int_a^{\pi} \left|\frac{\sin(nx)}{nx}\right|\mathrm{d}x \leq \int_a^{\pi} \frac{1}{nx}\,\mathrm{d}x = \frac{1}{n}\ln\frac{\pi}{a}$$

so,

$$\lim_{n\to\infty} \left|\int_a^{\pi} \frac{\sin(nx)}{nx}\,\mathrm{d}x\right| \leq \lim_{n\to\infty} \frac{1}{n}\ln\frac{\pi}{a} = 0$$

and hence

$$\lim_{n\to\infty}\int_a^\pi \frac{\sin(nx)}{nx}\,\mathrm{d}x = 0.$$

Now we tackle the case $a = 0$. The functions $f_n(x)$ on $[0,\pi]$ converge pointwise to the function

$$f(x) = \begin{cases} 0 & x \neq 0 \\ 1 & x = 0 \end{cases}$$

but convergence cannot be uniform (why?). Note that we also cannot evaluate

$$\frac{1}{n}\ln\frac{\pi}{a}$$

at $a = 0$. So, we cannot use either argument that was used for the case $a > 0$.

For simplicity we use the change of variable $y = nx$. So with

$$\int_0^\pi \frac{\sin(nx)}{nx}\,\mathrm{d}x = \frac{1}{n}\int_0^{n\pi}\frac{\sin y}{y}\,\mathrm{d}y$$

and by additivity

$$\int_0^{n\pi}\frac{\sin y}{y}\,\mathrm{d}y = \int_0^\pi \frac{\sin y}{y}\,\mathrm{d}y + \int_\pi^{n\pi}\frac{\sin y}{y}\,\mathrm{d}y.$$

Now on the interval $[\pi, n\pi]$,

$$\left|\int_\pi^{n\pi}\frac{\sin y}{y}\,\mathrm{d}y\right| \leq \int_\pi^{n\pi}\left|\frac{\sin y}{y}\right|\,\mathrm{d}y \leq \int_\pi^{n\pi}\frac{1}{y}\,\mathrm{d}y = \ln(n)$$

The function $\frac{\sin y}{y}$ is bounded by 1 on $(0,\pi]$ and so

$$\int_0^\pi \frac{\sin y}{y}\,\mathrm{d}y \leq \pi.$$

This yields

$$\left|\int_0^\pi \frac{\sin(nx)}{nx}\,\mathrm{d}x\right| \leq \left|\frac{1}{n}\int_0^\pi\frac{\sin y}{y}\,\mathrm{d}y\right| + \left|\frac{1}{n}\int_\pi^{n\pi}\frac{\sin y}{y}\,\mathrm{d}y\right| \leq \frac{1}{n}(\ln(n)+\pi)$$

and hence

$$\lim_{n\to\infty}\left|\int_0^\pi \frac{\sin(nx)}{nx}\,\mathrm{d}x\right| \leq \lim_{n\to\infty}\frac{1}{n}(\ln(n)+\pi) = 0.$$

■ **Example 5.33** Let the sequence of functions $\{f_n(x)\}_{n=2}^\infty$ be defined by

$$f_n(x) = \begin{cases} n^2 x & \text{if} \quad 0 \leq x \leq \frac{1}{n} \\ 2n - n^2 x & \text{if} \quad \frac{1}{n} \leq x \leq \frac{2}{n} \\ 0 & \text{if} \quad \frac{2}{n} \leq x \leq 1 \end{cases}$$

Let's find the integral of the pointwise limit and the limit of the integral. The pointwise limit is $f(x) = 0$ since $f_n(x) = 0$ for all $n > \frac{2}{x}$, and for $x = 0$, then $f_n(0) = 0$ for all n.

The integral of this pointwise limit is

$$\int_0^1 \lim_{n\to\infty} f_n(x)\mathrm{d}x = \int_0^1 0\,\mathrm{d}x = 0.$$

However,

$$\int_0^1 f_n(x)\,\mathrm{d}x = 1$$

and the limit of this as $n \to \infty$ is 1. So, we conclude that

$$\lim_{n\to\infty} \int_0^1 f_n(x)\,\mathrm{d}x \neq \int_0^1 \lim_{n\to\infty} f_n(x)\,\mathrm{d}x.$$

This example shows us that *pointwise convergence does not necessarily commute with integration.* Generally, for some sequences of functions,

$$\lim_{n\to\infty} \int_a^b f_n(x)\,\mathrm{d}x \neq \int_a^b \lim_{n\to\infty} f_n(x)\,\mathrm{d}x.$$

5.4 Interchanging Limits and Derivatives

A fundamental result of uniform continuity tells us that if we have a sequence of functions, $\{f_n\}_{n=1}^{\infty}$, which are uniformly continuous and they converge uniformly, then the function they converge to, f, must also be uniformly continuous. Now, we will extend this further to the situation where the f_n's are *differentiable functions*, provided the derivatives also converge uniformly.

Theorem 5.34 — Convergence of differentiable functions. Let $\{f_n\}_{n=1}^{\infty}$ be a sequence of continuously differentiable functions defined on a finite interval $[a,b]$. If $f_n' \to g$ uniformly on $[a,b]$ and the real sequence $\{f_n(p)\}_{n=1}^{\infty}$ converges for some $p \in [a,b]$ then we have the following

(a) $\{f_n\}_{n=1}^{\infty}$ converges uniformly to a function f on $[a,b]$.

(b) f is continuously differentiable and $f' = g$

Proof. Applying the Second Fundamental Theorem of Calculus to the functions f_n, we have

$$f_n(x) - f_n(p) = \int_p^x f_n'(t)\,\mathrm{d}t.$$

Now, $\{f_n(p)\}_{n=1}^{\infty}$ converges and we will denote its limit by $f(p)$. We are also given that $f_n' \to g$ uniformly on $[a,b]$, which means that g is continuous on $[a,b]$ as uniform convergence preserves continuity. Since $[p,x] \subset [a,b]$, we get

$$f(x) = \lim_{n\to\infty} f_n(x) = f(p) + \int_p^x g(t)\,\mathrm{d}t.$$

By using the First Fundamental Theorem of Calculus, we know that f is continuously differentiable with $f' = g$.

What remains is to show that $\{f_n\}_{n=1}^{\infty}$ converges uniformly on $[a,b]$. To this end, consider the difference $f_n(x) - f(x)$. We have

$$\begin{aligned}
|f_n(x) - f(x)| &= \left|(f_n(p) - f(p)) + \int_p^x (f_n'(t) - g(t))\,\mathrm{d}t\right| \\
&\leq |f_n(p) - f(p)| + \int_p^x |f_n'(t) - g(t)|\,\mathrm{d}t \\
&\leq |f_n(p) - f(p)| + |x-p|\|f_n' - g\|_S \\
&\leq |f_n(p) - f(p)| + (b-a)\|f_n' - g\|_S
\end{aligned}$$

where we have used the triangle inequality, a property we previously used about the definite integral, and that $|x - p| \leq b - a$. Since the last bound is independent of x, we have

$$\|f_n - f\|_S = \sup_{x\in[a,b]} |f_n(x) - f(x)| \leq |f_n(p) - f(p)| + (b-a)\|f_n' - g\|_S$$

where the expression on the right $\to 0$ as $n \to \infty$ from the convergence of the sequence $f_n(p) \to f(p)$ and the uniform convergence $f_n' \to g$. ■

▪ **Example 5.35** Let $g_n(x) : [0, \infty)$ be defined as

$$g_n(x) = \frac{e^{-nx}}{n}.$$

What can we say about the relation between the limit of g_n' and the derivative of the limit of g_n?

On $[0, \infty)$, $g_n(x) \leq \frac{1}{n}$ and so g_n converges uniformly to $g = 0$. However the derivative, $g_n'(x) = -e^{-nx}$, where we only have a pointwise limit to 0 if $x > 0$ and to 1 if $x = 0$. So the derivative of the limit $g' = 0$ does not equal to the limit of the derivatives.

However, this is not a contradiction to Theorem 5.34 as g_n' does not converge uniformly on any interval containing $x = 0$. This will also be the case if we had a finite interval $[0, a]$.